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Search for the $B \to X_s \nu\bar{\nu}$ decays at the Belle II experiment
(Belle II実験における$B \to X_s \nu\bar{\nu}$崩壊過程の探索)

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Department of Physics, Graduate School of Science,
The University of Tokyo
東京大学大学院理学系研究科
物理学専攻

Junewoo Park

Abstract

We present a search for the $B \to X_s \nu\bar{\nu}$ decay, where X_s is a hadronic system containing an s quark, using data collected with the Belle II detector at the SuperKEKB accelerator. The search uses a data sample corresponding to an integrated luminosity of 365 fb^{-1} collected at the $\Upsilon(4S)$ resonance. An additional off-resonance sample, collected at a center-of-mass energy 60 MeV below the $\Upsilon(4S)$ resonance, is used to correct and validate the modeling of the $e^+e^- \to q\bar{q}$ continuum background. We train a machine-learning classifier to separate signal from background and extract the signal yield from the two-dimensional distribution of the classifier output and the reconstructed mass of the X_s system, $M_{X_s}^{\rm reco}$. We find no significant signal and set upper limits on the partial branching fractions in three regions of the true X_s mass, M_{X_s}. For $0.0 < M_{X_s} < 0.6\,\mathrm{GeV}/c^2$, $0.6 < M_{X_s} < 1.0\,\mathrm{GeV}/c^2$, and $1.0\,\mathrm{GeV}/c^2 < M_{X_s}$, the upper limits at 90% confidence level are 2.2×10^{-5}, 9.3×10^{-5}, and 30.9×10^{-5}, respectively. Combining the three mass regions, we obtain an upper limit on the branching fraction of $B(B \to X_s \nu\bar{\nu}) < 3.3 \times 10^{-4}$ at 90% confidence level.

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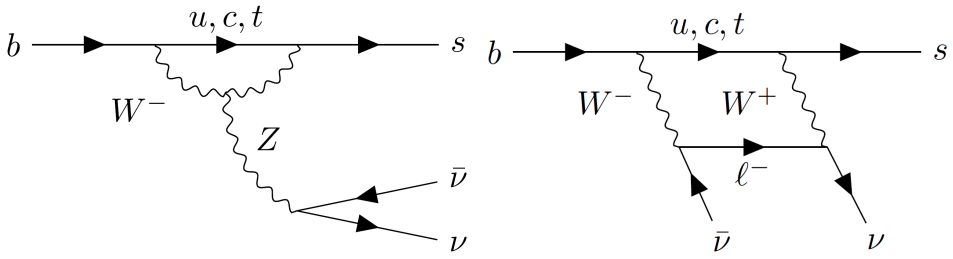
Chapter 1

Introduction

1.1 Theoretical Overview

$B \to X_s \nu\bar{\nu}$ is a flavour-changing neutral-current (FCNC) process and cannot occur at tree level. The dominant contributions arise from electroweak penguin and box diagrams, shown in Figure 1.1. The $b \to s\nu\bar{\nu}$ transition is described by the following effective



(a) penguin diagram (b) box diagram

Figure 1.1: Dominant Feynman diagrams for $B \to X_s \nu\bar{\nu}$ decay

Hamiltonian [3]:

$$\mathcal{H}_{\text{eff}} = -\frac{4G_F}{\sqrt{2}} V_{tb} V_{ts}^* (C_L^\nu \mathcal{O}_L^\nu + C_R^\nu \mathcal{O}_R^\nu) + \text{h.c.}, \tag{1.1}$$

where $C_{L,R}^\nu$ are the Wilson coefficients, and $\mathcal{O}_{L,R}^\nu$ are the operators defined as follows [3]:

$$\mathcal{O}_L^\nu = \frac{e^2}{16\pi^2}(\bar{s}\gamma_\mu P_L b)(\bar{\nu}\gamma^\mu(1-\gamma_5)\nu), \quad \mathcal{O}_R^\nu = \frac{e^2}{16\pi^2}(\bar{s}\gamma_\mu P_R b)(\bar{\nu}\gamma^\mu(1-\gamma_5)\nu). \tag{1.2}$$

As shown in Eq. 1.2, the effective Hamiltonian description reduces the decay to a single point-like interaction. In the $b \to s\nu\bar{\nu}$ transition, this is a valid approximation because the b-quark mass is much smaller than the W-boson and top-quark masses. In the Standard model (SM), the weak interaction couples only to left-handed fermions. Therefore,

$$C_R^\nu = 0 \tag{1.3}$$

in the SM, the effective Hamiltonian reduces to

$$\mathcal{H}_{\text{eff}} = -\frac{4G_F}{\sqrt{2}} V_{tb} V_{ts}^* C_L^\nu \mathcal{O}_L^\nu + \text{h.c.}. \tag{1.4}$$

1.1.1 $B \to K\nu\bar{\nu}$

From Eq. 1.4, the decay amplitude for $B \to K\nu\bar{\nu}$ can be written as follows [4]:

$$\mathcal{M}(b \to s\nu\bar{\nu}) \propto f_+(s) \left\{ (p_B + p)_\mu - \frac{m_B^2 - m_K^2}{s} q_\mu \right\} (\bar{\nu}\gamma^\mu(1-\gamma_5)\nu)\,, \tag{1.5}$$

where $q = p_\nu + p_{\bar{\nu}}$ and $s = q^2$. The form factor f_+ can be parametrized using the z expansion [5, 6]:

$$f_+(q^2) = \frac{1}{1 - q^2/m_+^2} \left[\alpha_0 + \alpha_1 z(q^2) + \alpha_2 z^2(q^2) + \frac{z^3(q^2)}{3}(-\alpha_1 + 2\alpha_2) \right], \tag{1.6}$$

where

$$z(t) = \frac{\sqrt{t_+ - t} - \sqrt{t_+ - t_0}}{\sqrt{t_+ - t} + \sqrt{t_+ - t_0}}, \tag{1.7}$$

with $t_\pm = (m_B \pm m_K)^2$ and $t_0 = t_+(1 - \sqrt{1 - t_-/t_+})$.

1.1.2 $B \to K^*\nu\bar{\nu}$

From Eq. 1.4, the decay amplitude for $B \to K^*\nu\bar{\nu}$ can be written as follows [4, 7]:

$$\mathcal{M}(b \to s\nu\bar{\nu}) \propto \mathcal{T}_\mu \, (\bar{\nu}\gamma^\mu(1-\gamma_5)\nu)\,, \tag{1.8}$$

where

$$\mathcal{T}_\mu = (m_B + m_{K^*}) A_1(s) \epsilon_\mu^* - A_2(q^2) \frac{\epsilon^* \cdot q}{m_B + m_{K^*}} (p + p_{K^*})_\mu + i \frac{2V(s)}{m_B + m_{K^*}} \epsilon_{\mu\nu\rho\sigma} \epsilon^{*\nu} p^\rho p_{K^*}^\sigma, \tag{1.9}$$

with three form factors A_1, A_2, and V. These three form factors can be parametrized using the z expansion [8]:

$$i = \frac{q}{1 - q^2/m_{R,i}^2} \left[\alpha_0^i + \alpha_1^i (z - z_0) + \alpha_2^i (z - z_0)^2 \right], \tag{1.10}$$

(1.11)

with $i = A_1, A_{12}, V$. Here,

$$A_2 = \left[(m_B + m_{K^*})^2 (m_B^2 - m_{K^*}^2 - q^2) A_1 - 16 m_B m_{K^*} (m_B + m_{K^*}) A_{12} \right] / \lambda(q^2), \tag{1.12}$$

where

$$\lambda(q^2) = m_B^4 \lambda(1, m_{K^*}^2/m_B^2, q^2/m_B^2), \tag{1.13}$$

with the Källén function $\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$.

1.1.3 $B \to X_s \nu\bar{\nu}$

For the inclusive $B \to X_s \nu\bar{\nu}$ decay, the branching ratio can be written as follows [3]:

$$\frac{d\Gamma}{dq^2} \propto \sqrt{\lambda(1, \hat{m}_s^2, s_b)} \left[3s_b(1 + \hat{m}_s^2 - s_b - 4\hat{m}_s) + \lambda(1, \hat{m}_s^2, s_b)\right], \tag{1.14}$$

where $\hat{m}_s = m_s/m_b$ and $s_b = q^2/m_b^2$.

1.2 Theory beyond the Standard Model

1.2.1 Flavour Violating $\bar{b}sZ'$ Coupling

If there is flavour violating $\bar{b}sZ'$ coupling, the effective Lagrangian for the $b \to s\nu\bar{\nu}$ transition can be written as [3]:

$$\mathcal{L}_{\text{eff}}^{\bar{b}sZ'} = \frac{G_F}{\sqrt{2}} \frac{e}{\pi^2} m_{Z'}^2 \cos\theta_w \sin\theta_w V_{tb}^* V_{ts} Z'^{\mu} (Z'_L \bar{b}\gamma_\mu P_L s + Z'_R \bar{b}\gamma_\mu P_R s), \tag{1.15}$$

and then the Wilson coefficients in Eq. 1.2 become [3]

$$\begin{aligned} C_L^\nu &= (C_L^\nu)^{\text{SM}} - \frac{g'^\nu_V}{2} Z'_L, \\ C_R^\nu &= -\frac{g'^\nu_V}{2} Z'_R. \end{aligned} \tag{1.16}$$

On the other hand, the Wilson coefficients affecting $b \to s\ell^+\ell^-$ become [3]:

$$\begin{aligned} C_{10} &= C_{10}^{\text{SM}} + \frac{g'^\ell_A}{2} Z'_L, \\ C'_{10} &= +\frac{g'^\ell_A}{2} Z'_R, \\ C_9 &= C_9^{\text{SM}} - \frac{g'^\ell_V}{2} Z'_L, \\ C'_9 &= -\frac{g'^\ell_V}{2} Z'_R. \end{aligned} \tag{1.17}$$

As seen in Eq. 1.16 and 1.17, the Wilson coefficients for $b \to s\nu\bar{\nu}$ and $b \to s\ell^+\ell^-$ are affected by different g' couplings. Therefore, in this model, the branching ratio for $B \to X_s\nu\bar{\nu}$ can be enhanced, while suppressing new-physics contributions to $B \to X_s\ell^+\ell^-$ decay. Consequently, $B \to X_s\nu\bar{\nu}$ can probe new-physics models that remain unconstrained even by precise measurements of the similar FCNC process, $B \to X_s\ell^+\ell^-$.

1.2.2 Invisible Light Scalar

For the invisible light scalar, S, the $b \to sSS$ transition produces the same visible final-state particles as the $b \to s\nu\bar{\nu}$ transition. For the $b \to sSS$ transition, the effective

514 Lagrangian can be written as follows [9]:

$$\mathcal{L}_{\text{eff}}^{b\to sSS} = \frac{1}{\Lambda^2}\left[C_{S_1} m_b \bar{s}_L b_R S^2 + C_{S_2} m_b \bar{b}_L s_R S^2 + \text{h.c.}\right], \tag{1.18}$$

515 where C_{S_1} and C_{S_2} are Wilson coefficients. Consequently, the search for the $B \to X_s \nu\bar{\nu}$
516 decay can constrain C_{S_1} and C_{S_2}. Additionally, an observed excess in the $B \to X_s \nu\bar{\nu}$
517 decay rate could provide evidence for an invisible light scalar.

Chapter 2

Belle II Experiment

In the Belle II experiment, the e^+e^- beams are accelerated by SuperKEKB, and collision events are detected by the Belle II detector. The center-of-mass energy of the Belle II experiment is mainly set to 10.58 GeV, corresponding to the $\Upsilon(4S)$ resonance. Data at energies below and above the $\Upsilon(4S)$ resonance are also collected. Because $e^+e^- \to q\bar{q}$, where q denotes a u, d, s, or c quark, and $e^+e^- \to \tau^+\tau^-$ processes also occur at this center-of-mass energy, we can study not only B physics, but also charm physics and tau physics.

2.1 SuperKEKB

This section summarizes the SuperKEKB accelerator based primarily on the SuperKEKB Design Report [10]. Figure 2.1 shows the schematic overview of the SuperKEKB accelerator [11]. The SuperKEKB main ring consists of the High Energy Ring (HER) and the Low Energy Ring (LER). In the HER and LER, 7 GeV electron beams and 4 GeV positron beams circulate, respectively. For the electron beam in the HER, a photo-cathode RF gun is used to obtain low-emittance and high-current electrons. For the positron beams in the LER, high-current primary electrons are first produced by a thermionic gun. Then positrons are obtained by making these primary electrons collide with a tungsten target. Electrons and positrons are accelerated to 7 GeV and 4 GeV, respectively, then injected into the main ring.

In a flat beam, where the beam size in x-axis is much larger than the beam size in y-axis, the luminosity $\mathcal{L}$ can be calculated by the following formula [12]:

$$\mathcal{L} \propto \frac{I_+ I_-}{\beta^*_{y+}\beta^*_{y-}}, \tag{2.1}$$

where I is the beam currents, β^*_y is the vertical beta function at the interaction point (IP), and the subscripts $+$ and $-$ denote the e^+ and e^- beams, respectively. To achieve higher luminosity, the KEKB accelerator, the predecessor of SuperKEKB, was upgraded. Table 2.1 shows the parameter values and luminosity for KEKB and SuperKEKB acceler-

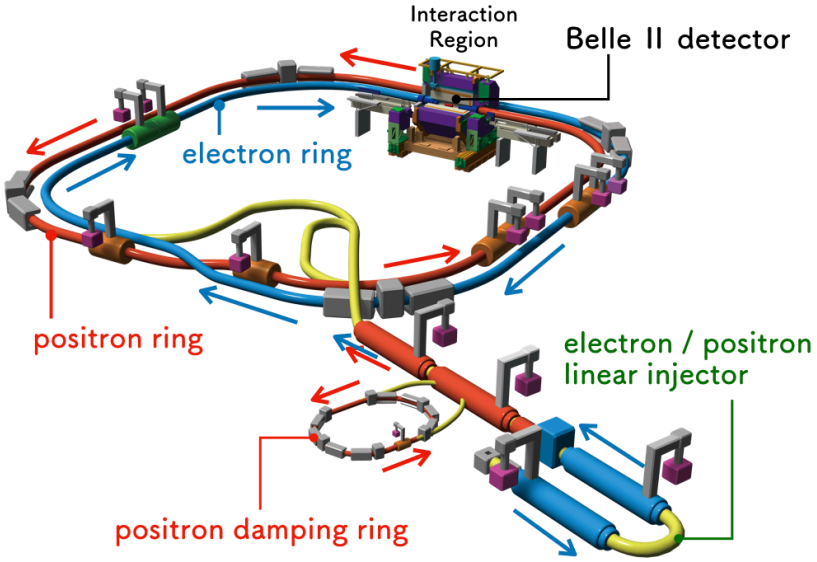


Figure 2.1: Schematic overview of the SuperKEKB accelerator.

Table 2.1: Comparison of parameters and luminosity between KEKB and SuperKEKB accelerators

	KEKB Achieved		SuperKEKB Design	
	LER	HER	LER	HER
I_{beam} [A]	1.637	1.188	3.6	2.6
β_y^* [mm]	5.9	5.9	0.27	0.30
$\mathcal{L}$ [10^{34} cm^{-2}s^{-1}]	2.11		80	
$\int \mathcal{L}dt$ [ab^{-1}]	1.04		50	

ator [12]. As seen in Eq. 2.1 and Table 2.1, the beam currents are increased to achieve higher luminosity during the upgrade from KEKB to SuperKEKB. According to Eq. 2.1, decreasing β_y^* can also improve the luminosity. However, because of the interaction between beams in head-on collision, there is a constraint on β_y^* [10]:

$$\beta_y^* > \sigma_z, \tag{2.2}$$

where σ_z is the bunch length. In SuperKEKB accelerator, the beams collide with a crossing angle. Therefore, the overlapped length between beams are shorter than head-on collision, allowing SuperKEKB to achieve smaller β_y^* value.

2.2 Belle II detector

Figure 2.2 shows a schematic overview of the Belle II detector [13]. The Belle II detector consists of seven sub-detectors: the Pixel Detector (PXD), the Silicon Vertex Detector (SVD), the Central Drift Chamber (CDC), the Time-of-Propagation (TOP) counter, the Aerogel Ring-Imaging Cherenkov counter (ARICH), the Electromagnetic Calorimeter (ECL), and the K_L^0 and μ detector (KLM). In this section, the concepts of the sub-detector systems, trigger system, and data acquisition system are discussed.

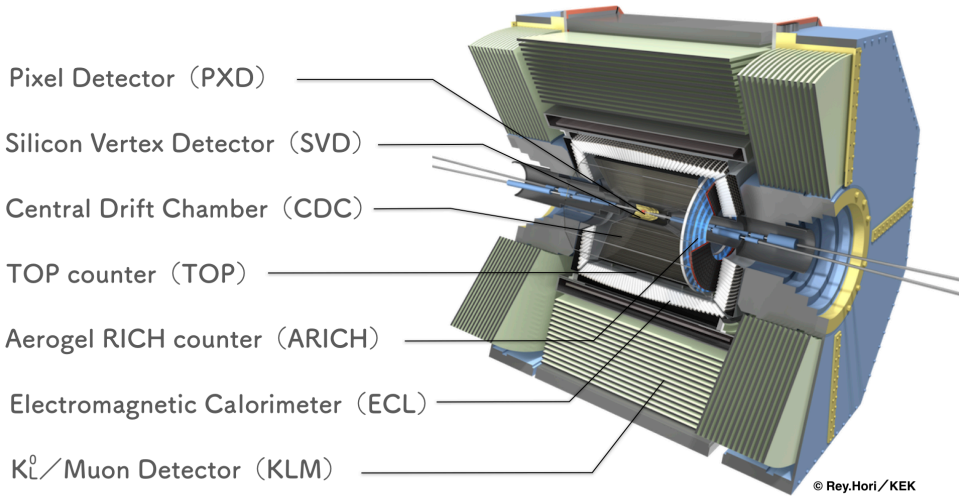


Figure 2.2: Schematic overview of the Belle II detector.

In the Belle II experiment, a right-handed Cartesian coordinate system, (x, y, z), is used. The origin is defined as the interaction point (IP). The x-axis points horizontally outward from the center of the main ring. The y-axis points vertically upward. The z-axis is aligned with the direction of the solenoidal magnetic field and coincides with the bisector of the LER and HER beams. The azimuthal angle ϕ is defined by $\sin\phi = y/\sqrt{x^2+y^2}$ and $\cos\phi = x/\sqrt{x^2+y^2}$. The polar angle θ is defined by $\sin\theta = \sqrt{x^2+y^2}/\sqrt{x^2+y^2+z^2}$ and $\cos\theta = z/\sqrt{x^2+y^2+z^2}$.

2.2.1 Pixel Detector (PXD)

Figure 2.3 shows a schematic overview of the PXD [14]. The PXD consists of two layers. In each layer, PXD modules are arranged in a windmill structure [15]. The inner layer has a radius of 14 mm and consists of 8 planar modules. The outer layer has a radius of 22 mm and consists of 12 planar modules.

The PXD sensor uses the DEPleted Field Effect Transistor (DEPFET) technology [16]. DEPFET sensors have low power consumption, and therefore require only air cooling [14]. Additionally, DEPFET sensors also have internal amplification. These

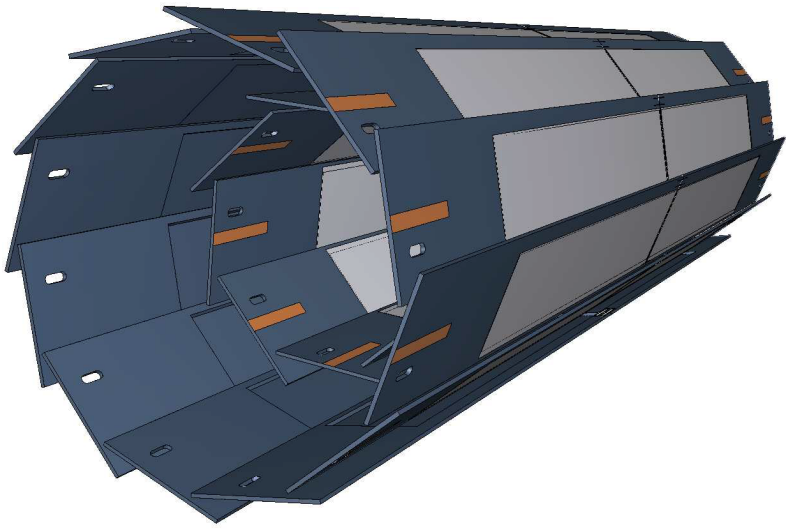


Figure 2.3: Schematic overview of the PXD.

features allow DEPFET sensors to be thin (75 μm), resulting in a low material budget. Each PXD module consists of 250×768 DEPFET pixel sensors.

The resolution in the xy plane is 13.64 ± 0.11 μm. The resolution along the z-axis is 14.92 ± 0.07 μm [15].

2.2.2 Silicon Vertex Detector (SVD)

Figure 2.4 shows a schematic overview of the SVD [17]. The SVD consists of four layers.

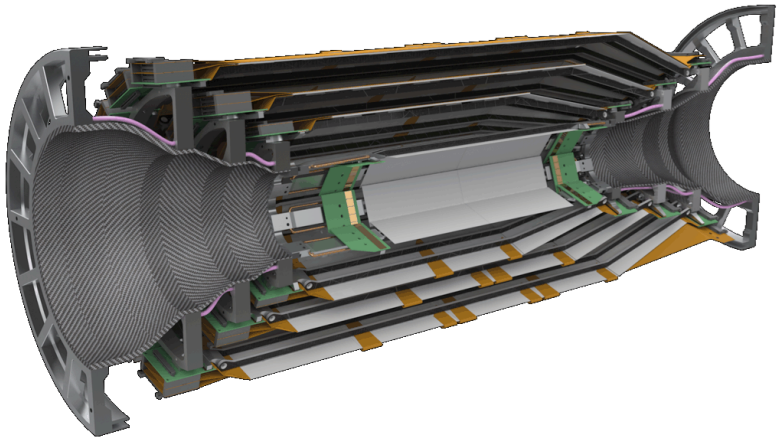


Figure 2.4: Schematic overview of the SVD.

Each layer consists of ladders, which are independent submodules arranged in a windmill structure. Table 2.2 shows the geometrical configuration of the SVD [14]. The SVD uses double-sided silicon strip detector (DSSD) technology.

The SVD adopts the origami chip-on-sensor concept [18]. In the origami scheme, a

Table 2.2: geometrical configuration of the SVD.

Layer	Radius (mm)	Ladders	Sensors/ladder	Sensors
6	140	17	5	85
5	115	14	4	56
4	80	10	3	30
3	38	8	2	16

readout chip is placed on one side of the DSSD. The other side of the DSSD is connected to the readout chip by flexible electronic circuits. This origami structure requires a single cooling channel, which leads to a low material budget [17].

2.2.3 Central Drift Chamber (CDC)

Figure 2.5 shows a schematic overview of the longitudinal cross section of the CDC [19]. A mixture of 50% He and 50% C_2H_6 is used because of its long radiation length, high

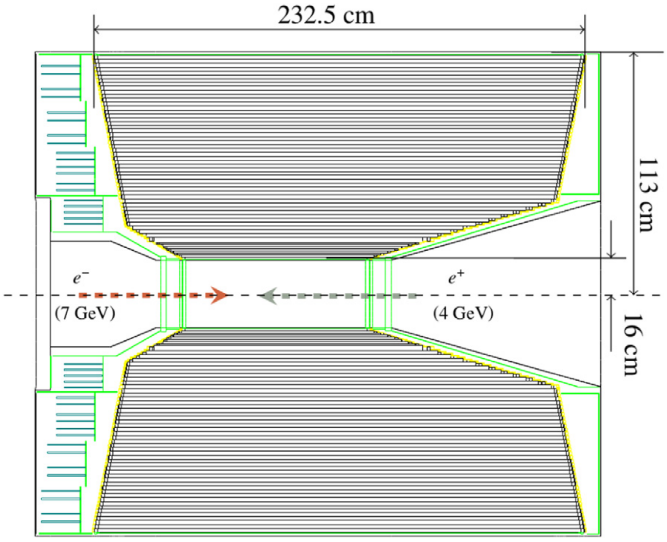
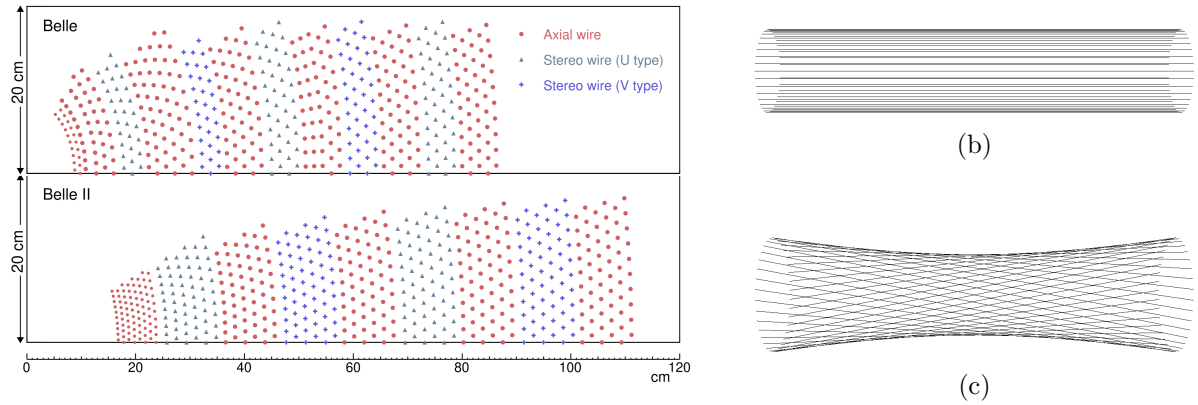


Figure 2.5: Longitudinal cross section of the CDC.

drift velocity, and good energy loss resolution [14].

The CDC consists of 56 wire layers. The layers can be classified into two types: axial and stereo layers. In the axial layers, wires are parallel to the z-axis. On the other hand, in the stereo layer, wires are skewed with respect to the z-axis. These different types of layers are needed to determine the three-dimensional positions of tracks. Adjacent layers of the same type are grouped into superlayers, and there are five axial superlayers and four stereo superlayers. Figure 2.6 shows the layer layouts of the CDC in the Belle and Belle II experiments [19], together with schematic illustrations of axial and stereo layers [20]. The U and V types of stereo wires denote the skew directions of the wires.



(a) (b) (c)

Figure 2.6: (a) Layer layouts of the CDC in the Belle and Belle II experiments. The U and V types of stereo wires denote the skew directions of the wires. (b) Schematic illustration of an axial layer. (c) Schematic illustration of a stereo layer.

2.2.4 Time-of-Propagation (TOP) counter

The Time-of-Propagation (TOP) detector is installed for particle identification in the barrel region of the Belle II detector. Figure 2.7 shows a schematic overview of the TOP detector and a TOP module [21]. The TOP detector consists of 16 modules.

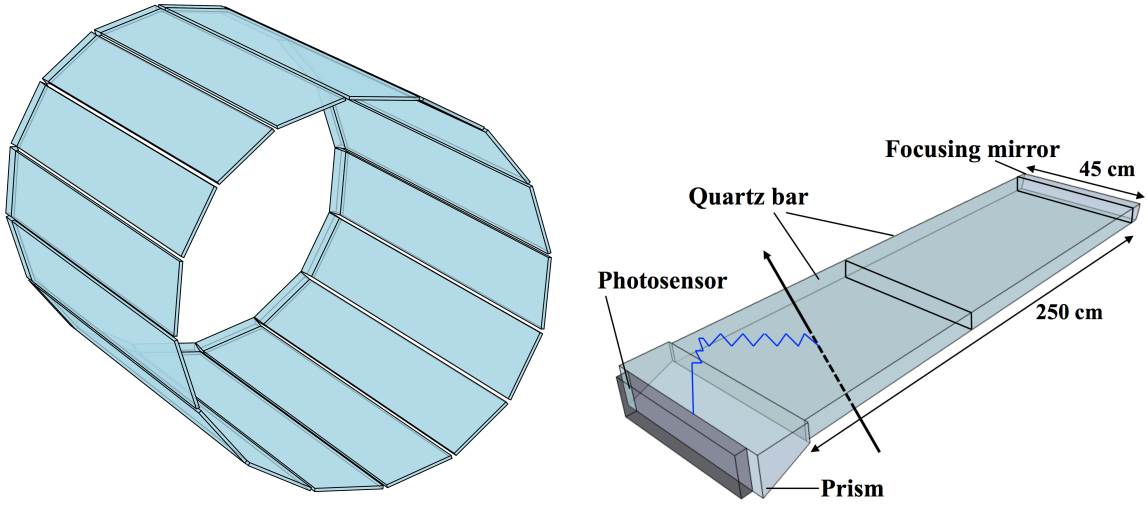


Figure 2.7: (Left) Schematic overview of the TOP detector and (right) a TOP module.

The TOP detector uses Cherenkov radiation to identify particles. When a particle with speed $c\beta$ passes through a material, Cherenkov radiation is emitted at an angle θ_c [22]:

$$\cos\theta_c = \frac{1}{n\beta}, \tag{2.3}$$

where n is the refractive index of the material. For the same momentum, the Cherenkov angle θ_c is larger for π than for K. Consequently, Cherenkov photons from π arrive at the

photon sensors earlier than those from K. This difference in time of propagation enables particle identification. Figure 2.8 shows a schematic overview of Cherenkov radiation for K and π [23], and simulated time-of-propagation distributions [24]. As seen in the figure, there is a difference in the time of propagation between π and K.

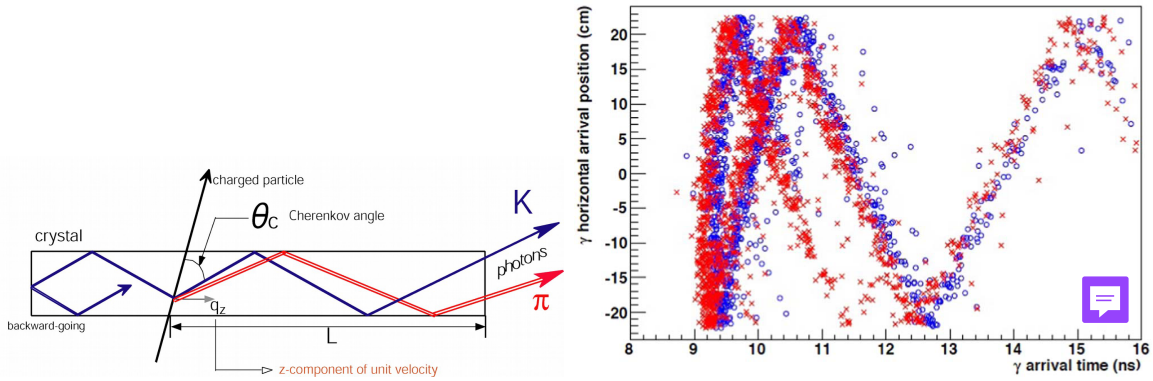


Figure 2.8: (left) Schematic overview of Cherenkov radiation for K and π. (right) the x-axis shows the arrival time of the radiation, and y-axis is horizontal detection position. The blue circles correspond to K, and the red crosses correspond to π.

2.2.5 Aerogel Ring-Imaging Cherenkov counter (ARICH)

For particle identification in the end-cap region, the Aerogel Ring-Imaging Cherenkov (ARICH) detector is used. Similar to the TOP detector, the ARICH detector uses Cherenkov radiation. Instead of using the propagation time, the ARICH detector uses ring images of Cherenkov radiation.

In the ARICH detector, aerogel layers are used as radiators. To increase the resolution, there are two aerogel layers whose refractive indices are 1.0467 and 1.0592, respectively. To detect photons, hybrid avalanche photo-detectors (HAPDs) are used. Figure 2.9 shows schematic illustrations of the two aerogel layers and the ring image produced by the ARICH detector [25]. From this ring image, the Cherenkov angle can be measured directly. Figure 2.10 shows a schematic overview of the arrangement of the photon detectors and aerogel radiators, together with the ARICH detector [26].

2.2.6 Electromagnetic Calorimeter (ECL)

In the Belle II experiment, a homogeneous electromagnetic calorimeter is used. The ECL consists of CsI(Tl) crystals, whose cross section is 6×6 cm^2 on average and whose length is 30 cm ($16.2X_0$) [27]. There are 8736 crystals in total: 1152 in the forward endcap, 960 in the backward endcap, and 6624 in the barrel region. Figure 2.11 shows a schematic overview of the ECL and a CsI(Tl) crystal [27].

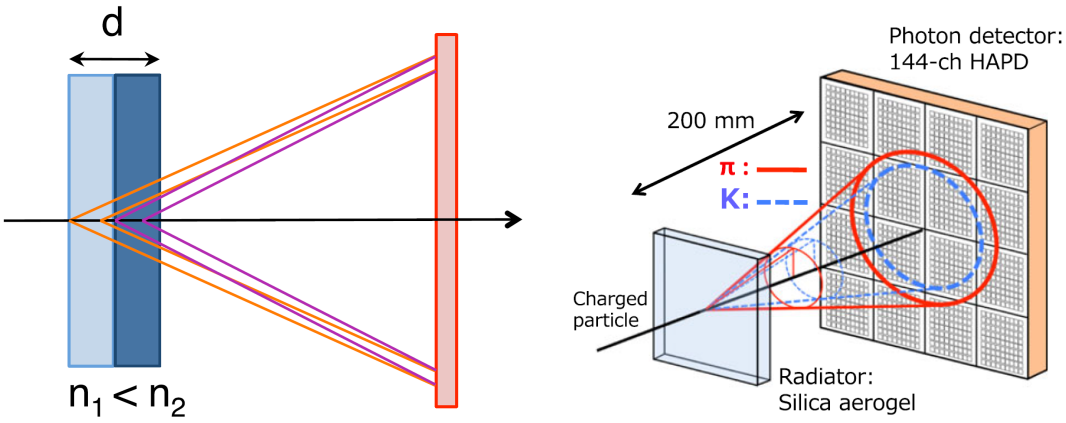


Figure 2.9: (Left) Schematic overview of the two aerogel layers. (Right) Schematic overview of a ring image produced by the ARICH detector.

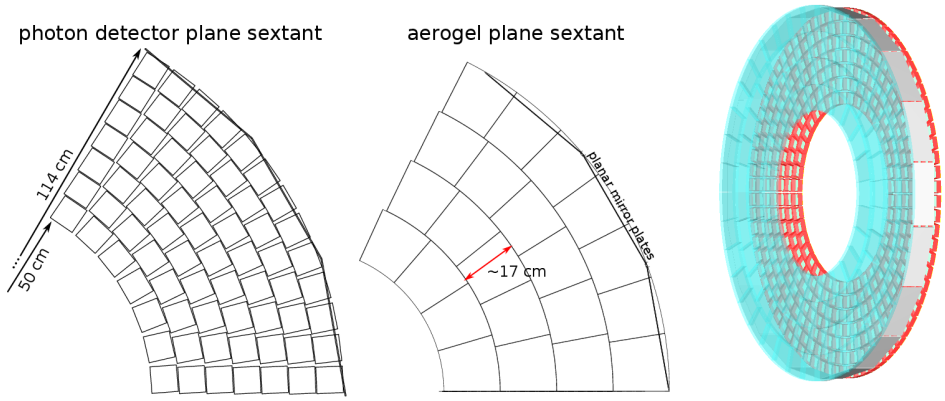


Figure 2.10: (Left) Schematic overview of the arrangement of the photon detectors and aerogel radiators. (Right) Schematic overview of the ARICH detector.

Because the Belle II experiment has higher luminosity than the Belle experiment, more fake clusters are produced by beam particles [27]. Using the signal arrival time, these fake clusters can be suppressed. In the Belle II experiment, the ECL signals are continuously digitized to obtain waveform information. The waveform is fitted using a known response function to obtain the signal arrival time.

2.2.7 K_L^0 and μ detector (KLM)

The KLM detector was designed to detect K_L^0 mesons and identify muons. In the end-cap region, the KLM consists of scintillator strips [28]. The scintillation light is collected by wavelength-shifting (WLS) fibers, and the readout is performed by silicon photomultipliers. Two orthogonal scintillator layers compose a superlayer, which is covered with an aluminum sheet. These superlayers are installed between iron plates serving as the magnetic flux return for the solenoid. There are 28 superlayers in total in the end-cap region: 14 in the forward endcap and 14 in the backward endcap. Figure 2.12 shows a

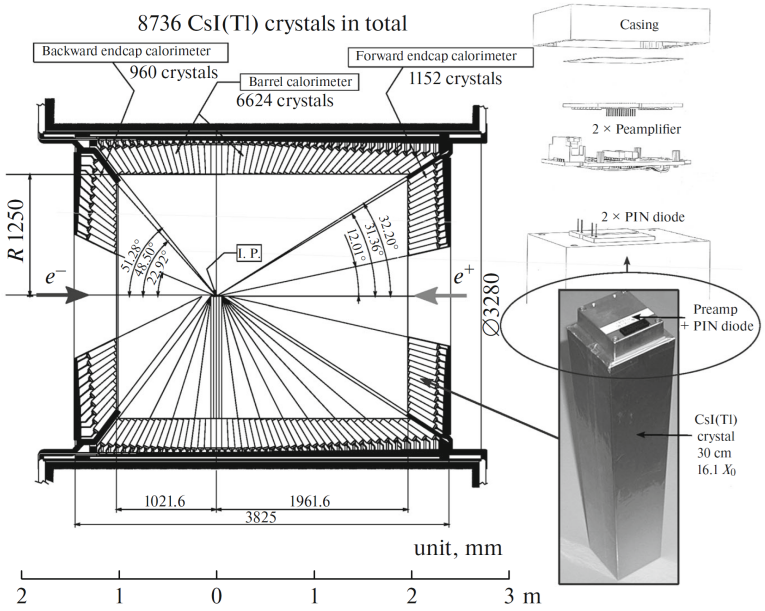


Figure 2.11: (Left) Schematic overview of the ECL. (Right) Schematic overview of the CsI(TI) crystal.

schematic overview of the scintillator strip and superlayer [28].

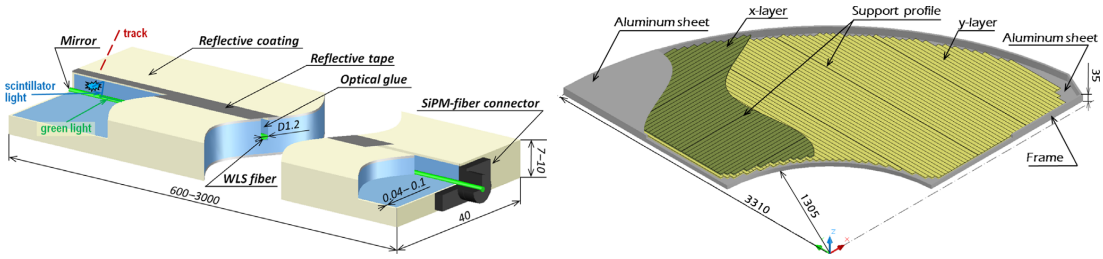


Figure 2.12: (Left) Schematic overview of a scintillator strip. (Right) Schematic overview of a superlayer.

In the barrel region, resistive plate counters (RPCs) used in the Belle experiment are mainly adopted [29]. During the Belle II upgrade, the two innermost RPC layers were replaced by scintillator layers. There are 13 RPC layers in the Belle II experiment.

2.2.8 Trigger and data acquisition systems

To meet the hardware limitations, data must be recorded selectively while keeping physically interesting events. The first stage of the trigger system is called the Level-1 trigger. In the Level-1 trigger, a trigger efficiency close to 100% is required for hadronic events, while keeping the trigger rate below 30 kHz [30]. The Level-1 trigger consists of sub-triggers, global reconstruction logic (GRL), and global decision logic (GDL). Figure 2.13 shows a schematic overview of the Level-1 trigger [30]. There are four sub-triggers: CDC,

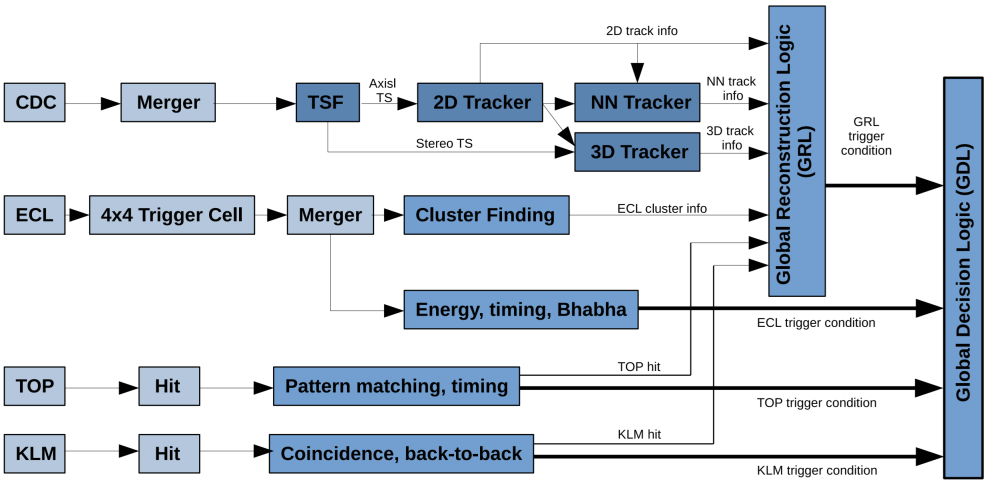


Figure 2.13: Schematic overview of the Level-1 trigger.

ECL, TOP, and KLM triggers. The GRL receives information from the four sub-triggers and analyzes the event topology. The GDL receives information from the GRL and the sub-triggers and produces output bits.

After an event is accepted by the Level-1 trigger, the detector signals are sent to the readout PCs. The event builder combines detector data to form one collision event [31]. A high-level trigger (HLT) uses the Belle II Analysis Software Framework [32] to reconstruct the events [33]. The HLT determines whether to retain or discard the reconstructed events.

Chapter 3

Software and Data Samples

3.1 Software

The Belle II software framework (BASF2) [32] is used to reconstruct events. The BASF2 version used in this analysis is `light-2305-korat`. BASF2 produces ROOT files that include kinematic and event information. To analyze these ROOT files, the ROOT data analysis framework [34] is used..

For Monte Carlo (MC) sample generation, EvtGen [35] is used to simulate B- and D-meson decay chains. KKMC [36] is used to simulate $e^+e^- \to q\bar{q}$ events. PHOTOS [37] is used to simulate photon radiation from charged particles. Pythia8 [38] is used to simulate quark fragmentation. Geant4 [39] is used to simulate the detector response.

3.2 Monte Carlo samples

To avoid human bias, the data collected from the Belle II experiment are not analyzed before the analysis procedure is determined. Before checking the data, the analysis procedure is optimized using MC samples.

3.2.1 Background Monte Carlo samples

The official MC15ri samples, the official MC15rd samples, and private MC samples at the $\Upsilon(4S)$ resonance are used. Additionally, the official MC15rd samples at a center-of-mass energy 60 MeV below resonance are used in the analysis. The on-resonance samples consist of $e^+e^- \to \Upsilon(4S) \to B\bar{B}$ and $e^+e^- \to q\bar{q}$ events. The off-resonance samples consist of $e^+e^- \to q\bar{q}$ events. The private MC samples consist of $B^+ \to K^+n\bar{n}$, $B^+ \to K^{*+}n\bar{n}$, $B^0 \to K^0n\bar{n}$, and $B^0 \to K^{*0}n\bar{n}$ events.

The official MC15ri on-resonance samples are used to train boosted decision tree (BDT) [40] and check for overfitting. The official MC15rd samples and the private MC

Table 3.1: Integrated luminosities or numbers of events of the background MC samples used in the analysis.

Background type	Version	Energy	Luminosity (fb^{-1})	Number of events
$e^+e^- \to \Upsilon(4S) \to B\bar{B}$	MC15ri	on-resonance	2800	-
$e^+e^- \to q\bar{q}$	MC15ri	on-resonance	1000	-
$B^+ \to K^+ n\bar{n}$	-	on-resonance	-	20000
$B^+ \to K^{*+} n\bar{n}$	-	on-resonance	-	20000
$B^0 \to K^0 n\bar{n}$	-	on-resonance	-	20000
$B^0 \to K^{*0} n\bar{n}$	-	on-resonance	-	20000
$e^+e^- \to \Upsilon(4S) \to B\bar{B}$	MC15rd	on-resonance	1458.959	-
$e^+e^- \to q\bar{q}$	MC15rd	on-resonance	1458.959	-
$e^+e^- \to q\bar{q}$	MC15rd	off-resonance	169.328	-

samples at the on-resonance are used to obtain the histogram templates for signal and background events.

The off-resonance MC15rd samples are used to estimate the correction factor and its systematic uncertainty for $e^-e^+ \to q\bar{q}$ ($q = u, d, s, c$) events.

Table 3.1 shows the integrated luminosities or numbers of events for the background MC samples.

The official $B \to X_s n\bar{n}$ MC samples are produced using a PHSP model. With the PHSP model, the daughter particles are distributed uniformly in phase space. However, studies of $B \to K^{(*)} p\bar{p}$ show $M_{p\bar{p}}$ distributions that differ from the flat phase-space distribution [41, 42, 43]. Because $B \to K^{(*)} n\bar{n}$ has a detector response similar to that of $B \to K^{(*)} \nu\bar{\nu}$ decays, dedicated modeling is needed. Therefore, direct $B \to K^{(*)} n\bar{n}$ decays without the J/ψ resonance are removed from the official MC15rd samples and private $B \to K^{(*)} n\bar{n}$ samples are produced. For the private $B \to K^{(*)} n\bar{n}$ samples, a flat $M_{n\bar{n}}$ distribution is required. The details of the correction for the $M_{n\bar{n}}$ distribution are discussed in Section 8.9.

3.2.2 Signal Monte Carlo samples

For the signal samples, $B \to K\nu\bar{\nu}$, $B \to K^*\nu\bar{\nu}$, and the remaining $B \to X_s\nu\bar{\nu}$ samples are produced separately.

For the $B \to K\nu\bar{\nu}$ sample, the following parameters are used in Eq. 1.6 [6]:

$$\begin{aligned} \alpha_0 &= 0.432 \pm 0.011, \\ \alpha_1 &= -0.664 \pm 0.096, \\ \alpha_2 &= -1.20 \pm 0.69. \end{aligned} \tag{3.1}$$

During the analysis, updated parameter values were published [5]:

$$\begin{aligned} \alpha_0 &= 0.2545 \pm 0.0090, \\ \alpha_1 &= -0.71 \pm 0.14, \\ \alpha_2 &= 0.32 \pm 0.59. \end{aligned} \tag{3.2}$$

Therefore, the generated $B \to K\nu\bar{\nu}$ sample is reweighted. This reweighting is applied to the $d\Gamma/dq^2$ distribution [6]:

$$\frac{d\Gamma}{dq^2} \propto \left[\lambda(m_B^2, m_K^2, q^2)\right]^{3/2} \left[f_+(q^2)\right]^2. \tag{3.3}$$

For the $B \to K^*\nu\bar{\nu}$ sample, the following parameters are used in Eq. 1.10 [8]:

$$\begin{aligned} \alpha_0^V &= 0.39 \pm 0.03, \quad & \alpha_0^{A_1} &= 0.30 \pm 0.03, \quad & \alpha_0^{A_{12}} &= 0.27 \pm 0.02, \\ \alpha_1^V &= -1.17 \pm 0.26, \quad & \alpha_1^{A_1} &= 0.39 \pm 0.19, \quad & \alpha_1^{A_{12}} &= 0.53 \pm 0.13, \\ \alpha_2^V &= 2.42 \pm 1.53, \quad & \alpha_2^{A_1} &= 1.19 \pm 1.03, \quad & \alpha_2^{A_{12}} &= 0.48 \pm 0.66. \end{aligned} \tag{3.4}$$

For the remaining $B \to X_s\nu\bar{\nu}$ sample, Eq. 1.14 is used to simulate the q^2 distribution. However, this equation does not determine the mass of X_s. Therefore, the Fermi motion model [44] is adopted to produce the $B \to X_s\nu\bar{\nu}$ sample. In the Fermi motion model, inside the B meson, b quark and spectator quark have four-momenta p and $p_{\text{spectator}}$, respectively. The probability density function for the momentum is

$$\phi(p) = \frac{4}{\sqrt{\pi} p_F^3} \exp\left(\frac{-p^2}{p_F^2}\right), \tag{3.5}$$

where p_F is the Fermi motion momentum, $p = |\vec{p}|$ is the magnitude of the momentum of the b quark [44], which satisfies $\int_0^\infty dp\, p^2 \phi(p) = 1$. The Fermi motion momentum satisfies the following equation for a spectator-quark mass of $m_{\text{sp}} \sim 0$ GeV/c^2 [45]:

$$\mu_\pi^2 = 6p_F^2 - 3\left\langle p \right\rangle_{B-rest}^2, \tag{3.6}$$

where $\mu_\pi^2 = 0.464 \pm 0.076$ GeV^2 [46] is the average kinetic energy of b–quark. From Eq. 3.5,

$$\left\langle p \right\rangle_{B-rest} = \int_0^\infty p^2 \phi(p) p dp = \frac{2}{\sqrt{\pi}} p_F. \tag{3.7}$$

By substituting Eq. 3.7 into Eq. 3.6 with $\mu_\pi^2 = 0.464 \pm 0.076$ GeV^2, we obtain

$$p_F = 0.461^{+0.036}_{-0.039} \text{ GeV}/c. \tag{3.8}$$

With this value of the Fermi motion momentum, the momentum of the s-quark is determined from the momentum of b-quark. Then the mass of X_s can be calculated as follows:

$$M_{X_s}^2 = \left(p_{\text{s}} + p_{\text{spectator}}\right)^2, \tag{3.9}$$

where p_s is the momentum of s-quark and $p_{\text{spectator}}$ is the momentum of spectator quark in the B meson. Here, the mass of X_s is required to satisfy $M_{X_s} > 1.1$ GeV/c^2. This mass threshold is motivated by the X_s mass distribution in $B \to X_s\gamma$ decays [47]. In addition, studies of $B^+ \to K^+\pi^+\pi^-\mu^+\mu^-$ and $B^0 \to K^+\pi^-\mu^+\mu^-$ decays show that no significant contribution is observed below this mass threshold, except for the K^* resonance [48, 49]. After determining the kinematics of the $X_s\nu\bar{\nu}$ decay, the hadronization is performed using PYTHIA8 [38].

Figures 3.1 and 3.2 show the q^2 distribution for the MC samples and theoretical calculations. The q^2 distribution for $B \to K^*\nu\bar{\nu}$ shows a discrepancy between MC and

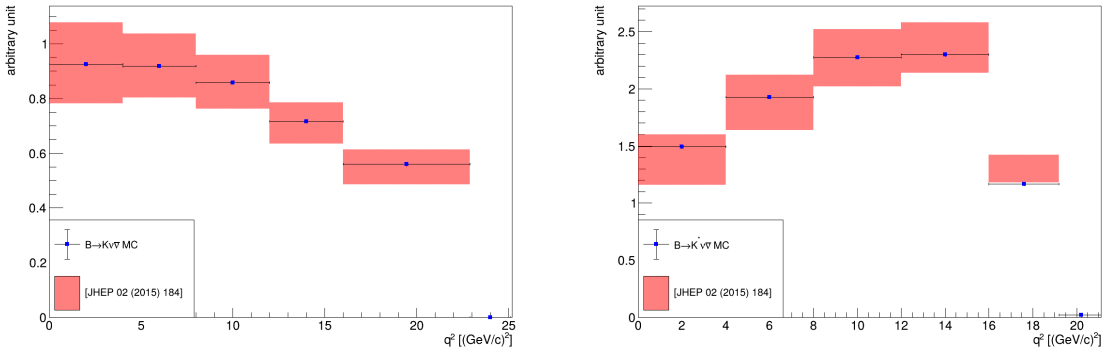


Figure 3.1: q^2 distributions for the (left) $B \to K\nu\bar{\nu}$ and (right) $B \to K^*\nu\bar{\nu}$ MC samples and the corresponding theoretical calculations.

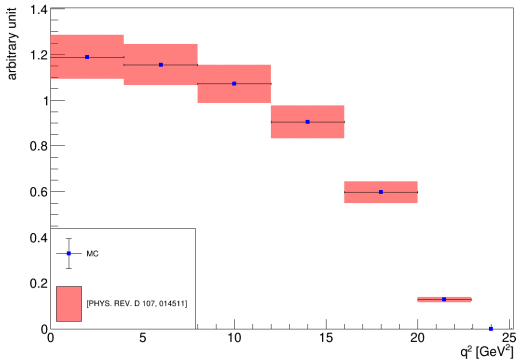


Figure 3.2: q^2 distribution for the reweighted $B \to K\nu\bar{\nu}$ MC sample and the corresponding theoretical calculation.

the theoretical calculation. This discrepancy comes from the fact that the theoretical calculation is based on a fixed K^* mass, while the MC sample is generated with a finite K^* mass width. Consequently, in this analysis, we do not apply any additional correction for $B \to K^*\nu\bar{\nu}$ samples.

When combining B^+B^- and $B^0\bar{B}^0$ samples, we use the following relation:

$$\begin{aligned} N_{B^+B^-} &= N_{\Upsilon(4S)} \cdot f^{+-}, \\ N_{B^0\bar{B}^0} &= N_{\Upsilon(4S)} \cdot f^{00}, \end{aligned} \tag{3.10}$$

where $f^{+-} = 0.5113^{+0.0073}_{-0.0108}$, $f^{00} = 0.4861^{+0.0074}_{-0.0080}$ [50], $N_{\Upsilon(4S)}$ is the number of $\Upsilon(4S)$ events, $N_{B^+B^-}$ is the number of B^+B^- pairs, and $N_{B^0\bar{B}^0}$ is the number of $B^0\bar{B}^0$ pairs. Also, when combining the $B \to K\nu\bar{\nu}$, $B \to K^*\nu\bar{\nu}$, and remaining $B \to X_s\nu\bar{\nu}$ samples, we use $BR(B^+ \to K^+\nu\bar{\nu}) = 5.044 \times 10^{-6}$, $BR(B^0 \to K^{*0}\nu\bar{\nu}) = 9.19 \times 10^{-6}$, and $BR(B \to X_s\nu\bar{\nu}) = 2.9 \times 10^{-5}$ [6, 51], with the following relation [6]:

$$\frac{BR(B^+ \to K^{(*)+}\nu\bar{\nu})}{BR(B^0 \to K^{(*)0}\nu\bar{\nu})} = \frac{\tau_{B^+}}{\tau_{B^0}}, \tag{3.11}$$

where τ_{B^+} and τ_{B^0} are the lifetimes of the B^+ and B^0 mesons, respectively. Consequently, the following remaining branching fractions are used:

$$\begin{aligned} BR(B^+ \to X_s\nu\bar{\nu})_{\text{remain}} &= BR(B \to X_s\nu\bar{\nu}) - BR(B^+ \to K^+\nu\bar{\nu}) - BR(B^+ \to K^{*+}\nu\bar{\nu}) \\ &= 2.9 \times 10^{-5} - 5.044 \times 10^{-6} - 9.91 \times 10^{-6} \\ &= 1.405 \times 10^{-5}, \end{aligned} \tag{3.12}$$

$$\begin{aligned} BR(B^0 \to X_s\nu\bar{\nu})_{\text{remain}} &= BR(B \to X_s\nu\bar{\nu}) - BR(B^0 \to K^0\nu\bar{\nu}) - BR(B^0 \to K^{*0}\nu\bar{\nu}) \\ &= 2.9 \times 10^{-5} - 3.69 \times 10^{-6} - 9.19 \times 10^{-6} \\ &= 1.513 \times 10^{-5}. \end{aligned} \tag{3.13}$$

Figure 3.3 shows the fragmentation of X_s in the entire $B \to X_s\nu\bar{\nu}$ signal MC sample. K in the legend denotes charged and neutral kaons. π in the legend denotes charged pions only. The "other" category includes decay modes such as $B \to K\eta\nu\bar{\nu}$.

Table 3.2 shows the numbers of events in the signal MC samples. The MC15ri samples are used to train BDT and check for overfitting. The MC15rd samples are used to obtain the histogram templates.

3.3 Data

In this analysis, we use on-resonance and off-resonance data samples corresponding to integrated luminosities of $\mathcal{L} = 365$ fb^{-1} and $\mathcal{L} = 43$ fb^{-1}, respectively [52]. These data were collected from 2019 to 2022. The off-resonance sample is used to estimate the correction and its systematic uncertainty for $e^+e^- \to q\bar{q}$ events. To avoid bias, the data in the signal region are not examined until the analysis procedure is finalized. The definition of the signal region is given in Section 5.1.

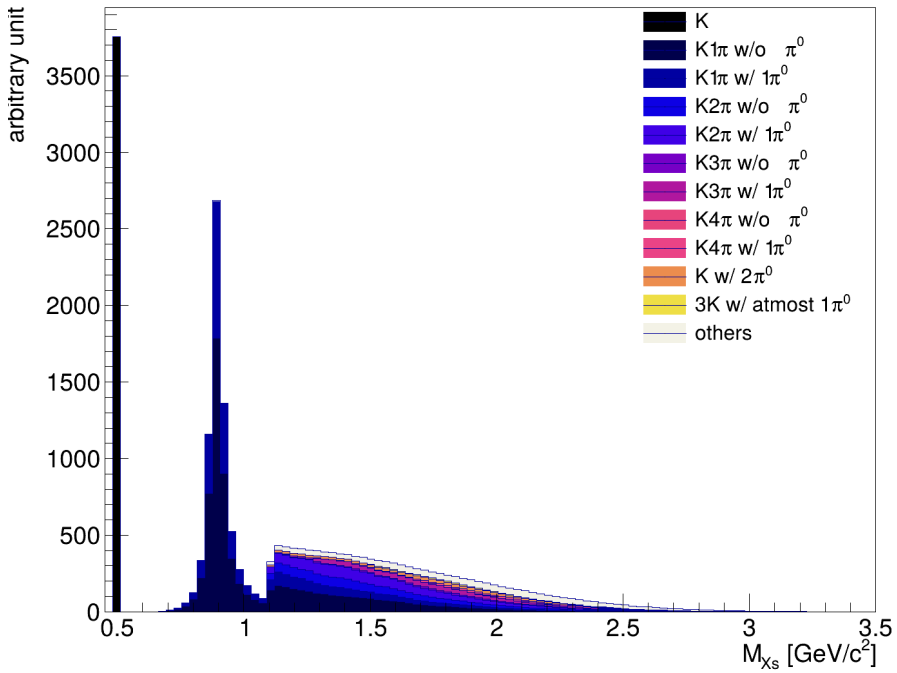


Figure 3.3: Fragmentation of X_s in the entire $B \to X_s \nu\bar{\nu}$ signal MC sample. K in the legend denotes charged and neutral kaons. π in the legend denotes charged pions only. The "other" category includes decay modes such as $B \to K\eta\nu\bar{\nu}$.

Table 3.2: Numbers of events in the signal MC samples used in the analysis.

Signal type	Version	Energy	Number of events
$B^+ \to K^+\nu\bar{\nu}$	MC15ri	on-resonance	10^7
$B^+ \to K^{*+}\nu\bar{\nu}$	MC15ri	on-resonance	10^7
Remaining $B^+ \to X_s\nu\bar{\nu}$	MC15ri	on-resonance	5×10^7
$B^0 \to K^0\nu\bar{\nu}$	MC15ri	on-resonance	10^7
$B^0 \to K^{*0}\nu\bar{\nu}$	MC15ri	on-resonance	10^7
Remaining $B^0 \to X_s\nu\bar{\nu}$	MC15ri	on-resonance	5×10^7
$B^+ \to K^+\nu\bar{\nu}$	MC15rd	on-resonance	3962022
$B^+ \to K^{*+}\nu\bar{\nu}$	MC15rd	on-resonance	3962022
Remaining $B^+ \to X_s\nu\bar{\nu}$	MC15rd	on-resonance	15846594
$B^0 \to K^0\nu\bar{\nu}$	MC15rd	on-resonance	3962022
$B^0 \to K^{*0}\nu\bar{\nu}$	MC15rd	on-resonance	3962022
Remaining $B^0 \to X_s\nu\bar{\nu}$	MC15rd	on-resonance	15846594

Chapter 4

Reconstruction and Selection

4.1 Tagging Method

The missing energy carried by neutrinos in the $B \to X_s \nu\bar{\nu}$ decay makes the analysis challenging. Therefore, we use the full event interpretation (FEI) algorithm [53] to reconstruct the other B meson in the event, namely, the tag-side B meson, or $B_{\rm tag}$. In this analysis, the hadronic tagging method is used. In the hadronic tagging method, $B_{\rm tag}$ is reconstructed in hadronic decay modes. The advantage of hadronic tagging is that there are no undetected neutrinos from the $B_{\rm tag}$ decay, resulting in a higher purity than that of leptonic tagging. On the other hand, hadronic tagging has a lower signal efficiency than leptonic tagging.

4.2 Skimming

To save computing resources, samples skimmed using the FEI algorithm are used. This skimming requires that the beam-constrained mass to satisfy $M_{\rm bc} > 5.2$ GeV/c^2, the energy difference to satisfy $|\Delta E| < 0.3$ GeV, and the FEI signal probability to satisfy $\mathcal{P}_{\rm FEI} > 0.001$. Here,

$$\begin{aligned} M_{\rm bc} &= \sqrt{(\sqrt{s}/2)^2 - |\vec{p}^{\,*}_B c|^2}, \\ \Delta E &= E^*_B - \sqrt{s}/2, \end{aligned} \tag{4.1}$$

where $\sqrt{s}$ is the SuperKEKB center-of-mass (c.m.) energy, and $\vec{p}^{\,*}_B$ and E^*_B are the momentum and energy of the B meson in the c.m. system, respectively. The FEI signal probability $\mathcal{P}_{\rm FEI}$ is the output of the multivariate classifier in the FEI algorithm. $\mathcal{P}_{\rm FEI}$ ranges from 0 to 1, and a value closer to 1 indicates that the $B_{\rm tag}$ candidate is more likely to be correctly reconstructed.

4.3 Particle Selections

For pion and kaon candidates used to reconstruct the signal side, the requirements $dr < 0.5$ cm and $|dz| < 2$ cm are applied, where dr and dz are the distances of closest approach of a track to the interaction point (IP) in the transverse plane and along the z-axis, respectively. The number of CDC hits is required to be greater than 20, `nCDCHits` > 20, following the recommendation of the PID performance group. The number of PXD hits is required to be greater than 0, `nPXDHits` > 0, motivated by the $B^+ \to K^+\nu\bar{\nu}$ study to reduce the effects of SVD resolution mismodeling [54]. After these requirements are applied, the tracks are identified as pion or kaon candidates according to the particle-identification (PID) likelihood criteria. For the pion candidates, `pionID` > 0.6 is applied. For the kaon candidates, `kaonID` > 0.6 is applied.

For the signal side, A K_S^0 candidate is reconstructed from two oppositely charged tracks. The vertex fit [55] is required to converge for these two tracks. The `significanceOfDistance` variable for the K_S^0 candidate, defined as the distance from the IP to the decay vertex divided by its uncertainty, is required to be greater than 50. This significance of distance criterion is applied to suppress the background from the pion tracks near IP. The invariant mass of K_S^0 should be within 10 MeV/c^2 of the nominal K_S^0 mass [22], $|dM| < 10$ MeV/c^2.

For the π^0 candidates, photon candidates are required. The standard photon selection used in the Belle II experiment is applied. The number of ECL crystals associated with the photon candidate, `clusterNHits`, is required to be greater than 1.5. The value of `clusterNHits` can be non-integer when ECL clusters overlap. The photon candidate is required to lie within the CDC acceptance region to ensure that the ECL cluster is not associated with a charged particle. To suppress ECL background noise, energy requirements depending on the detector region are applied. The photon energy is required to exceed 80 MeV, 30 MeV, and 60 MeV in the forward, barrel, and backward ECL regions, respectively.

Each π^0 candidate is reconstructed from two photon candidates. The invariant mass is required to be $0.120 < M_{\gamma\gamma} < 0.145$ GeV/c^2. The difference in azimuthal angle between the two photons should be $-1.5 < \Delta\phi_{\gamma\gamma} < 1.5$ rad. The opening angle between the two photons should be $\Delta\theta_{\gamma\gamma} < 1.4$ rad. To suppress combinatorial background, the momentum should satisfy $p_{\pi^0} > 0.4$ GeV/c.

Table 4.1 shows the summary of particle selection criteria used for the signal side reconstruction.

4.4 Reconstruction

In this analysis, the sum-of-exclusive method is used. In the sum of exclusive method, we construct the exclusive decay channels and combine them to obtain the branching

Table 4.1: Particle selection criteria for signal side reconstruction.

Particle	Selections
$\pi^\pm$	$dr < 0.5$ cm, $\lvert dz\rvert < 2$ cm, `nCDCHits` > 20, `nPXDHits` > 0, `pionID` > 0.6
$K^\pm$	$dr < 0.5$ cm, $\lvert dz\rvert < 2$ cm, `nCDCHits` > 20, `nPXDHits` > 0, `kaonID` > 0.6
K^0_S	reconstructed from two oppositely charged tracks `significanceOfDistance` > 50, $\lvert dM\rvert < 10$ MeV/c^2
γ	`clusterNHits` > 1.5, within the CDC acceptance region $E > 80$ MeV (forward), $E > 30$ MeV (barrel), $E > 60$ MeV (backward)
π^0	$0.120 < M_{\gamma\gamma} < 0.145$ GeV/c^2, $-1.5 < \Delta\phi_{\gamma\gamma} < 1.5$ rad $\Delta\theta_{\gamma\gamma} < 1.4$ rad, $p_{\pi^0} > 0.4$ GeV/c

Table 4.2: List of reconstructed X_s decays.

	X^0_{sd}			$X^\pm_{su}$		
K	K^0_S			$K^\pm$		
$K\pi$	$K^\pm\pi^\mp$	$K^0_S\pi^0$		$K^\pm\pi^0$	$K^0_S\pi^\pm$	
$K2\pi$	$K^\pm\pi^\mp\pi^0$	$K^0_S\pi^\pm\pi^\mp$	$K^0_S\pi^0\pi^0$	$K^\pm\pi^\mp\pi^\pm$	$K^0_S\pi^\pm\pi^0$	$K^\pm\pi^0\pi^0$
$K3\pi$	$K^\pm\pi^\mp\pi^\pm\pi^\mp$	$K^0_S\pi^\pm\pi^\mp\pi^0$	$K^\pm\pi^\mp\pi^0\pi^0$	$K^\pm\pi^\mp\pi^\pm\pi^0$	$K^0_S\pi^\pm\pi^\mp\pi^\pm$	$K^0_S\pi^\pm\pi^0\pi^0$
$K4\pi$	$K^\pm\pi^\mp\pi^\pm\pi^\mp\pi^0$	$K^0_S\pi^\pm\pi^\mp\pi^\pm\pi^\mp$	$K^0_S\pi^\pm\pi^\mp\pi^0\pi^0$	$K^\pm\pi^\mp\pi^\pm\pi^\mp\pi^\pm$	$K^0_S\pi^\pm\pi^\mp\pi^\pm\pi^0$	$K^\pm\pi^\mp\pi^\pm\pi^0\pi^0$
$3K$	$K^\pm K^\mp K^0_S$			$K^\pm K^\mp K^\pm$		
$3K\pi$	$K^\pm K^\mp K^\pm\pi^\mp$	$K^\pm K^\mp K^0_S\pi^0$		$K^\pm K^\mp K^\pm\pi^0$	$K^0_S K^\pm K^\mp\pi^\pm$	

fraction of the inclusive decay mode. A total of 30 decay modes are reconstructed for X_s: $Kn\pi$ ($0 \leq n \leq 4$) with at most two π^0, and $3Kn\pi$ ($0 \leq n \leq 1$) with at most one π^0 and one K^0_S, where K indicates $K^\pm$ or K^0_S and π indicates $\pi^\pm$ or π^0. These decay modes cover 93% of the entire $B \to X_s\nu\bar{\nu}$ signal MC sample, assuming that K^0 mesons decay into K^0_S and K^0_L with equal probabilities. These decay modes also cover 83% of the remaining $B \to X_s\nu\bar{\nu}$ signal MC sample, excluding the separately generated $B \to K\nu\bar{\nu}$ and $B \to K^*\nu\bar{\nu}$ samples, as described in Section 3.2.2. Table 4.2 summarizes the reconstructed X_s decay modes.

After the X_s candidate is reconstructed, an $\Upsilon(4S)$ candidate is formed from B_{tag} and X_s: $B^+_{\text{tag}}X^-_{su}$, $B^0_{\text{tag}}X^0_{sd}$, $\bar{B}^0_{\text{tag}}X^0_{sd}$, and their charge conjugates.

4.5 Preselections

After reconstruction, loose selection criteria are applied to save computing resources. These criteria are referred to as preselections because they are applied before the output files are written by `basf2`; otherwise, they are no different from the selection criteria discussed in Section 4.6. The only difference is that the preselection is applied before the

output files are written by `basf2`.

To reduce combinatorial background, we require

$$M_{X_s}^{\mathrm{reco}} < 2.0 \text{ GeV}/c^2, \tag{4.2}$$

where $M_{X_s}^{\mathrm{reco}}$ is the reconstructed mass of the X_s candidate. At the generator level, there are 9% of the generated $B \to X_s \nu\bar{\nu}$ events have $M_{X_s} > 2.0$ GeV/c^2, where M_{X_s} is the true mass of X_s. Figure 4.1 shows the $M_{X_s}^{\mathrm{reco}}$ distributions for the background and signal MC samples.

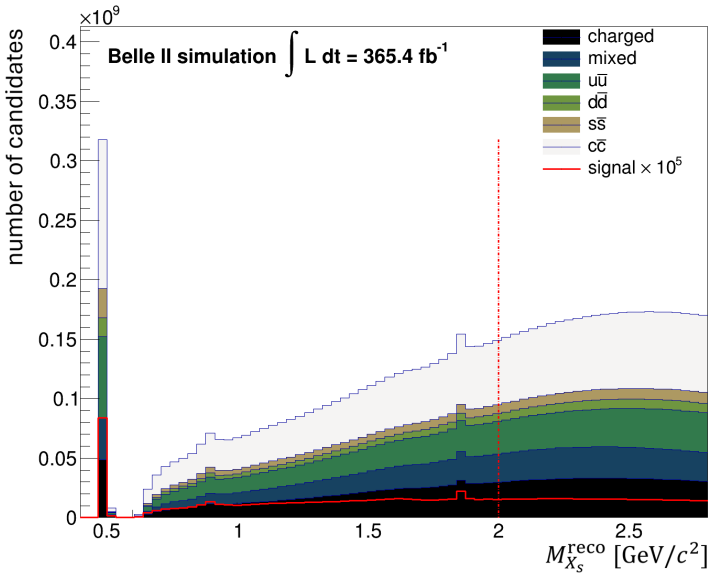


Figure 4.1: $M_{X_s}^{\mathrm{reco}}$ distribution for the backgrounds and signal MC samples.

Ideally, no tracks, π^0 candidates, or K_S^0 candidates should remain after reconstructing the $\Upsilon(4S)$ candidate. We require that no π^0 or K_S^0 candidates defined in Section 4.3 remain. In addition, no loose tracks are allowed to remain. Loose tracks are defined by $dr < 2$ cm and $|dz| < 4$ cm. For these loose tracks, other criteria, such as `nCDCHits` and `nPXDHits`, are not applied.

Another preselection criterion is based on vertex-fit information. In the FEI algorithm, a vertex fit is performed for the B_{tag} candidate. We require that the vertex fit converge for the B_{tag} candidate. The decay-chain vertex fit, `TreeFit` [55], is performed for the X_s candidates. This fit is also required to converge for X_s whose decay channels have sufficient degrees of freedom for the vertex fit.

Table 4.3 shows the cut flow and the numbers of MC events at the preselection stage.

Table 4.3: Number of events surviving the preselection requirements described in Section 4.5. Cross-feed events are included in the signal category. The signal and background samples are scaled to an equivalent luminosity of 365.37 fb^{-1}. The calibration and correction factors listed in Section 8 are not applied.

Selection	SIGNAL	CHG	MIX	UUBAR	DDBAR	SSBAR	CHARM
Reconstruction and $M_{X_s}^{\rm reco} < 2.0$ GeV/c^2	206.7	1.47×10^7	1.01×10^7	2.34×10^7	5.41×10^6	7.44×10^6	3.86×10^7
No remaining π^0	205.3	1.35×10^7	9.28×10^6	1.91×10^7	4.43×10^6	6.19×10^6	3.32×10^7
No remaining loose tracks	105.2	1.86×10^5	1.13×10^5	5.52×10^5	1.35×10^5	2.69×10^5	9.53×10^5
No remaining K_S^0	104.9	1.81×10^5	1.10×10^5	5.39×10^5	1.33×10^5	2.59×10^5	9.24×10^5
X_s vertex fit converges	104.9	1.81×10^5	1.09×10^5	5.37×10^5	1.32×10^5	2.58×10^5	9.22×10^5
$B_{\rm tag}$ vertex fit converges	87.0	1.51×10^5	8.94×10^4	4.48×10^5	1.07×10^5	2.05×10^5	6.98×10^5

4.6 Selections

After the preselections, several selections are applied to suppress backgrounds, while retaining as much signal as possible. The selection strategy is to apply loose one-dimensional cut-based selections and a best-candidate selection, followed by a cut on the output of an optimized multivariate analysis (MVA) classifier. We use this strategy because optimization at each selection stage does not guarantee global optimization. Consequently, we try to optimize the selection at the most powerful selection stage, MVA output, and use loose selection criteria before the optimization.

4.6.1 One-dimensional cut-based selection

We require the B_{tag} candidate to satisfy

$$\begin{aligned} M_{\text{bc}} &> 5.27 \text{ GeV}/c^2, \\ |\Delta E| &< 0.2 \text{ GeV}. \end{aligned} \tag{4.3}$$

Ideally, there should be no remaining energy deposited in the ECL after reconstructing $\Upsilon(4S)$. The remaining energy deposited in the ECL, E_{ecl}, is required to be

$$E_{\text{ecl}} < 1.3 \text{ GeV}. \tag{4.4}$$

E_{ecl} is defined as the sum of the energies from the ECL clusters which satisfy

$$\begin{cases} E > 100 \text{ MeV} & \text{in the forward region} \\ E > 60 \text{ MeV} & \text{in the barrel region} \\ E > 150 \text{ MeV} & \text{in the backward region}, \end{cases} \tag{4.5}$$

$$\texttt{minC2TDist} > 50 \text{ cm}, \tag{4.6}$$

and lie within the CDC acceptance region, where `minC2TDist` is the distance between the ECL cluster and its nearest track. These ECL cluster criteria come from the hadronic $B^+ \to K^+\nu\bar{\nu}$ study [56]. Good data-MC agreement is observed for these criteria, according to ~~Figures~~ D.2 and E.2.

Particles that escape through the beam pipe can mimic missing momentum from neutrinos. Therefore, the missing momentum is required to satisfy

$$0.297 < \theta_{\text{missing}} < 2.618 \text{ rad}, \tag{4.7}$$

where θ_{missing} is the polar angle of the missing momentum in the laboratory frame. The missing momentum is calculated by the following equation:

$$\vec{p}_{\text{miss}} = \vec{p}_{e^+e^-} - \sum \vec{p}_{\text{track}} - \sum \vec{p}_{\gamma}, \tag{4.8}$$

where $\vec{p}_{\text{miss}}$, $\vec{p}_{\text{track}}$, and $\vec{p}_{\gamma}$ are the momenta of the e^+e^- system, tracks, and photons, respectively. For tracks used to calculate the missing momentum, the following conditions are required:

- transverse momentum $p_t > 0.1$ GeV/c
- within the CDC acceptance region
- $dr < 0.5$ cm
- $|dz| < 3$ cm

The pion-mass hypothesis is assigned to these tracks. For photons used to calculate the missing momentum, the following requirements are applied:

- $E > 0.05$ GeV
- within the CDC acceptance region
- $|\texttt{clusterTiming}| < 200$ ns,

where `clusterTiming` is the difference between the ECL cluster time, obtained from a waveform fit to the highest-energy crystal in the cluster, and the event collision time. These track and photon requirements are the default options in `basf2`.

Especially for the low-multiplicity X_s decay modes, particles from $B_{\rm tag}$ tend to have lower momenta than the particles forming X_s. Therefore, if particles from $B_{\rm tag}$ are used to reconstruct X_s, the X_s momentum becomes lower. In addition, $B \to X_s \nu\bar{\nu}$ is a three-body decay and the X_s momentum has an upper bound. We require the momentum of the X_s candidates to satisfy

$$0.5 < p^*_{X_s} < 2.96 \text{ GeV}/c \tag{4.9}$$

in the c.m. system. This requirement is motivated by the $p^*_{X_s}$ distribution shown in Figure 4.2 and by the $B \to h\nu\bar{\nu}$ study.

To remove D mesons that are misidentified as X_s, a D veto is applied based on the mass of X_s. Candidates satisfying

$$1.84 < M^{\rm reco}_{X_s} < 1.89 \text{ GeV}/c^2 \tag{4.10}$$

are rejected. This veto region is determined from the $M^{\rm reco}_{X_s}$ distribution shown in Figure 4.2.

4.6.2 Best candidate selection

After the selections discussed in Section 4.6.1, the multiplicity of $\Upsilon(4S)$ candidates per event is 1.71 in the signal MC. Figure 4.3 shows the multiplicity of $\Upsilon(4S)$ candidates for the signal and background MC samples. The best candidate selection (BCS) is performed in two steps. First, we keep the $\Upsilon(4S)$ candidates whose $B_{\rm tag}$ has the highest $\mathcal{P}_{\rm FEI}$ value. For the signal sample, the fraction of $B_{\rm tag}$ candidates with `isSignal` == 1 increases from 30% to 46% after the first step of BCS, where the `isSignal` value is unity when the particle is correctly reconstructed in `basf2`. After the first step of BCS, the multiplicity becomes 1.06 in the signal MC. There can be more than one $\Upsilon(4S)$ candidate, because

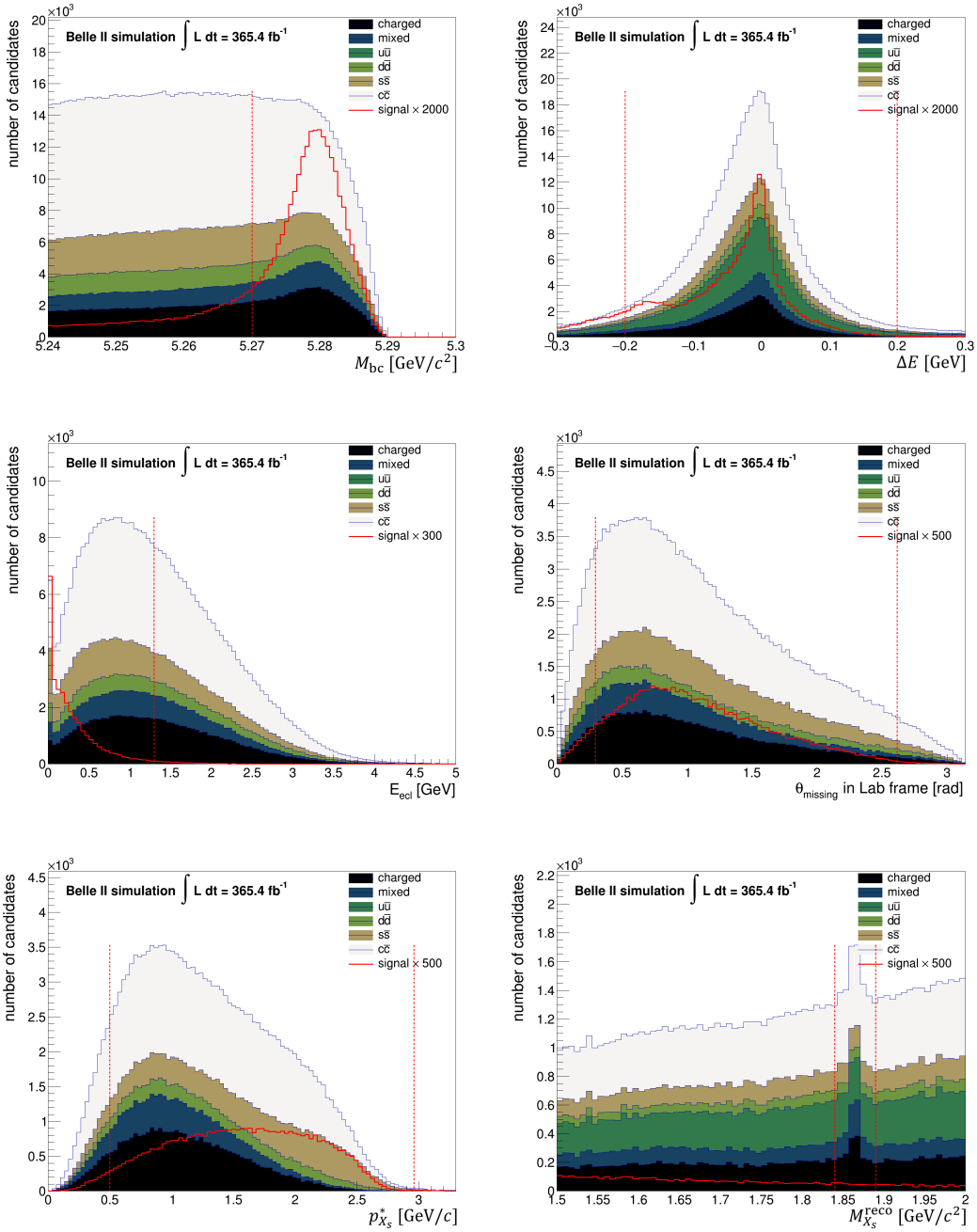


Figure 4.2: Cut variable distributions

there can be exchange between photon candidates in the rest of the event (ROE) and photon candidates in π^0 used to reconstruct X_s. In addition, ROE can have tracks that are not identified as loose tracks defined in Section 4.5. These additional tracks can also be exchanged with tracks in K^0 candidates used to reconstruct X_s. Second, we keep only one $\Upsilon(4S)$ candidate randomly. The reason why the random selection is used is that other powerful variables will be used in the MVA stage.

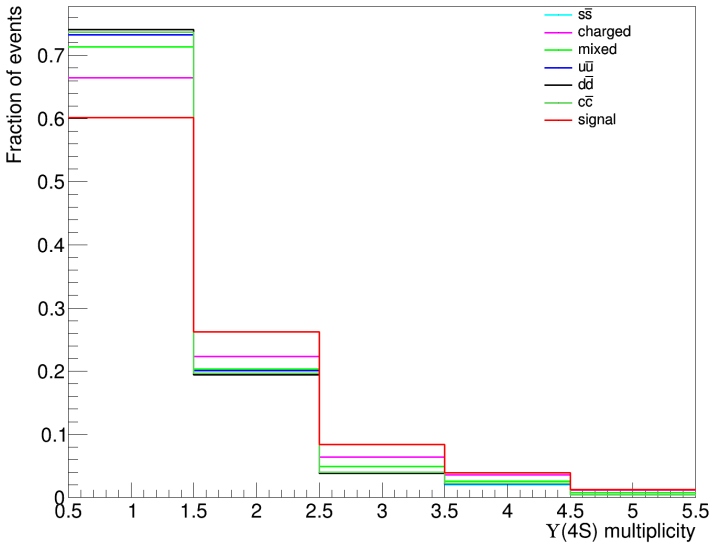


Figure 4.3: The $\Upsilon(4S)$ multiplicity before the best candidate selection stage.

4.6.3 D veto variables

Even though D veto is applied, as discussed in Section 4.6.1, there is still background from D mesons. To suppress these events, we try to reconstruct D meson candidates in the event, and use their information in the MVA technique. In this section, the reconstruction procedure for these D-meson candidates is discussed.

The reconstruction of D mesons is performed using the following procedure:

1. Select $K^{\pm}$ or K^0_S candidates from X_s
2. Select $\pi^{\pm}$ candidates from the entire event. Selection criteria are
   - $dr < 0.5$ cm, $|dz| < 2$ cm, and `nPXDHits` > 0
3. Select π^0 candidates from the entire event. The selection criteria are the same as those in Table 4.1
4. Reconstruct D meson candidates using the particles selected in steps 1, 2, and 3. There are four categories of reconstructed decay modes. The first category is

- $D^+ \to K^0_S \pi^+$
- $D^+ \to K^- \pi^+ \pi^+$

The second category is:

- $D^0 \to K^+ \pi^-$
- $D^0 \to K^0_S \pi^+ \pi^-$
- $D^0 \to K^+ \pi^- \pi^+ \pi^-$

The third category is:

- $D^+ \to K^0_S \pi^+$
- $D^+ \to K^- \pi^+ \pi^+$
- $D^+ \to K^0_S \pi^+ \pi^0$
- $D^+ \to K^- \pi^+ \pi^+ \pi^0$
- $D^+ \to K^0_S \pi^+ \pi^0 \pi^0$

The fourth category is:

- $D^0 \to K^+ \pi^-$
- $D^0 \to K^0_S \pi^+ \pi^-$
- $D^0 \to K^+ \pi^- \pi^+ \pi^-$
- $D^0 \to K^+ \pi^- \pi^0$
- $D^0 \to K^0_S \pi^+ \pi^- \pi^0$
- $D^0 \to K^+ \pi^- \pi^0 \pi^0$

As shown above, the third category includes the first category, and the fourth category includes the second category. Because fake D-meson candidates can arise from ECL background in decay modes containing a π^0, we deliberately reconstruct with and without π^0. Charge-conjugate decay modes are also reconstructed.

5. Perform a vertex fit. Candidates for which the vertex fit does not converge are not treated as D-meson candidates.

In this analysis, the following variables are used as MVA input:

- mass of the $D^\pm$ candidate with the highest vertex-fit p-value in the first category
- standard deviation of p-value of vertex fitting for $D^\pm$ in the first category
- mass of the D^0 candidate with the highest vertex-fit p-value in the second category
- standard deviation of p-value of vertex fitting for D^0 in the second category
- mass of the D^0 candidate with the highest vertex-fit p-value in the fourth category
- standard deviation of p-value of vertex fitting for D^0 in the fourth category

As noted above, the third category is not used for MVA. The reason is that the third category shows some data-MC discrepancy.

4.6.4 Multivariate Analysis

This section discusses the MVA performed after the best-candidate selection. We use FastBDT as the MVA method [40]. FastBDT is a boosted-decision-tree-based classifier. There are some advantages to using FastBDT in the analysis. First, FastBDT has a shorter training time than deep-learning models or classifiers implemented in the Toolkit for Multivariate Data Analysis (TMVA). This shorter training time not only facilitates a smoother analysis workflow but also provides sufficient time for hyperparameter optimization. Second, for tabular data, tree-based models generally show better performance than deep learning.

To avoid overfitting in FastBDT, we need to constrain the number of input variables. Two criteria are used to select the input variables of FastBDT: variable importance and data-MC agreement. The variable importance shows the discriminating power between signal and background samples, which is evaluated using the `GetVariableRanking()` method provided by FastBDT. Among powerful variables, we remove the variables that show some data-MC discrepancy in the off-resonance sample and $M_{\rm bc}$ sideband. Here, the $M_{\rm bc}$ sideband is defined by $M_{\rm bc} < 5.26$ GeV/c^2 for the $B_{\rm tag}$ candidate. Figure 4.4 shows the variable importance for FastBDT input variables. Appendices D-E show distributions of several variables for the off-resonance sample and the sideband region. A detailed list and descriptions for the input variables are provided in Table A.1. Figure 4.5 shows two important variables: $E_{\rm ecl}$ and `harmonicMomentThrust2`. Figures C.1-C.2 show the MC distributions of all FastBDT input variables.

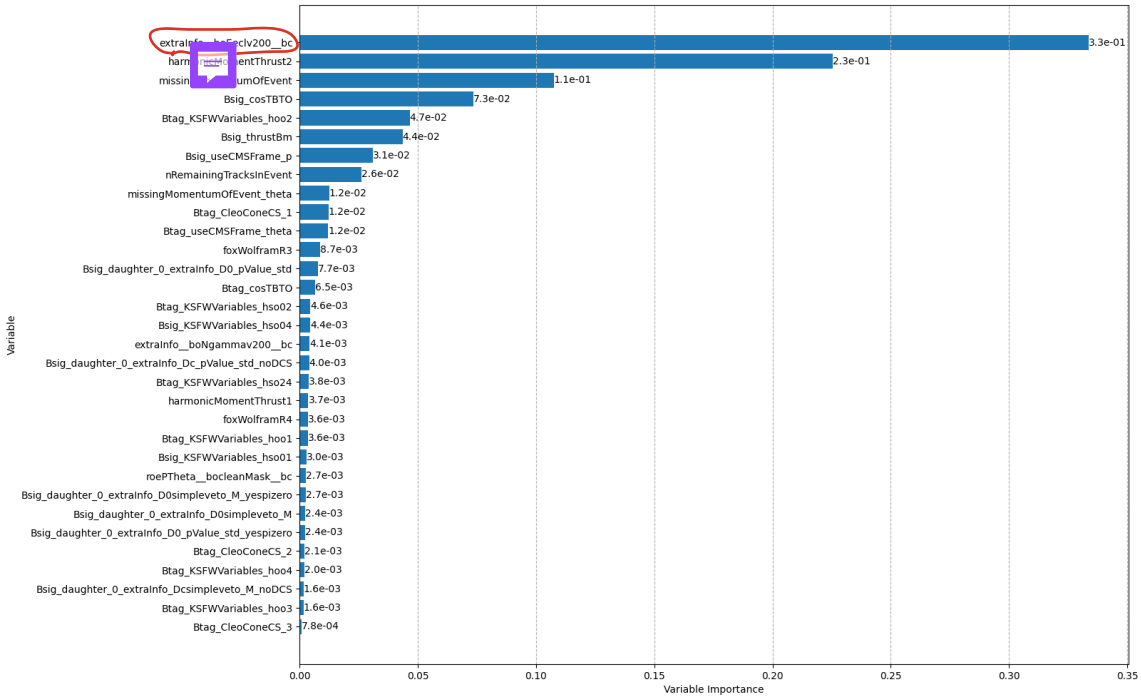


Figure 4.4: Variable importance obtained using the `GetVariableRanking()` method provided by FastBDT.

Table 4.4: Name of hyperparameters, descriptions, and their values.

Name	Description	Tested values	Selected value
`nTrees`	Number of trees	100, 500, 1000, 1500, 2000	2000
`depth`	Maximum tree depth	2, 3, 4	3
`shrinkage`	Learning rate	0.05, 0.1, 0.15, 0.2	0.05
`subsample`	Sampling fraction	0.3, 0.4, 0.5, 0.6, 0.7	0.5
`binning`	Number of binning levels	6, 7, 8, 9	6

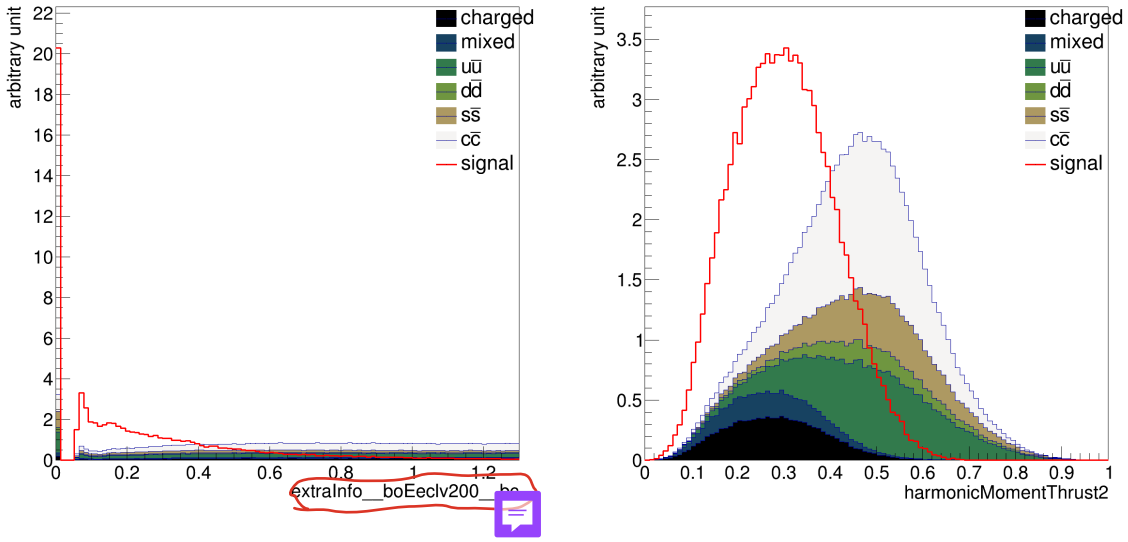


Figure 4.5: Two important FastBDT input variables. The left plot shows the E_{ecl} distribution. The right plot shows `harmonicMomentThrust2` distribution.

To increase the available training statistics, the MC15ri sample is used to train and test FastBDT. MC15rd sample is used to obtain histogram templates. Appendices ??-?? show the comparison between MC15ri and MC15rd for signal and background samples.

FastBDT has several hyperparameters. Some of these parameters are optimized by the grid search method. Table 4.4 shows the names of the hyperparameters and their descriptions, tested values in the grid search, and the selected value. In the grid search, the AUC is used as the optimization criterion. Figure 4.6 shows the result of the grid search. For each hyperparameter combination in the grid search, there are two AUC values: AUC from training and testing samples. We want to keep the AUC value as large as possible, while keeping the testing AUC close to the training AUC. The x-axis of the figure is the rank of the AUC value for the testing sample. A lower rank indicates a higher AUC value. The blue dots show the AUC values for the testing sample. The red dots show the AUC values for the training sample. With the selected hyperparameters in Table 4.4, the AUC values for training and testing samples are 0.966473 and 0.963756, respectively.

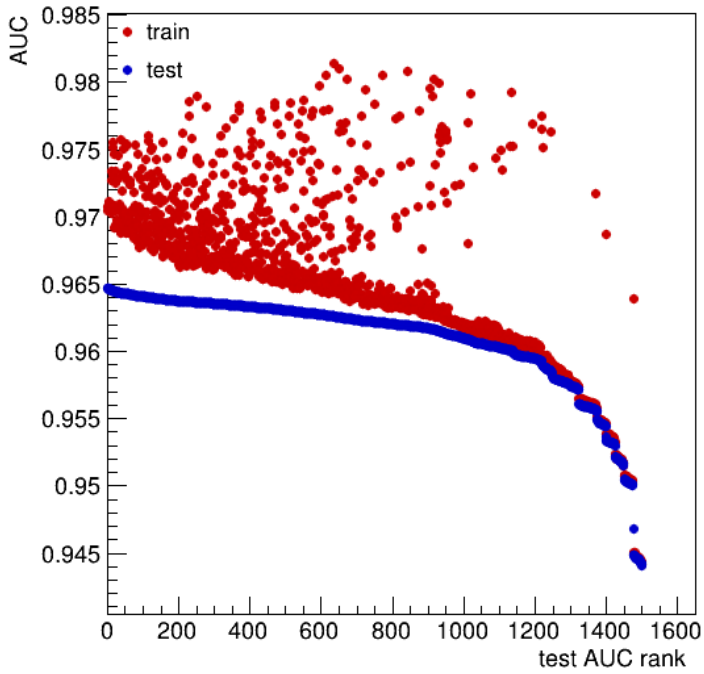


Figure 4.6: Result of grid search. x-axis shows rank of AUC value for the testing sample. Lower rank means higher AUC value. y-axis shows AUC value. The blue dots show the AUC values for the testing sample. The red dots show the AUC values for the training sample.

The difference in the AUC value is smaller than 0.3%. Therefore, we conclude that the overfitting is negligible.

Figure 4.7 shows the FastBDT output distribution for the training signal MC, testing signal MC, training background MC, and testing background MC samples.

The cut criterion on the FastBDT output is determined by the figure of merit (FOM):

$$\mathrm{FOM} = \frac{S}{\sqrt{S+B}}, \tag{4.11}$$

where S and B are the numbers of signal and background events, respectively. Figure 4.8 shows the FOM value as a function of the FastBDT cut. The signal yield extraction will be performed using the FastBDT output variable. Consequently, we do not apply the FastBDT cut which maximizes FOM values. Instead, we apply a loose requirement on the FastBDT output, $\mathcal{O}_{\mathrm{BDT}} > 0.86$, and use the signal-depleted region to estimate the background yields.

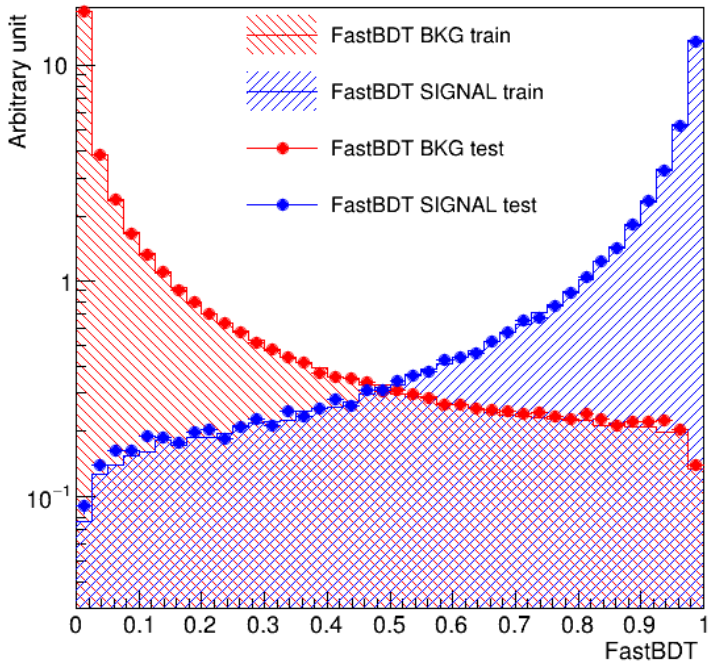
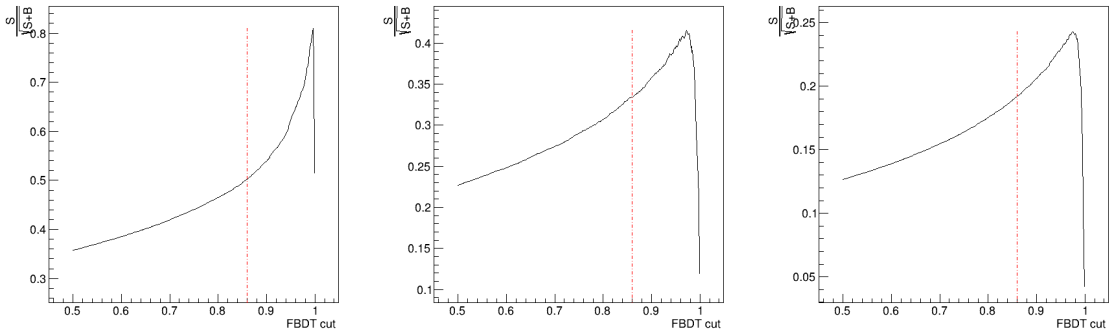


Figure 4.7: FastBDT output distribution.

4.7 Summary

Table 4.5 shows the cut flow described in Section 4.6. The signal and background samples are scaled to an equivalent integrated luminosity of 365.37 fb^{-1}. The efficiency as a function of M_{X_s} is shown in Section B. Figure 4.9 shows the proper reconstruction efficiency for each X_s decay mode. Figure 4.10 shows the cross-feed matrix. The cross-feed matrix is defined as $\epsilon_{\text{mode}}/\epsilon_{\text{true}}$, where ϵ_{mode} is the efficiency for events to be reconstructed in a given mode, and ϵ_{true} is the efficiency for events to be reconstructed in the correct decay mode.



(a) $0.0 < M_{X_s}^{\text{reco}} < 0.6$ GeV/c^2 (b) $0.6 \leq M_{X_s}^{\text{reco}} < 1.0$ GeV/c^2 (c) $1.0 \leq M_{X_s}^{\text{reco}} < 2.0$ GeV/c^2

Figure 4.8: FOM distribution for (a) $0.0 < M_{X_s}^{\text{reco}} < 0.6$ GeV/c^2, (b) $0.6 \leq M_{X_s}^{\text{reco}} < 1.0$ GeV/c^2, and (c) $1.0 \leq M_{X_s}^{\text{reco}} < 2.0$ GeV/c^2. The FOMs are maximized near FBDT > 0.996, FBDT > 0.973, and FBDT > 0.975, respectively.

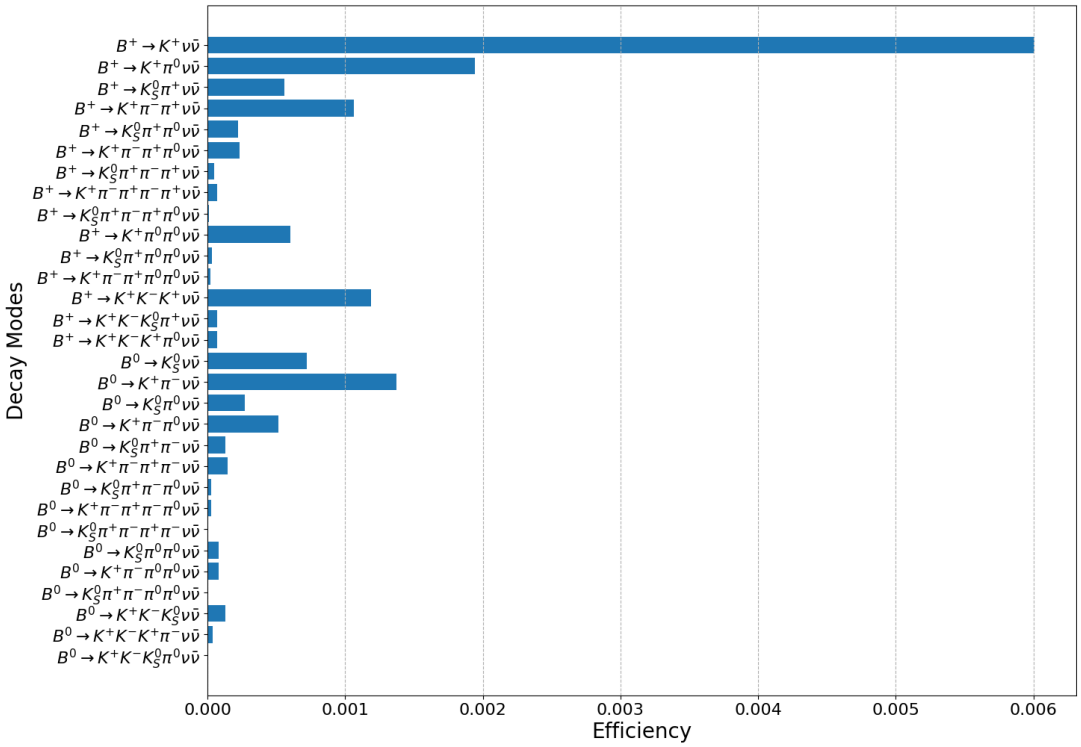


Figure 4.9: Proper reconstruction efficiencies after all selection requirements are applied. For example, the efficiency for $B^+ \to K^+\nu\bar{\nu}$ events to be reconstructed as $B^+ \to K^+\nu\bar{\nu}$ is 6.01×10^{-3}.

Table 4.5: Numbers of events surviving the selection requirements described in Section 4.6. The signal and background samples are scaled to an equivalent integrated luminosity of 365.37 fb^{-1}. Cross-feed events are included in the signal category. The calibration and correction factors listed in Section 8 are not applied.

Selection	SIGNAL	CHG	MIX	UUBAR	DDBAR	SSBAR	CHARM
Generated	22451.8	365.37 fb^{-1}					
Skim	689.1	3.44×10^7	2.62×10^7	9.11×10^7	2.19×10^7	1.73×10^7	9.64×10^7
Reconstruction and preselections	86.8	1.51×10^5	8.94×10^4	4.48×10^5	1.07×10^5	2.05×10^5	6.98×10^5
$M_{\rm bc} > 5.27$ GeV/c^2 for $B_{\rm tag}$	61.4	40488.1	23369.8	83729.7	20103.7	39879.0	1.24×10^5
$\lvert\Delta E\rvert < 0.2$ GeV for $B_{\rm tag}$	57.6	39106.5	22283.9	78875.1	18935.4	37449.6	1.17×10^5
$E_{\rm ecl} < 1.3$ GeV	56.1	21913.3	12983.9	41440.5	9974.0	22868.4	64568.4
$0.297 < \theta_{\rm miss} < 2.618$ rad	52.5	19354.9	11321.3	34360.5	8337.9	20314.0	55395.6
$0.5 < p^*_{X_s} < 2.96$ GeV/c	50.8	17532.5	10174.3	31450.0	7658.2	19136.3	50693.7
Reject $1.84 < M^{\rm reco}_{X_s} < 1.89$ GeV/c^2	50.6	16452.4	9521.7	30161.8	7350.9	18498.2	48501.4
BCS	50.6	16452.4	9521.7	30161.8	7350.9	18498.2	48501.4
$\mathcal{O}_{\rm BDT} > 0.86$	34.5	1604.3	796.4	471.6	113.9	475.8	641.6

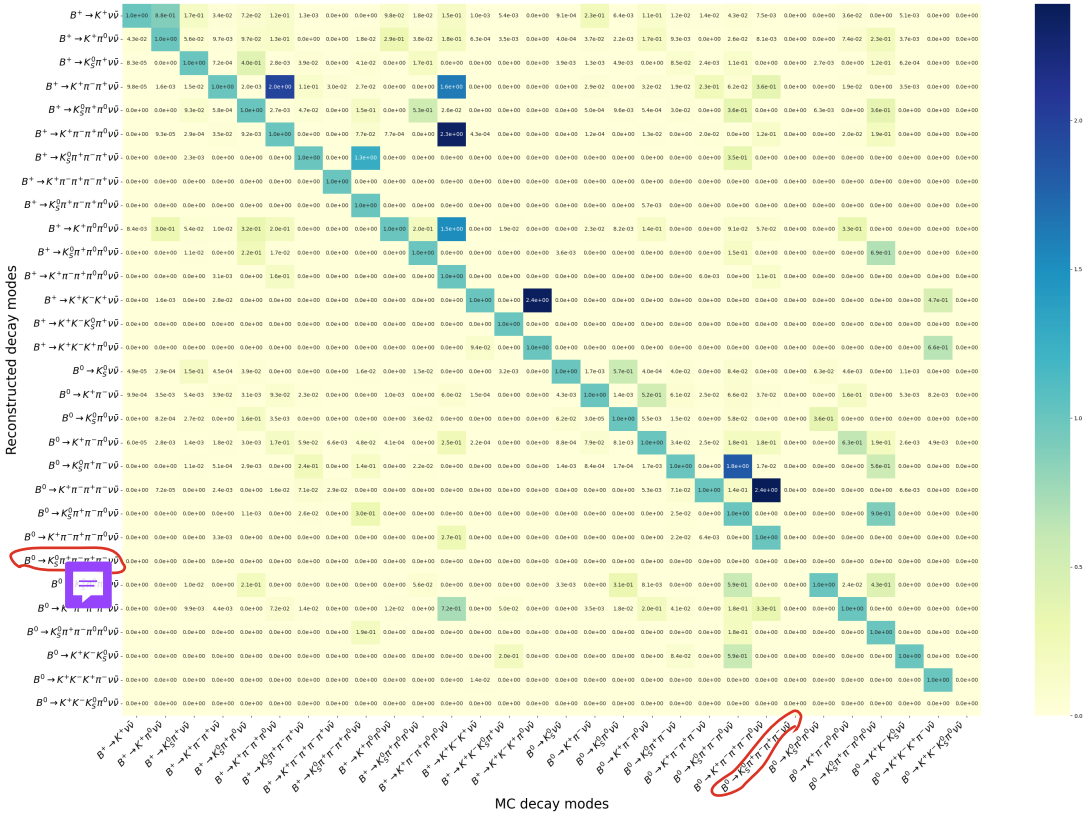


Figure 4.10: Cross-feed matrix. It is defined as $\epsilon_{\text{mode}}/\epsilon_{\text{true}}$, where ϵ_{mode} is the efficiency for events to be reconstructed in a given mode, and ϵ_{true} is the efficiency for events to be reconstructed in the correct decay mode.

Chapter 5

Signal yield extraction strategy

5.1 Signal region definition

In this analysis, we define the signal region in the two-dimensional $M_{X_s}^{\mathrm{reco}}$–$\mathcal{O}_{\mathrm{BDT}}$ plane. Because $\mathcal{O}_{\mathrm{BDT}}$ has different distributions depending on the value of $M_{X_s}^{\mathrm{reco}}$, we use this two-dimensional region. The signal region is defined by $0.0 < M_{X_s}^{\mathrm{reco}} < 2.0$ GeV/c^2 and $0.86 < \mathcal{O}_{\mathrm{BDT}} \leq 1.0$.

We divide the signal region into 3×5 bins. The bin boundaries along the $M_{X_s}^{\mathrm{reco}}$ axis are $[0.0, 0.6, 1.0, 2.0]$ GeV/c^2. These three bins correspond to the regions dominated by K, K^*, and the remaining X_s states. Hereafter, we label them MR1, MR2, and MR3, respectively. When the signal MC is produced, the mass threshold, $M_{X_s} > 1.1$ GeV/c^2, is required, as discussed in Section 3.2.2. The bin boundary between MR2 and MR3 is set to 1.0 GeV/c^2, slightly below this threshold, to minimize sensitivity to the modeling of this threshold. In each $M_{X_s}^{\mathrm{reco}}$ region, the bin boundaries along $\mathcal{O}_{\mathrm{BDT}}$ are defined as follows:

$$\text{Bin boundary} = \begin{cases} [0.86, 0.9195, 0.979, 0.986, 0.993, 1.0] & \text{(in MR1)}, \\ [0.86, 0.8925, 0.925, 0.95, 0.975, 1.0] & \text{(in MR2)}, \\ [0.86, 0.9, 0.94, 0.96, 0.98, 1.0] & \text{(in MR3)}. \end{cases} \tag{5.1}$$

These bin boundaries are determined to maximize the sensitivity while avoiding fit bias. If small bin widths are used, the sensitivity improves. However, if the number of events in the bin is small, the pull distribution of the signal strength does not follow a Gaussian distribution, as seen in Figure 5.1. Therefore, the number and size of bins are determined to fulfill these conditions.

For simplicity, we define the variable "bin index" and the mapping from $M_{X_s}^{\mathrm{reco}}$-$\mathcal{O}_{\mathrm{BDT}}$ plane to the bin index value. The bin index ranges from 1 to 5 in MR1, from 6 to 10 in MR2, and from 11 to 15 in MR3, increasing along the positive $\mathcal{O}_{\mathrm{BDT}}$ axis within each mass region. Figure 5.2 shows the mapping between the $M_{X_s}^{\mathrm{reco}}$-$\mathcal{O}_{\mathrm{BDT}}$ bins and the bin

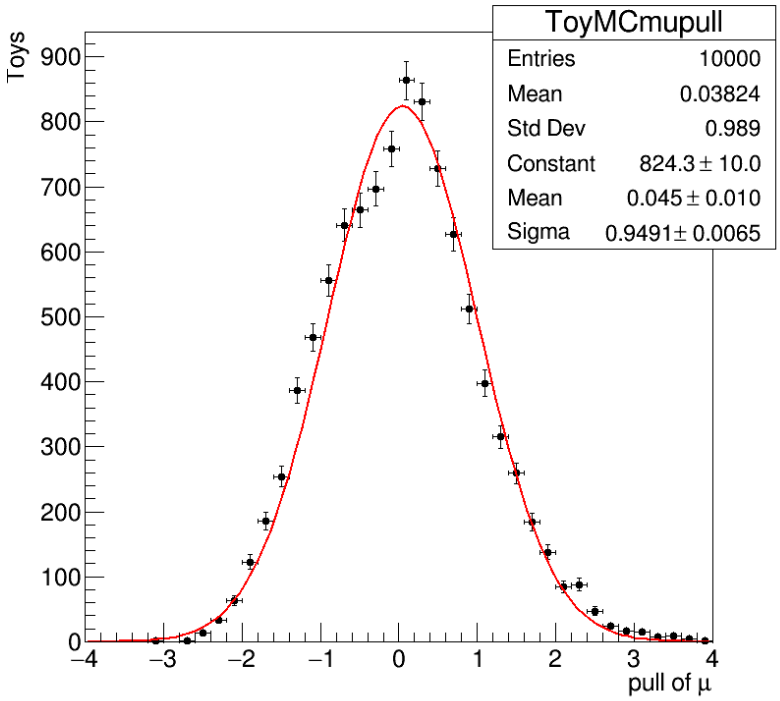


Figure 5.1: Pull distribution when the number of events in the bin is small.

indices. The number inside each bin indicates the bin index value. The bin index is used as the fitting variable.

Figure 5.3 shows the bin index distributions for the signal and background samples.

5.2 Fitting strategy

The tools `RooFit` and `RooStats` are used for fitting and limit setting. A binned maximum-likelihood method is used. To construct the probability density function, the tool `HistFactory`, provided within `RooStats`, is used [57]. `HistFactory` provides an easy way to construct a probability density function. It also gives a way to incorporate changes on shapes of histogram templates by systematic uncertainties.

`HistFactory` provides the following probability density function:

$$\mathcal{P}(n_{cb}, a_p|\phi_p, \alpha_p, \gamma_b) = \prod_{\text{c}\in\text{channels}} \prod_{\text{b}\in\text{bins}} \text{Pois}(n_{cb}|\nu_{cb}) \cdot G(L_0|\lambda, \Delta_L) \cdot \prod_{p\in\mathbb{S}+\mathbf{\Gamma}} f_p(a_p|\alpha_p), \quad (5.2)$$

where n_{cb} is the number of observed events in channel c and bin b, $f_p(a_p|\alpha_p)$ is a constraint term for an auxiliary measurement a_p with nuisance parameter α_p. In this analysis, all constraint terms, $f_p(a_p|\alpha_p)$, are Gaussian distributions. The number of expected events is

$$\nu_{cb}(\phi_p, \alpha_p, \gamma_b) = \lambda_{cs}\gamma_{cb}\phi_{cs}(\alpha)\eta(\boldsymbol{\alpha})\sigma_{csb}(\boldsymbol{\alpha}), \quad (5.3)$$

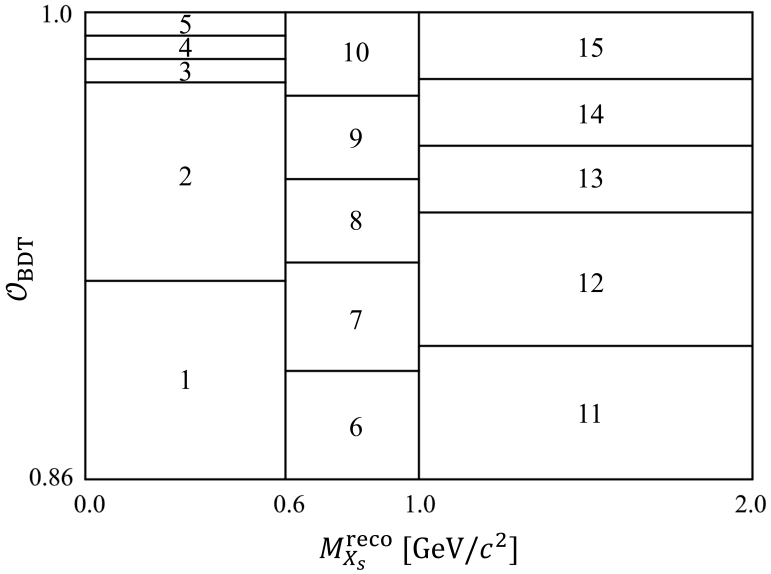
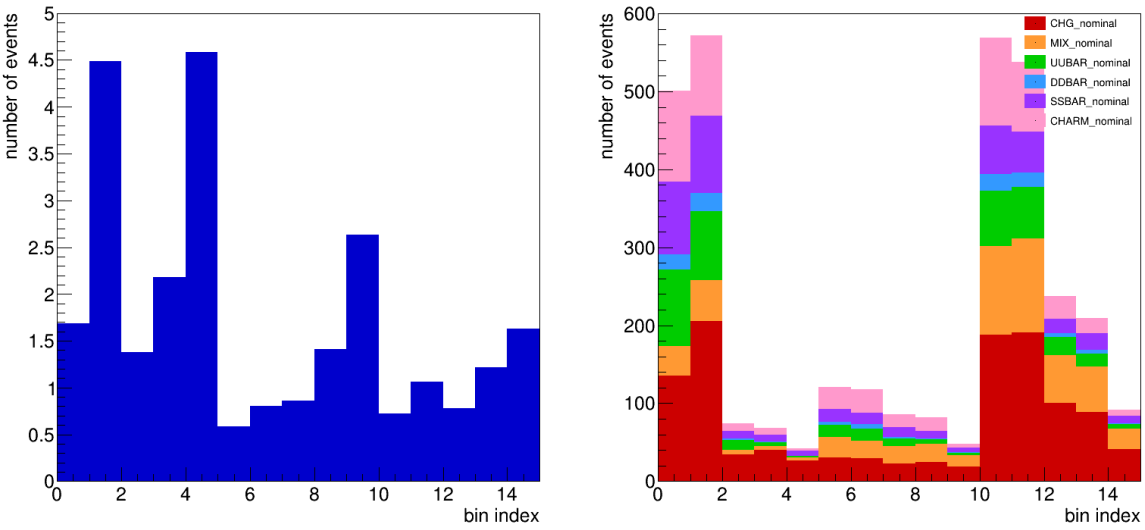


Figure 5.2: Mapping between the two-dimensional $M_{X_s}^{\mathrm{reco}}$–$\mathcal{O}_{\mathrm{BDT}}$ bins and the bin indices.



(a) Signal

(b) Background

Figure 5.3: Bin index distributions for the signal and background samples.

where

- λ_{cs}: luminosity parameter for a given channel and sample.
- γ_{cb}: bin-by-bin scale factor used for statistical uncertainties, bin-by-bin shape systematics (`ShapeSys`), and data-driven shape extrapolations (`ShapeFactor`). For statistical uncertainties, γ_{csb} is shared among all samples in the channel.
- ϕ_{cs}: product of unconstrained normalization factors for a given sample within a given channel. These typically include the parameter of interest.
- $\eta_{cs}(\boldsymbol{\alpha})$: the parametrized normalization uncertainties (`OverallSys`) for a given sample within a given channel.
- σ_{csb}: the parametrized histogram (`HistoSys`) for a given sample within a given channel.

The branching ratios of $B \to K^{(*)} n\bar{n}$ decay is determined by the $B \to K^{(*)} p\bar{p}$ decays [22]. The branching ratios of $B \to K^{(*)} n\bar{n}$ decay are estimated by isospin symmetry:

$$\mathrm{BR}(B^+ \to K^+ n\bar{n}) = \mathrm{BR}(B^0 \to K^0 p\bar{p}) \times \frac{\tau_{B^+}}{\tau_{B^0}}, \tag{5.4}$$

$$\mathrm{BR}(B^+ \to K^{*+} n\bar{n}) = \mathrm{BR}(B^0 \to K^{*0} p\bar{p}) \times \frac{\tau_{B^+}}{\tau_{B^0}}, \tag{5.5}$$

$$\mathrm{BR}(B^+ \to K^0 n\bar{n}) = \mathrm{BR}(B^+ \to K^+ p\bar{p}) \times \frac{\tau_{B^0}}{\tau_{B^+}}, \tag{5.6}$$

$$\mathrm{BR}(B^+ \to K^{*0} n\bar{n}) = \mathrm{BR}(B^+ \to K^{*+} p\bar{p}) \times \frac{\tau_{B^0}}{\tau_{B^+}}, \tag{5.7}$$

where τ_{B^+} and τ_{B^0} are the life time of B^+ and B^0, respectively. The weights are applied to satisfy the estimated branching ratios of $B \to K^{(*)} n\bar{n}$ decay.

In this analysis, three parameters of interest are defined: μ_1, μ_2, and μ_3. These parameters are signal strengths defined as follows:

$$\begin{cases} \mu_1 = & \dfrac{\text{Branching ratio in the } 0.0{<}M_{X_s}{<}0.6 \text{ GeV}/c^2 \text{ region, obtained from the fit}}{\text{Branching ratio in the } 0.0{<}M_{X_s}{<}0.6 \text{ GeV}/c^2 \text{ region in SM}}, \\ \mu_2 = & \dfrac{\text{Branching ratio in the } 0.6{<}M_{X_s}{<}1.0 \text{ GeV}/c^2 \text{ region, obtained from the fit}}{\text{Branching ratio in the } 0.6{<}M_{X_s}{<}1.0 \text{ GeV}/c^2 \text{ region in SM}}, \\ \mu_3 = & \dfrac{\text{Branching ratio in the } 1.0 \text{ GeV}/c^2{<}M_{X_s} \text{ region, obtained from the fit}}{\text{Branching ratio in the } 1.0 \text{ GeV}/c^2{<}M_{X_s} \text{ region in SM}}. \end{cases} \tag{5.8}$$

As shown in Eq. 5.8, each signal-strength parameter is in one-to-one correspondence with the branching ratio in the corresponding mass region.

Alternatively, a single parameter of interest, μ, can be defined as

$$\mu = \frac{\text{Branching ratio over the entire } M_{X_s} \text{ region, obtained from the fit}}{\text{Branching ratio over the entire } M_{X_s} \text{ region in SM}}. \tag{5.9}$$

However, this method assumes fixed relative signal strengths among the three M_{X_s} regions and therefore relies on the signal modeling. Consequently, the single-parameter approach is not used in this analysis.

5.3 Limit setting strategy

To set an upper limit on the branching ratio of $B \to X_s \nu\bar{\nu}$ decays, the CL_s method is used [58]. The CL_s method provides a conservative procedure for setting upper limits. In this section, the CL_{s+b} method is first described, followed by the motivation for using the CL_s method.

Let μ be the parameter of interest, such as a branching ratio or a signal strength. The goal is to test whether a specific value, μ_{test}, is compatible with the observed data. If μ_{test} is too large to describe the observed data, this value can be excluded.

For this purpose, a test statistic $q_\mu(x)$ is defined, which maps a dataset x to a single real number. The convention adopted here is that a larger value of $q_\mu(x)$ corresponds to greater incompatibility between the tested value of μ and the dataset x.

Consequently, a large value of $q_{\mu_{\text{test}}}(x_{\text{obs}})$ indicates that μ_{test} is incompatible with the observed data x_{obs}. To quantify this incompatibility, toy MC samples are generated under the hypothesis $\mu = \mu_{\text{test}}$: $x_{\text{toy},1}$, $x_{\text{toy},2}$, ... From these toys, we can obtain the distribution of $q_{\mu_{\text{test}}}(x)$. The corresponding p-value, denoted by CL_{s+b}, is then defined as

$$CL_{s+b} = \mathcal{P}(q > q_\mu(x_{\text{obs}}) | \mu = \mu_{\text{test}}). \tag{5.10}$$

For example, if $CL_{s+b} = 0.03$, then, assuming μ_{test} is the true value, the probability of obtaining data that are more incompatible than the observed data is 0.03.

To obtain an upper limit, CL_{s+b} is scanned as a function of μ_{test}. The 90% confidence-level upper limit, μ_{UL}, is obtained from the condition

$$CL_{s+b}(\mu_{\text{UL}}) = 0.1. \tag{5.11}$$

However, even in the absence of sensitivity, the CL_{s+b} method can exclude the signal because of a downward statistical fluctuation of the background. To avoid such false exclusions, the p-value under the background-only hypothesis, CL_b, is also calculated:

$$CL_b = \mathcal{P}(q > q_\mu(x_{\text{obs}}) | \mu = 0). \tag{5.12}$$

The CL_s value is then defined as

$$CL_s = \frac{CL_{s+b}}{CL_b}. \tag{5.13}$$

The use of CL_s helps avoid false exclusions. For example, if the experiment has no sensitivity to the signal, the signal-plus-background and background-only hypotheses give identical distributions of the test statistic. In this case, the CL_{s+b} method would still exclude the signal with 10% probability at the 90% confidence level. However, since $CL_{s+b} = CL_b$, one obtains $CL_s = 1$, and the signal is not excluded.

In this analysis, the one-sided negative log profile likelihood ratio is used as the test statistic:

$$q_\mu = \begin{cases} -\ln \lambda(\mu), & \hat{\mu} < \mu, \\ 0, & \hat{\mu} \geq \mu, \end{cases} \tag{5.14}$$

where $\hat{\mu}$ is the signal strength obtained from the best fit to the data.

Chapter 6

Validation

6.1 Off-resonance

The off-resonance sample is used for two purposes. First, it is used to check the data–MC agreement of the FastBDT input variables. Second, it is used to obtain correction factors and their uncertainties. For this study, we use off-resonance data and off-resonance $e^+e^- \to q\bar{q}$ MC samples.

We use a cut flow similar to that shown in Table 4.5. However, $M_{\rm bc}$ distribution is shifted because of the lower center-of-mass energy. Therefore, we correct the $M_{\rm bc}$ values for both data and MC:

$$M_{\rm bc,\ corrected} = \sqrt{(\sqrt{s^{\rm off}}/2)^2 - |\vec{p}^{\,*}_B c|^2} + (\sqrt{s^{\rm on}} - \sqrt{s^{\rm off}})/2, \tag{6.1}$$

where $\sqrt{s^{\rm on}}$ and $\sqrt{s^{\rm off}}$ are the SuperKEKB c.m. energies for the on-resonance and off-resonance samples, respectively. In this analysis, $\sqrt{s^{\rm on}}$ is set to the center-of-mass energy used for the run-independent MC sample. No correction is applied to ΔE, because its distribution is not shifted.

Figures D.1–D.2 show the distributions of the FastBDT input variables. MC distributions are scaled to the data in the figures. The overall event yields in data and MC show a discrepancy. After the FastBDT requirement, the data-to-MC ratios are

$$\frac{\rm Data}{\rm MC} = \begin{cases} 1.21 \pm 0.13 & (0.0 < M_{X_s}^{\rm reco} < 0.6\ {\rm GeV}), \\ 0.90 \pm 0.20 & (0.6 \le M_{X_s}^{\rm reco} < 1.0\ {\rm GeV}), \\ 0.78 \pm 0.10 & (1.0 \le M_{X_s}^{\rm reco} < 2.0\ {\rm GeV}). \end{cases} \tag{6.2}$$

These ratios are obtained after applying the calibration and correction factors listed in Section 8, except for the $q\bar{q}$ correction factor.

Table 6.1 shows the cut flow for the off-resonance sample. Figure 6.1 shows the FastBDT output distribution.

Table 6.1: Cut flow for off-resonance sample. The entries show the numbers of events surviving each selection requirement. MC samples are scaled to 42.74 fb^{-1}. Calibration and correction factors listed in Section 8 are not applied in this table.

Selection	data	MC	data/MC
Generated	42.74 fb^{-1}	42.74 fb^{-1}	1.0
Skim	$2.35 \cdot 10^7$	$2.20 \cdot 10^7$	1.0693
Reconstruction and preselections	113586	123254	0.9216
M_{bc} correction	113586	123254	0.9216
$M_{\mathrm{bc,\ corrected}} > 5.27$ GeV/c^2 for B_{tag}	31402	35143	0.8935
$\lvert\Delta E\rvert < 0.2$ GeV for B_{tag}	29851	33328	0.8957
$E_{\mathrm{ecl}} < 1.3$ GeV	16423	18155	0.9046
$0.297 < \theta_{miss} < 2.618$	14092	15485	0.9100
$0.5 < p^*_{X_s} < 2.96$ GeV/c	13034	14229	0.9160
Reject $1.84 < M^{\mathrm{reco}}_{X_s} < 1.89$ GeV/c^2	12418	13645	0.9187
BCS	12418	13645	0.9187
$\mathcal{O}_{\mathrm{BDT}} > 0.86$	208	225.4	0.9228

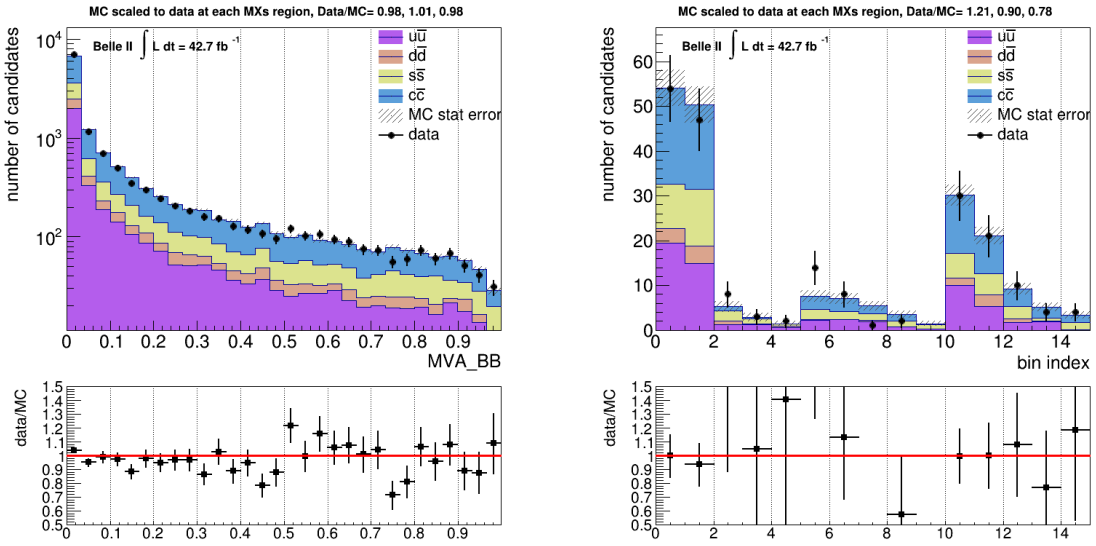


Figure 6.1: The left plot shows the FastBDT output distribution for the off-resonance sample. The right plot shows the distribution of the bin index defined in Section 5.1. Calibration and correction factors listed in Section 8 are applied, except for the $q\bar{q}$ correction factor.

Table 6.2: Cut flow for the on-resonance $M_{\rm bc}^{\rm tag}$ sideband sample. The entries show the numbers of events surviving each selection requirement. The MC samples are scaled to an integrated luminosity of 365.4 fb^{-1}. Calibration and correction factors listed in Section 8 are not applied in this table.

Selection	data	MC	data/MC
Generated	365.4 fb^{-1}	365.4 fb^{-1}	1.0
Skim	$3.01 \cdot 10^8$	$2.87 \cdot 10^8$	1.0469
Reconstruction, preselections, and $M_{\rm bc} < 5.26$ GeV/c^2	1115436	1233668	0.9042
$M_{\rm bc} > 5.235$ GeV/c^2	531606	586715	0.9061
$\lvert\Delta E\rvert < 0.2$ GeV	504049	556703	0.9054
$E_{\rm ecl} < 1.3$ GeV	277954	305240	0.9106
$0.297 < \theta_{miss} < 2.618$	235494	257390	0.9149
$0.5 < p^*_{X_s} < 2.96$ GeV/c	216586	236273	0.9167
Reject $1.84 < M^{\rm reco}_{X_s} < 1.89$ GeV/c^2	207587	225963	0.9187
BCS	207587	225963	0.9187
$\mathcal{O}_{\rm BDT} > 0.86$	4767	5067	0.9407

6.2 On-resonance $M_{\rm bc}$ sideband

During the selection procedure in Section 4.6, we apply the following requirement on $M_{\rm bc}$:

$$M_{\rm bc} > 5.27 \text{ GeV}/c^2. \tag{6.3}$$

We define an $M_{\rm bc}$ sideband as

$$5.235 < M_{\rm bc} < 5.26 \text{ GeV}/c^2. \tag{6.4}$$

The region $5.26 < M_{\rm bc} < 5.27$ GeV/c^2 is not included in the sideband to ensure sufficient separation from the signal region. Table 6.2 shows the cut flow for the sideband study.

Figures E.1-E.2 show the distributions of the FastBDT input variables. The event yields in data and MC show a discrepancy. After the FastBDT requirement, the data-to-MC ratio is

$$\frac{\text{Data}}{\text{MC}} = \begin{cases} 1.12 \pm 0.03 & (0.0 < M^{\rm reco}_{X_s} < 0.6 \text{ GeV}), \\ 1.04 \pm 0.05 & (0.6 \leq M^{\rm reco}_{X_s} < 1.0 \text{ GeV}), \\ 1.10 \pm 0.03 & (1.0 \leq M^{\rm reco}_{X_s} < 2.0 \text{ GeV}). \end{cases} \tag{6.5}$$

These ratios are obtained after applying the calibration and correction factors listed in Section 8, including the $q\bar{q}$ correction factor. The $q\bar{q}$ correction factor is calculated using the off-resonance sample, following the cut flow in Table 6.2, with the $M_{\rm bc}$ correction

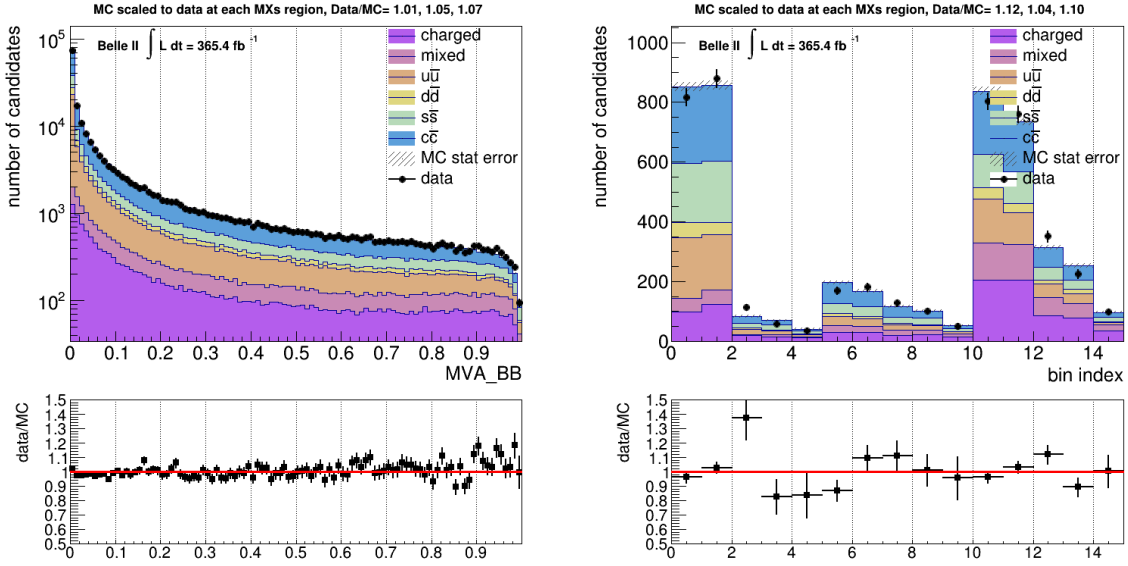


Figure 6.2: The left plot shows the FastBDT output distribution for the on-resonance sideband sample. The right plot shows the distribution of the bin index defined in Section 5.1. In both plots, calibration and correction factors listed in Section 8 are applied, including $q\bar{q}$ correction factor.

defined in Eq. 6.1. The MC distributions are normalized to the data in Figure E.1-E.2. $q\bar{q}$ correction factor is not applied in Figure E.1-E.2.

Figure 6.2 shows the distribution of the bin index defined in Section 5.1. There seems to be some deviation between data and MC in the region $0.0 < M_{X_s}^{\rm reco} < 0.6$ GeV. If the assigned systematic uncertainties account for this discrepancy, no additional systematic uncertainty or correction is required. To evaluate this quantitatively, a fit is performed. Figure 6.3 shows the FastBDT output distribution after the fit. After the fit, the χ^2 for the first region is

$$\chi^2_5 = 8.34 \ (0.0 < M_{X_s}^{\rm reco} < 0.6 \text{ GeV}). \tag{6.6}$$

Then, the p-value is

$$\text{p-value} = \mathcal{P}(\chi^2_5 > 8.34) = 0.14 \ (0.0 < M_{X_s}^{\rm reco} < 0.6 \text{ GeV}), \tag{6.7}$$

which is an acceptable value. Consequently, the assigned systematic uncertainties can account for the data-MC discrepancy.

6.3 FastBDT sideband

For validation, the FastBDT sideband is also examined. This sideband is defined as

$$0.8 < \mathcal{O}_{\rm BDT} < 0.86. \tag{6.8}$$

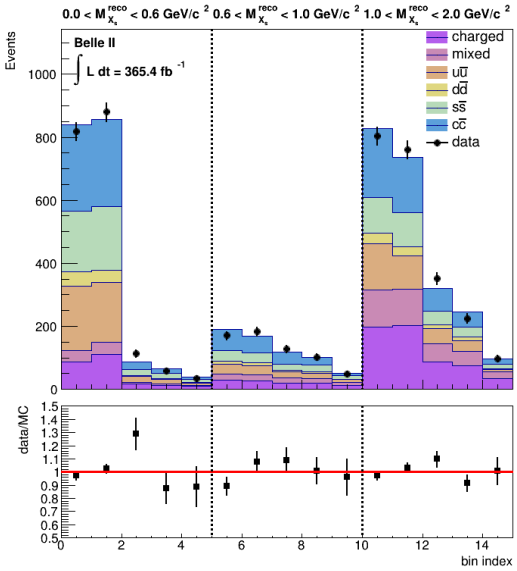
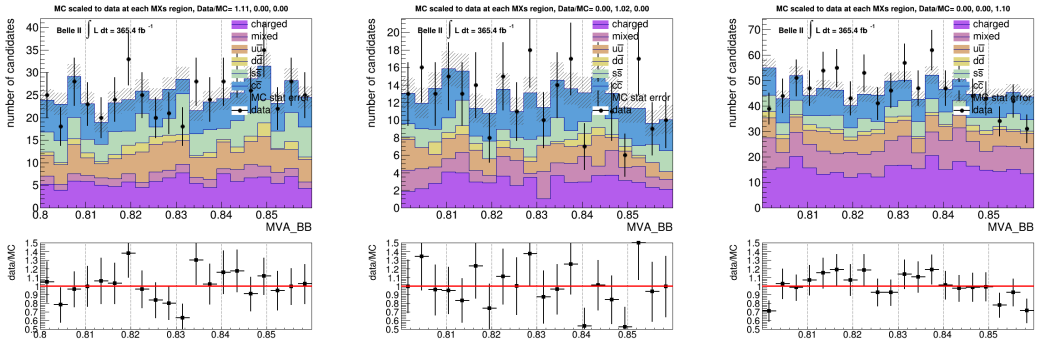


Figure 6.3: The bin index distribution after the fit in the M_{bc} sideband. The p-value is 0.14 in the region $0.0 < M_{X_s}^{\mathrm{reco}} < 0.6$ GeV/c^2.



(a) $0.0 < M_{X_s}^{\mathrm{reco}} < 0.6$ GeV/c^2 (b) $0.6 \leq M_{X_s}^{\mathrm{reco}} < 1.0$ GeV/c^2 (c) $1.0 \leq M_{X_s}^{\mathrm{reco}} < 2.0$ GeV/c^2

Figure 6.4: FastBDT output distributions in the FastBDT sideband.

This region is dominated by background.

The cut flow shown in Table 4.5 is used. However, the requirement $0.8 < \mathcal{O}_{\mathrm{BDT}} < 0.86$ is applied instead of $\mathcal{O}_{\mathrm{BDT}} > 0.86$. Additionally, the off-resonance ~~sample~~ sample is analyzed to obtain the $q\bar{q}$ correction factor for the FastBDT sideband.

Figure 6.4 shows the FastBDT output distribution for each $M_{X_s}^{\mathrm{reco}}$ region. The distributions show good agreement between data and MC. χ^2 values are

$$\chi^2_{19} = \begin{cases} 11.97 & (0.0 < M_{X_s}^{\mathrm{reco}} < 0.6 \text{ GeV}/c^2), \\ 13.41 & (0.6 \leq M_{X_s}^{\mathrm{reco}} < 1.0 \text{ GeV}/c^2), \\ 17.31 & (1.0 \leq M_{X_s}^{\mathrm{reco}} < 2.0 \text{ GeV}/c^2), \end{cases} \tag{6.9}$$

with 19 degrees of freedom. Here, the number of degrees of freedom is 19 rather than 20 because the MC distributions are normalized to the data in Figure 6.4. Since an overall normalization uncertainty of $\pm 20\%$ is assigned to the background samples, only the shape agreement is evaluated. The corresponding p-values are

$$p\text{-value} = \begin{cases} \mathcal{P}(\chi^2_{19} > 11.97) = 0.89 & (0.0 < M^{\text{reco}}_{X_s} < 0.6 \text{ GeV}/c^2), \\ \mathcal{P}(\chi^2_{19} > 13.41) = 0.82 & (0.6 \leq M^{\text{reco}}_{X_s} < 1.0 \text{ GeV}/c^2), \\ \mathcal{P}(\chi^2_{19} > 17.31) = 0.57 & (1.0 \leq M^{\text{reco}}_{X_s} < 2.0 \text{ GeV}/c^2), \end{cases} \tag{6.10}$$

indicating acceptable agreement between data and MC.

Chapter 7

Control sample study

In this analysis, $B \to X_s J/\psi$ is selected as a control sample. This decay mode has a large branching fraction and contains an X_s system. We use the control sample to estimate the correction factor for the FastBDT output requirement and its uncertainty.

We do not produce $B \to X_s J/\psi$ MC samples separately. Instead, we extract $B \to X_s J/\psi$ decays from the official $B\bar{B}$ background samples. The official $B\bar{B}$ background samples contain double counting in some $B \to X_s J/\psi$ decay channels, such as $B^+ \to J/\psi K^+\pi^+\pi^-$. Therefore, we reweight those double-counted decay channels.

The $B \to X_s J/\psi$ analysis procedure is similar to that of the $B \to X_s \nu\bar{\nu}$ analysis. However, the following additional procedures are applied at the reconstruction stage:

1. Reconstruct $B_{\rm tag}$ and X_s candidates. The reconstruction method is the same as that used in the $B \to X_s \nu\bar{\nu}$ analysis described in Section 4.4.

2. Reconstruct $J/\psi(\to \mu^+\mu^-)$. Selection criteria are

   - μ: $dr < 2$ cm, $|dz| < 4$ cm, and `muonID` > 0.1
   - J/ψ: reconstructed from two muons, with the requirement $|dM| < 0.05$ GeV/c^2.

3. Reconstruct $B_{\rm sig}(\to X_s J/\psi)$. Selection criteria are

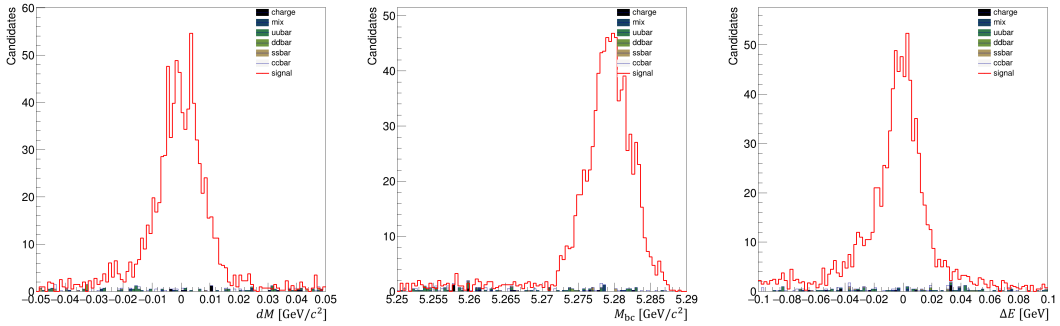
   - $M_{bc} > 5.25$ GeV/c^2 and $|\Delta E| < 0.1$ GeV

4. Reconstruct $\Upsilon(4S)(\to B_{\rm sig} B_{\rm tag})$

5. Perform the BCS for the J/ψ that is a daughter of the reconstructed $\Upsilon(4S)$ candidates in (4). Random selection is used for BCS.

6. Select the $B_{\rm tag}$ and X_s candidates reconstructed in step 1 that do not contain the muons from the J/ψ selected in step 5.

7. Apply the $B \to X_s \nu\bar{\nu}$ analysis procedure using the selected $B_{\rm tag}$ and X_s candidates.



(a) dM of J/ψ (b) $M_{\rm bc}$ of $B_{\rm sig}$ (c) ΔE of $B_{\rm sig}$

Figure 7.1: Distributions of variables for $B \to X_s J/\psi$ analysis. dM distribution is for J/ψ which is a granddaughter of $\Upsilon(4S)$ candidates. The $M_{\rm bc}$ and ΔE distributions are for B_{sig} that is a daughter of $\Upsilon(4S)$ candidates. These plots are made after BCS of $\Upsilon(4S)$.

- The muons from the J/ψ in step 4 are excluded when calculating the kinematic variables and the D-veto information.

Table 7.1 shows the cut flow for $B \to X_s J/\psi$ MC studies.

Figure 7.1 shows the distributions of the J/ψ mass difference dM, $M_{\rm bc}$, and ΔE. These distributions are obtained after the BCS of the $\Upsilon(4S)$ candidates.

Figure 7.2 shows the FastBDT output distribution for $B \to X_s J/\psi$ analysis. The third bin in the right figure shows a large discrepancy between data and MC. This large discrepancy had not been found in the previous iteration. In Appendix **??**, this discrepancy is investigated. We conclude that this large deviation can occur when the look-elsewhere effect associated with FastBDT retraining is taken into account. Figures F.1–F.2 show the distributions of the FastBDT input variables. The MC distributions are normalized to the data in Figures F.1–F.2.

The data-to-MC ratios are

$$\frac{N_{Data}}{N_{MC}} = \begin{cases} 0.82 \pm 0.10 & (0.0 < M_{X_s}^{\rm reco} < 0.6 \text{ GeV}), \\ 1.17 \pm 0.21 & (0.6 \leq M_{X_s}^{\rm reco} < 1.0 \text{ GeV}), \\ 0.61 \pm 0.12 & (1.0 \leq M_{X_s}^{\rm reco} < 2.0 \text{ GeV}), \end{cases} \tag{7.1}$$

where only the statistical uncertainties in data and MC are included. The calibration and correction factors listed in Section 8 are applied, except for the $q\bar{q}$ correction factor.

Table 7.1: Cut flow for the $B \to X_s J/\psi$ MC study. The entries show the numbers of surviving events, except in the last column, which shows the data-to-MC ratio. The signal and background samples are scaled to an integrated luminosity of 365.37 fb^{-1}. Cross-feed events are included in the signal category. The calibration and correction factors listed in Section 8 are not applied.

Selection	SIGNAL	CHG	MIX	UUBAR	DDBAR	SSBAR	CHARM	data/MC
Generated				365.37 fb^{-1}				1.0000
Skim	2.71×10^5	3.42×10^7	2.60×10^7	9.11×10^7	2.19×10^7	1.73×10^7	9.64×10^7	1.0475
Reconstruction and preselections	368.6	1.0	0.8	6.8	0.8	1.3	3.8	0.6398
$M_{\rm bc} > 5.27$ GeV/c^2 for $B_{\rm tag}$	332.3	0.8	0.8	3.0	0.3	0.5	1.5	0.6282
$\lvert\Delta E\rvert < 0.2$ GeV for $B_{\rm tag}$	310.0	0.8	0.8	2.3	0.3	0.5	0.8	0.6026
$E_{\rm ecl} < 1.3$ GeV	308.0	0.8	0.8	2.3	0.3	0.5	0.8	0.6065
$0.297 < \theta_{\rm miss} < 2.618$ rad	295.3	0.8	0.8	1.8	0.3	0.5	0.8	0.6100
$0.5 < p^*_{X_s} < 2.96$ GeV/c	295.3	0.8	0.8	1.8	0.3	0.5	0.8	0.6100
Reject $1.84 < M^{\rm reco}_{X_s} < 1.89$ GeV/c^2	292.5	0.8	0.8	1.8	0.3	0.5	0.8	0.6055
BCS	292.5	0.8	0.8	1.8	0.3	0.5	0.8	0.6055
$\mathcal{O}_{\rm BDT} > 0.86$	219.6	0.3	0.3	0.3	0	0	0	0.6262

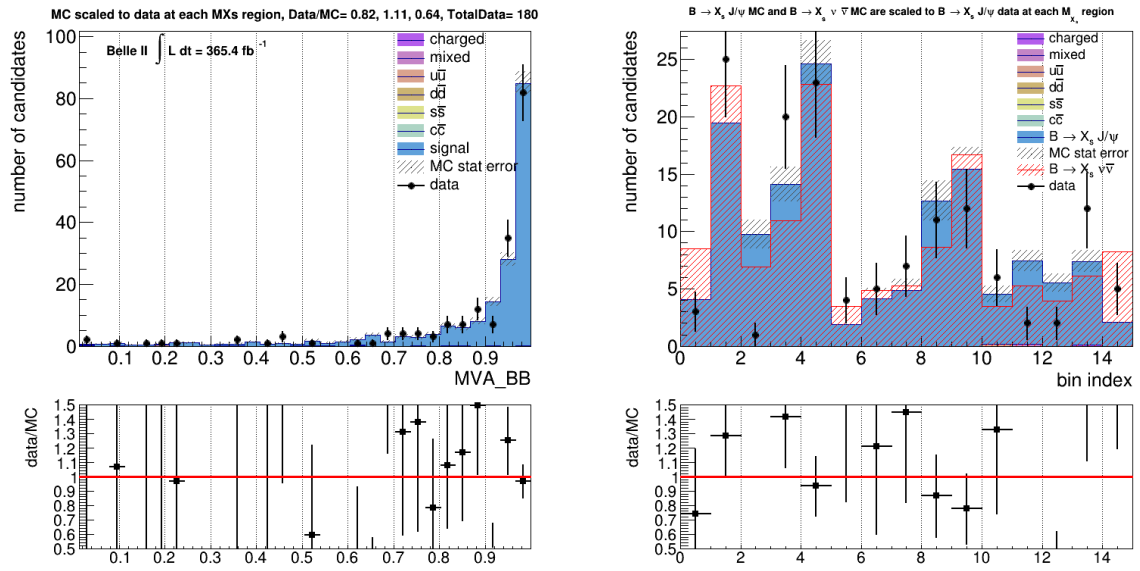


Figure 7.2: The left plot shows the FastBDT output distribution for $B \to X_s J/\psi$ analysis. The right plot shows the bin index distribution defined in Section ??. In both plots, calibration and correction factors in Section 8 are applied, except for the $q\bar{q}$ correction factor. For the comparison, the right plot also includes the $B \to X_s \nu\bar{\nu}$ signal MC distribution ~~for comparison~~.

For the FastBDT selection efficiency, the data-to-MC ratios are

$$\frac{\epsilon_{Data}^{FBDT}}{\epsilon_{MC}^{FBDT}} = \begin{cases} 1.00 \pm 0.04 & (0.0 < M_{X_s}^{\text{reco}} < 0.6 \text{ GeV}/c^2), \\ 1.05 \pm 0.08 & (0.6 \leq M_{X_s}^{\text{reco}} < 1.0 \text{ GeV}/c^2), \\ 0.97 \pm 0.14 & (1.0 \leq M_{X_s}^{\text{reco}} < 2.0 \text{ GeV}/c^2), \end{cases} \tag{7.2}$$

where $\epsilon_{\text{Data}}^{\text{FBDT}}$ and $\epsilon_{\text{MC}}^{\text{FBDT}}$ are the FastBDT selection efficiencies in data and MC, respectively.

Chapter 8

Calibration and corrections

8.1 Hadron PID

In this analysis, we apply corrections to the hadron PID efficiencies in the MC samples. These correction factors are calculated in the two-dimensional plane of momentum and polar angle.

For the kaon and pion efficiencies, we use the Belle II systematic framework with the $D^{*+} \to D^0(\to K^-\pi^+)\pi^+$ decay mode [59]. The kaon–pion misidentification rates are also corrected.. Figures G.1 and G.7 show the data-to-MC ratio for the kaon and pion PID efficiencies. Figure G.2 shows the data-to-MC ratio for the misidentification rate for pions identified as kaons. Figure G.8 shows the data-to-MC ratio for the misidentification rate for kaons identified as pions.

Leptons misidentified as kaons or pions constitute one source of background. In Section H, the particle compositions of the kaon and pion candidates are shown. We correct the rates at which electrons and muons are misidentified as pions or kaons. Figures G.3 and G.4 show the data-to-MC ratios for the rates at which electrons are misidentified as kaons. Figures G.5 and G.6 show the data-to-MC ratios for the rates at which muons are misidentified as kaons. Figures G.9 and G.10 show the data-to-MC ratios for the rates at which electrons are misidentified as pions. Figures G.11 and G.12 show the data-to-MC ratios for the rates at which muons are misidentified as pions.

If a PID correction factor has an unphysical value, the correction is not applied. Figure 8.1 shows the ratio of events for which the PID correction factor has an unphysical value in the signal sample. In addition, if the momentum or polar angle lies outside the calibrated range, the correction is not applied.

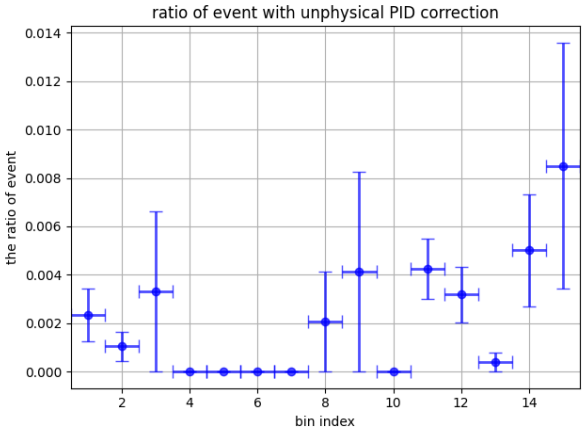


Figure 8.1: The ratio of events whose PID correction is unphysical

8.2 Photon multiplicity

It is known that the Belle II simulation does not model the beam background perfectly. To address this issue, an ECL energy-scale correction [60] or a photon-multiplicity correction [56] is applied in other analyses. In this analysis, a photon-multiplicity correction is applied. To determine the photon-multiplicity correction factor, two wrong-charge sidebands are defined. The wrong-charge sidebands are defined by reconstructing $\Upsilon(4S)$ with an incorrect charge combination of B_{tag} and X_s:

- Neutral-signal wrong-charge sideband: $\Upsilon(4S)$ is reconstructed from $B^{\pm}_{\text{tag}} X^{0}_{sd}$.
- Neutral-tag wrong-charge sideband: $\Upsilon(4S)$ is reconstructed from $B^{0}_{\text{tag}} X^{\pm}_{su}$.

The photon multiplicity correction factor is obtained from the neutral-signal wrong-charge sideband. The neutral-tag wrong-charge sideband is used to validate the photon multiplicity correction and estimate its uncertainty.

The multiplicity of the remaining photon candidates is examined. The correction factor for the photon multiplicity is obtained by

$$C_i = \frac{N_i^{data}}{N_i^{MC}} \times \frac{\sum N_j^{MC}}{\sum N_j^{data}}, \tag{8.1}$$

where C_i is the correction factor, N_i^{data} is the number of events in data, and N_i^{MC} is the number of events in MC when the number of remaining photon candidates is i.

Figure 8.2 shows the distribution of the number of remaining photon candidates for the neutral-signal wrong-charge sideband. The left plot shows the distribution before the photon multiplicity correction. The right plot shows the distribution after the correction.

Figure 8.3 shows the distribution of the number of remaining photon candidates for the neutral-tag wrong-charge sideband. After the correction, the data-MC agreement improves, especially in the high-photon-multiplicity region.

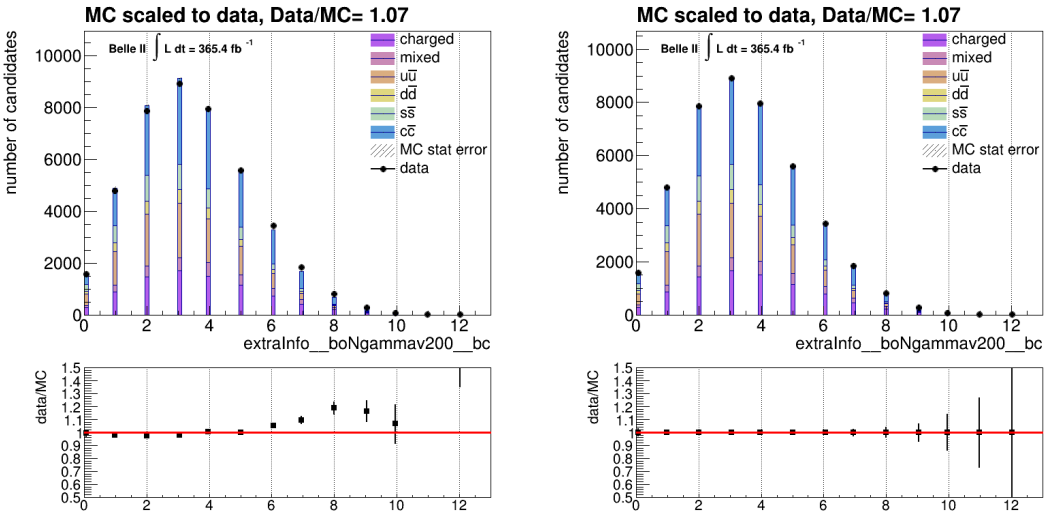


Figure 8.2: Distribution of the number of remaining photon candidates in the neutral-signal wrong-charge sideband. The left plot shows the distribution before the photon-multiplicity correction, and the right plot shows the distribution after the correction. By definition, the data-to-MC ratio is unity in each photon-multiplicity bin after the correction.

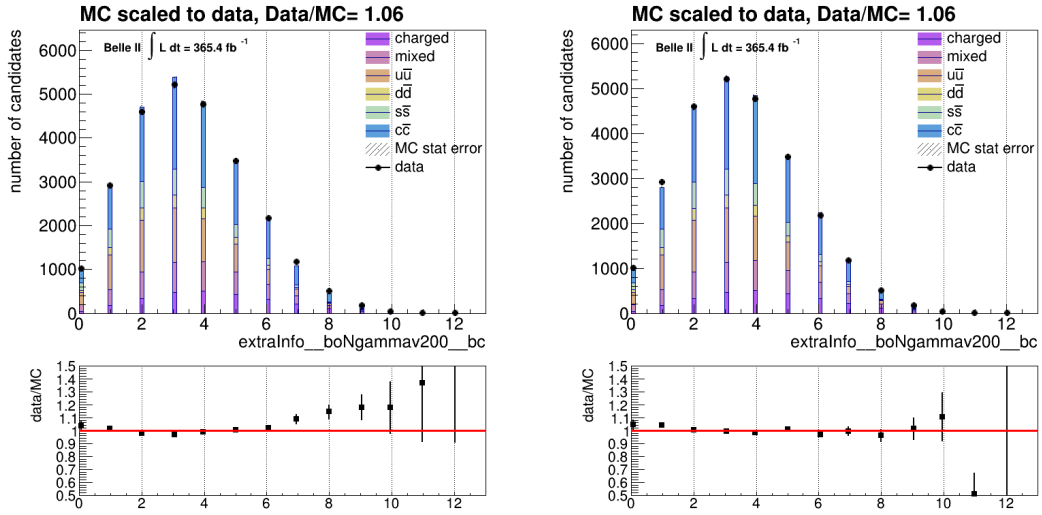


Figure 8.3: Distribution of the number of remaining photon candidates in the neutral-tag wrong-charge sideband. The left plot shows the distribution before the photon-multiplicity correction, and the right plot shows the distribution after the correction. The data-to-MC ratio is not exactly unity after the correction. These residual data-to-MC differences are used to estimate the systematic uncertainty associated with the photon-multiplicity correction.

8.3 π^0 efficiency

The official correction factor for the π^0 efficiency is applied. The correction factor is derived from $\tau \to h\pi^0\nu$ decays. This factor depends on the π^0 momentum and polar angle. Figure 8.4 shows the π^0 efficiency correction factors.

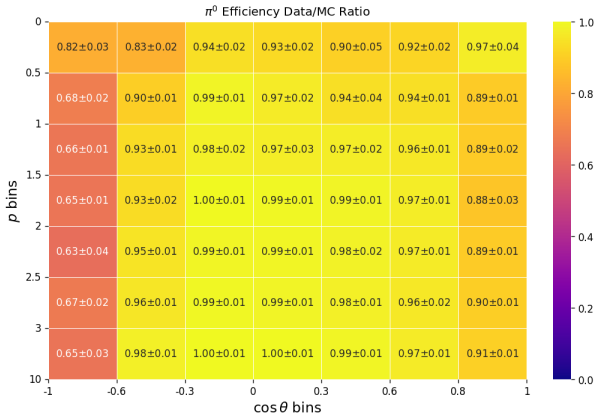


Figure 8.4: Data-to-MC ratio for the π^0 reconstruction efficiency as a function of momentum and polar angle.

8.4 K^0_S efficiency

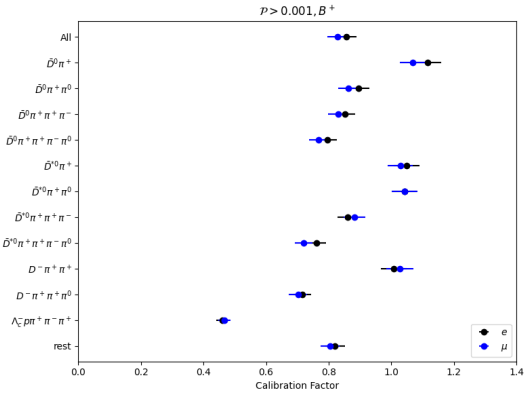
The official correction factor for the K^0_S efficiency is applied. The correction factor is derived using the $D^{*+} \to D^0(\to K^0_S\pi^+\pi^-)\pi^+$ decay mode. This correction factor depends on the K^0_S momentum, polar angle, and the distance between the IP and the K^0_S decay vertex.

8.5 FastBDT efficiency

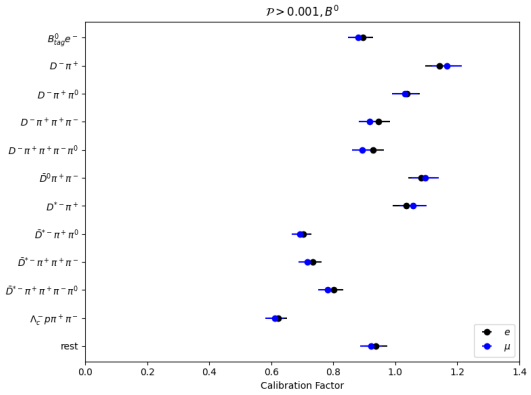
The FastBDT efficiency correction factors are obtained from the $B \to X_s J/\psi$ control-sample study described in Section 7. The data-to-MC ratios given in Eq. 7.2 are applied separately in each $M^{\text{reco}}_{X_s}$ bin.

8.6 FEI calibration for $B\bar{B}$

The official calibration factors for the hadronic FEI efficiency are applied. The calibration factors are obtained using inclusive $B \to X\ell\nu$ decays, where $\ell = e$ or μ. They are determined separately for charged and neutral B_{tag} candidates and for several B_{tag} decay modes. Figure 8.5 shows the FEI calibration factors obtained with the requirement



(a) FEI calibration factors for charged B_{tag} candidates.



(b) FEI calibration factors for neutral B_{tag} candidates.

Figure 8.5: FEI efficiency calibration factors obtained with the requirement $\mathcal{P}_{\text{FEI}} > 0.001$. The black and blue points show the results obtained using the $B \to Xe\nu$ and $B \to X\mu\nu$ samples, respectively.

$\mathcal{P}_{\text{FEI}} > 0.001$. The labels e and μ denote the calibration results obtained using the $B \to Xe\nu$ and $B \to X\mu\nu$ samples, respectively.

8.7 $q\bar{q}$ efficiency

The $q\bar{q}$ background efficiency is corrected using correction factors derived from the off-resonance sample described in Section 6.1. The data-to-MC ratios given in Eq. 6.2 are applied separately in each $M_{X_s}^{\text{reco}}$ bin.

8.8 Fragmentation of X_s

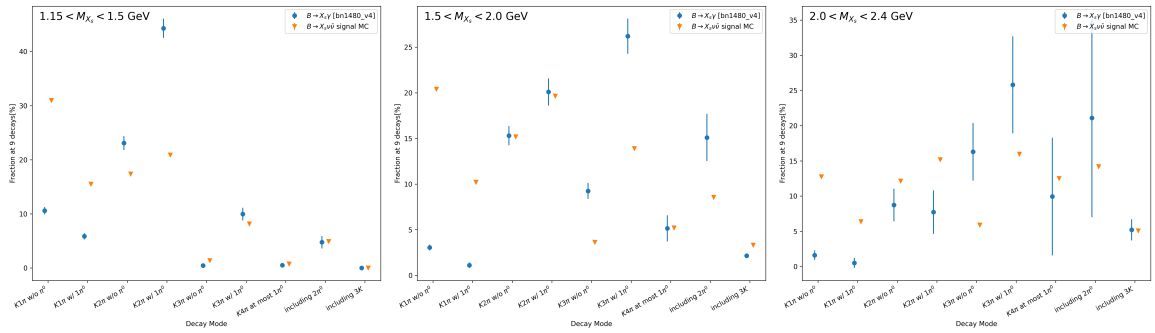
The fragmentation of X_s is simulated using PYTHIA8. The relative fractions of the decay modes should be corrected using experimental measurements.

Three possible samples are considered for the correction:

1. $B \to X_s\ell\bar{\ell}$
2. $B \to X_s\gamma$
3. $B \to X_sJ/\psi$

Each sample has advantages and disadvantages:

- The $B \to X_s\ell\bar{\ell}$ decay is the most similar to the $B \to X_s\nu\bar{\nu}$ decay. However, no sufficiently precise measurement is available to derive the correction factors.



(a) (b) (c)

Figure 8.6: Fragmentation fractions of the nine decay-mode categories. The fractions are normalized within these nine categories and therefore sum to 100%. The blue points show the results of the $B \to X_s\gamma$ study, and the orange points show the corresponding fractions in the $B \to X_s\nu\bar{\nu}$ signal MC sample.

- The $B \to X_s\gamma$ decay proceeds through a similar electroweak-penguin process. However, because the photon is a massless spin-1 particle, the spin of the X_s system cannot be zero. Therefore, the fragmentation may differ from that in $B \to X_s\nu\bar{\nu}$ decays.
- The $B \to X_s J/\psi$ decay proceeds through a different process dominated by strong interactions. Therefore, its fragmentation may also differ from that in $B \to X_s\nu\bar{\nu}$ decays.

In this analysis, the fragmentation of the remaining $B \to X_s\nu\bar{\nu}$ signal MC sample, defined in Section 3.2.2, is corrected using the results of the $B \to X_s\gamma$ study. The uncertainty in this correction is estimated from the difference between the results of the $B \to X_s\gamma$ and $B \to X_s J/\psi$ studies.

Figure 8.6 shows the fragmentation fractions of the nine decay-mode categories in each M_{X_s} region. The fractions are normalized within these nine categories, and therefore their sum is 100%. The blue points show the results of the $B \to X_s\gamma$ study [61, 62]. The orange points show the fragmentation fractions in the $B \to X_s\nu\bar{\nu}$ signal MC sample. Correction factors are derived and applied so that the fragmentation fractions in the $B \to X_s\nu\bar{\nu}$ signal MC sample reproduce those measured in the $B \to X_s\gamma$ study. The correction factors are not applied in the regions $M_{X_s} < 1.15$ GeV/c^2 and $M_{X_s} > 2.4$ GeV/c^2.

8.9 Mismodelling of $B \to X_s n\bar{n}$ decays

In the official Belle II simulation, $B \to K^{(*)}n\bar{n}$ decays are generated according to a flat phase-space model. However, experimental measurements of $B \to K^{(*)}p\bar{p}$ decays [63, 64, 65] show $M_{p\bar{p}}$ distributions that differ from a flat phase-space model. Figures 8.7–8.10 compare the measured distributions with those obtained from the official MC samples.

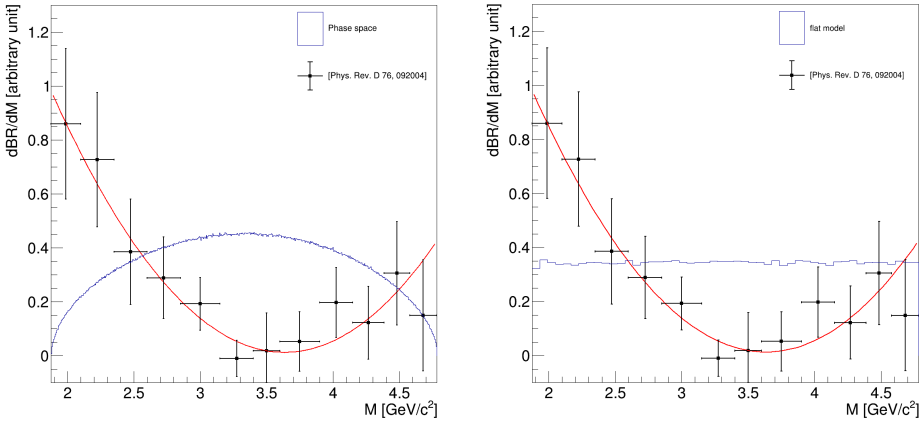


Figure 8.7: Distribution of the invariant mass of the baryon–antibaryon pair. The black points show the distribution of $B^0 \to K^0 p\bar{p}$ decays. The blue histogram shows the distribution of $B^+ \to K^+ n\bar{n}$ decays generated using (left) a flat phase-space model and (right) a model with a flat $M_{n\bar{n}}$ distribution.

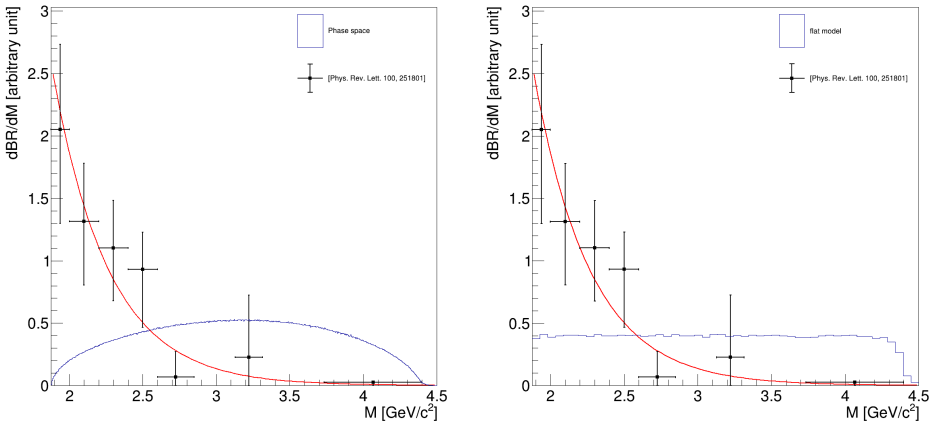


Figure 8.8: Distribution of the invariant mass of the baryon–antibaryon pair. The black points show the distribution of $B^0 \to K^{*0} p\bar{p}$ decays. The blue histogram shows the distribution of $B^+ \to K^{*+} n\bar{n}$ decays generated using (left) a flat phase-space model and (right) a model with a flat $M_{n\bar{n}}$ distribution.

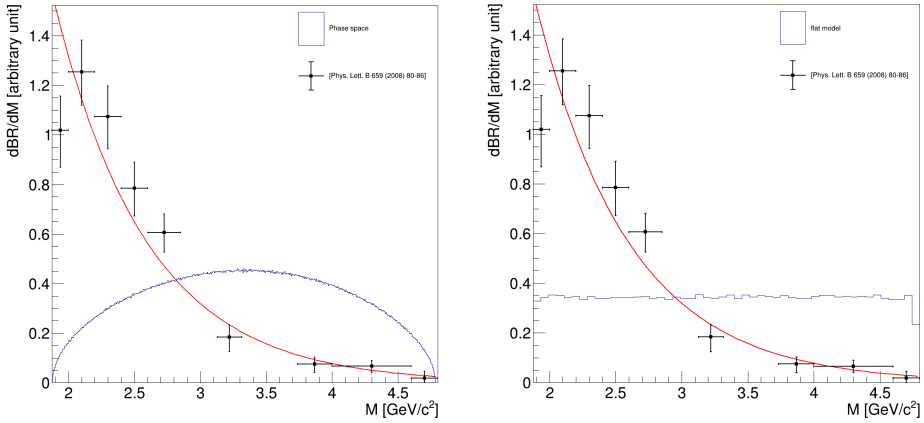


Figure 8.9: Distribution of the invariant mass of the baryon–antibaryon pair. The black points show the distribution of $B^+ \to K^+ p\bar{p}$ decays. The blue histogram shows the distribution of $B^0 \to K^0 n\bar{n}$ decays generated using (left) a flat phase-space model and (right) a model with a flat $M_{n\bar{n}}$ distribution.

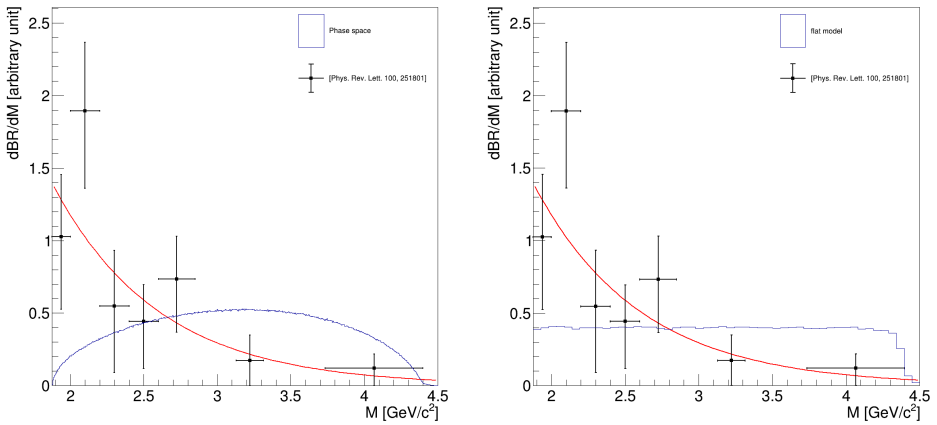


Figure 8.10: Distribution of the invariant mass of the baryon–antibaryon pair. The black points show the distribution of $B^0 \to K^{*0} p\bar{p}$ decays. The blue histogram shows the distribution of $B^0 \to K^{*0} n\bar{n}$ decays generated using (left) a flat phase-space model and (right) a model with a flat $M_{n\bar{n}}$ distribution.

As discussed in Section 3.2.1, the official $B \to K^{(*)} n\bar{n}$ samples are replaced with private MC samples generated with a flat $M_{n\bar{n}}$ distribution. Correction factors are then applied to these private samples to reproduce the observed $M_{p\bar{p}}$ distributions. The measured $M_{p\bar{p}}$ distributions are fitted to obtain the correction functions. An exponential function is used to fit the $B^+ \to K^+ p\bar{p}$, $B^+ \to K^{*+} p\bar{p}$, and $B^0 \to K^{*0} p\bar{p}$ distributions. A second-order polynomial is used for the $B^0 \to K^0 p\bar{p}$ distribution. The red curves in Figures 8.7–8.10 show the fitted functions. The correction factors are defined as the ratios of the fitted functions to the distributions generated with a flat $M_{n\bar{n}}$ model.

A correction is also required for the remaining $B \to X_s n\bar{n}$ decays. However, no previous experimental measurement of $B \to X_s p\bar{p}$ decays is available. Therefore, a study of $B \to X_s p\bar{p}$ decays is performed. To reconstruct $B \to X_s p\bar{p}$ decays, the X_s and B_{tag} candidates are reconstructed using the method described in Section 4. Additionally, proton candidates are selected by the following criteria:

- `protonID` > 0.6
- $dr < 0.5$ cm and $|dz| < 2$ cm
- `nCDCHits` > 20
- `nPXDHits` > 0

The selected proton and antiproton candidates are combined with the X_s candidate to reconstruct $B_{\text{sig}} \to X_s p\bar{p}$. Several selection requirements are applied after reconstruction. Table 8.1 shows the cut flow for the $B \to X_s p\bar{p}$ analysis. To suppress background, the Fox–Wolfram ratio R_2, the cosine of the angle between the thrust axis of B_{sig} and that of its ROE, and the event sphericity are used [66]. Figure 8.11 shows the M_{bc} distribution of B_{sig} candidates after the BCS.

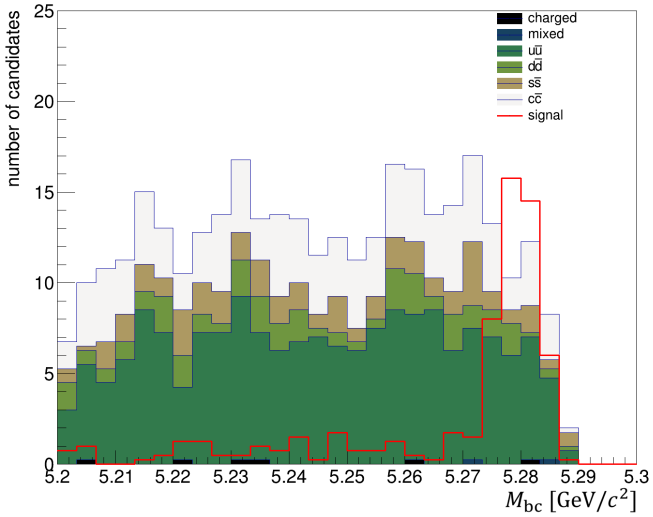


Figure 8.11: Distribution of M_{bc} for the B_{sig} candidates in the $B \to X_s p\bar{p}$ analysis.

Table 8.1: Cut flow for the $B \to X_s p\bar{p}$ analysis.

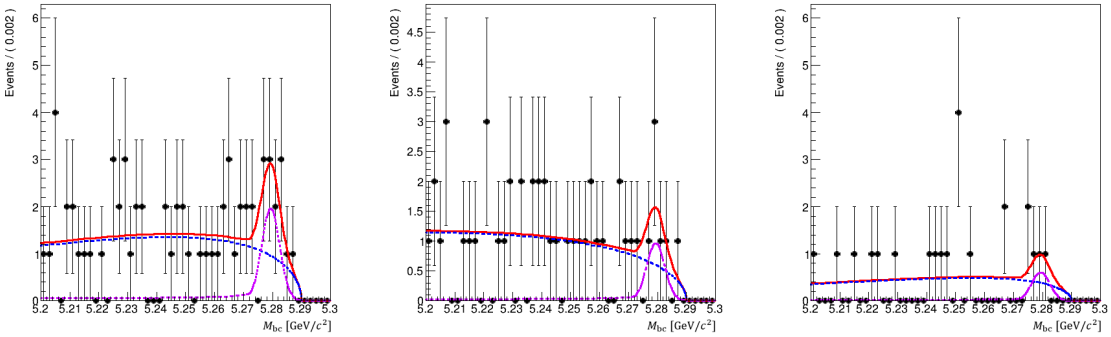
Selection
Generated
Skim
Reconstruction and preselections
$M_{\rm bc} > 5.2$ GeV/c^2 for $B_{\rm sig}$
$\lvert\Delta E\rvert < 0.05$ GeV for $B_{\rm sig}$
$\lvert\Delta E\rvert < 0.2$ GeV for $B_{\rm tag}$
$E_{\rm ecl} < 1.3$ GeV
`sphericity` > 0.2
`R2` < 0.4 for $B_{\rm sig}$
`cosTBTO` < 0.9 for $B_{\rm sig}$
$M_{X_s} > 1.0$ GeV/c^2
BCS

Table 8.2: Fit model used in the $B \to X_s p\bar{p}$ analysis.

Function	Parameter	
Crystal Ball	mean	fixed using $B \to K^{(*)} p\bar{p}$ data
	width	fixed using $B \to K^{(*)} p\bar{p}$ data
	alpha	fixed from MC
	n	fixed from MC
ARGUS	endpoint	fixed to the beam energy
	curvature	floated
	power	fixed to 0.5

To determine the $M_{p\bar{p}}$ distribution, fits are performed in several $M_{p\bar{p}}$ regions. Table 8.2 summarizes the fit model used in the $B \to X_s p\bar{p}$ analysis. Figure 8.12 shows the fit results in the different $M_{p\bar{p}}$ regions.

Figure 8.13 compares the $M_{p\bar{p}}$ distributions in data and MC. The blue points show the event yields in data, and the red points show the corresponding MC yields. For $M_{p\bar{p}} > 3.675$ GeV/c^2, no events are observed in the region $M_{\rm bc} > 5.27$ GeV/c^2, and only one event is observed in the sideband $5.2 < M_{\rm bc} < 5.27$ GeV/c^2, where $M_{\rm bc}$ is calculated for the $B_{\rm sig}$ candidate. Therefore, the yield is assumed to be zero in the region $M_{p\bar{p}} > 3.675$ GeV/c^2. The correction factors are defined as the ratios of the data yields to the MC yields.



(a) $1.800 < M_{p\bar{p}} < 2.425$ (b) $2.425 < M_{p\bar{p}} < 3.050$ (c) $3.050 < M_{p\bar{p}} < 3.675$

Figure 8.12: Fit results for the $B \to X_s p\bar{p}$ analysis in three $M_{p\bar{p}}$ intervals, where $M_{p\bar{p}}$ is expressed in GeV/c^2. The fit variable is $M_{\rm bc}$ of the $B_{\rm sig}$ candidate. The blue curve shows the background component, the purple curve shows the $B \to X_s p\bar{p}$ signal component, and the red curve shows their sum.

8.10 Mismodelling of $B \to X_s K^0_L K^0_L$ decays

In the $B^+ \to K^+ \nu\bar{\nu}$ analysis [56], $B^+ \to K^+ K^0_L K^0_L$ decays were found to be one of the major sources of systematic uncertainty. Therefore, corrections are applied to $B \to X_s K^0_L K^0_L$ decays.

To correct the $B^+ \to K^+ K^0_L K^0_L$ decay model, experimental results for $B^+ \to K^+ K^0_S K^0_S$ decays are used. According to Ref. [1], the dynamical amplitude for the $B^+ \to K^+ K^0_S K^0_S$ decay can be modeled as follows:

$$F^L_j(s_{12}, s_{23}) = R_j(m)\, X_L(|\overrightarrow{p}^*|\, r')\, X_L(|\overrightarrow{q}|\, r)\, T_j(L, \overrightarrow{p}, \overrightarrow{q}), \tag{8.2}$$

where s_{12} is the squared invariant mass of the $K^0_S K^0_S$ system, and s_{23} is the larger of the two squared invariant masses of the $K^0_S K^+$ combinations.

In the official Belle II simulation, the $B^+ \to K^+ K^0_L K^0_L$ sample contains both resonant and nonresonant components. The nonresonant component is generated using a flat phase-space model. In this analysis, the resonant $B^+ \to K^+ K^0_L K^0_L$ components are removed. More specifically, $B^+ \to K^+ K^0_L K^0_L$ decays from $B^+ \to K^+ f_0(980)$, $B^+ \to K^+ f_0(1370)$, $B^+ \to K^+ f_2(1270)$, $B^+ \to K^+ f'_2(1525)$, and $B^+ \to K^+ \chi_{c0}(1P)$ decays are removed. These are all resonant decay modes that contribute to $B^+ \to K^+ K^0_L K^0_L$ in the official simulation. A correction factor is then applied to the nonresonant $B^+ \to K^+ K^0_L K^0_L$ component. Figure 8.14 shows the Dalitz plots for $B^+ \to K^+ K^0_L K^0_L$ decays generated using a flat phase-space model and a model based on the experimental results of Ref. [1]. A small distortion is present in the flat-phase-space distribution because of `PHOTOS`. Because this effect is small, additional correction for `PHOTOS` is not considered. The correction factor is obtained from the bin-by-bin ratio of the two Dalitz-plot distributions shown in Figure 8.14. A correction factor for the branching ratio is also applied.

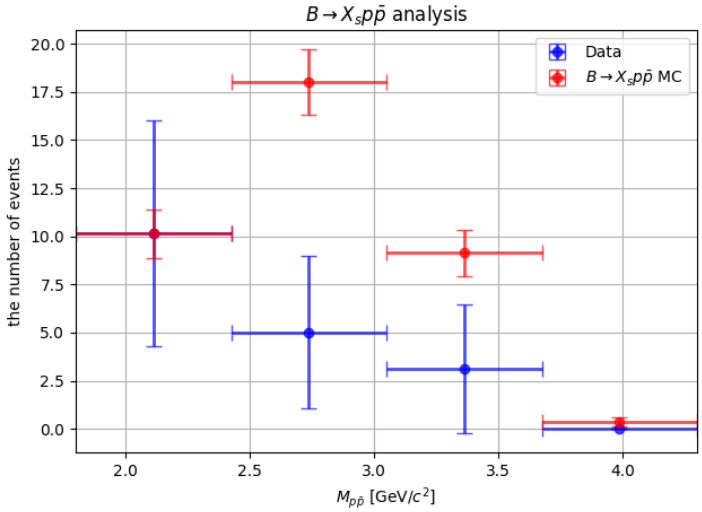


Figure 8.13: Comparison of the $M_{p\bar{p}}$ distributions in data and MC for $B \to X_s p\bar{p}$ decays. The blue points show the signal yields extracted from data, and the red points show the expected yields from MC.

It is calculated using the PDG value as follows [22]:

$$w = \frac{BR_{\rm PDG}(B^+ \to K^+ K_S^0 K_S^0)}{BR_{\rm MC}^{\rm nonres}(B^+ \to K^+ K_L^0 K_L^0)}. \tag{8.3}$$

The correction factor for $B^0 \to K_S^0 K_L^0 K_L^0$ is obtained using a similar method. However, a different procedure is required. The $B^0 \to K_S^0 K_L^0 K_L^0$ decay contains a resonant contribution from $\phi \to K_S^0 K_L^0$. In contrast, the $B^0 \to K_S^0 K_S^0 K_S^0$ decay contains no contribution from $\phi \to K_S^0 K_L^0$. Consequently, the Dalitz-plot distribution for $B^0 \to K_S^0 K_S^0 K_S^0$ is not identical to that for $B^0 \to K_S^0 K_L^0 K_L^0$.

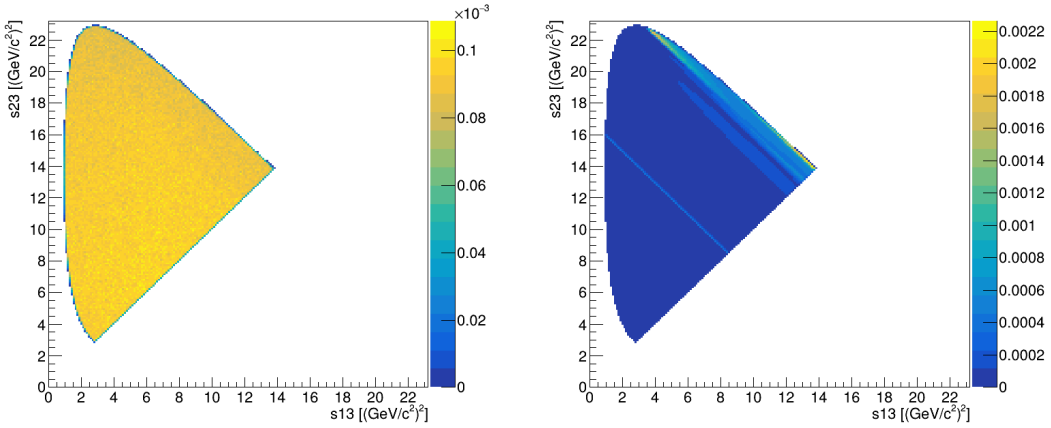
According to the calculation [67], the relation between $B^0 \to K_S^0 K_S^0 K_S^0$ and $B^0 \to K_S^0 K_L^0 K_L^0$ is

$$BR(B^0 \to K_S^0 K_S^0 K_S^0) = 3 \times BR(B^0 \to K_S^0 K_L^0 K_L^0), \tag{8.4}$$

where the ϕK contribution is excluded. Therefore, all resonant contributions are removed except for $B^0 \to \phi K_L^0$. More specifically, $B^0 \to K_S^0 K_L^0 K_L^0$ decays from $B^0 \to K^0 f_0(980)$, $B^0 \to K^0 f_0(1370)$, $B^0 \to K^0 f_2(1270)$, $B^0 \to K^0 f_2'(1525)$, and $B^0 \to K^0 \chi_{c0}(1P)$ decays are removed. These modes comprise all resonant contributions to $B^0 \to K_S^0 K_L^0 K_L^0$ included in the official simulation, except $B^0 \to \phi K_L^0$. For the $B^0 \to \phi K_L^0$ component, the branching fraction in the official Belle II simulation is corrected using the PDG value [22]:

$$w = \frac{BR_{\rm PDG}(B^0 \to \phi K_L^0)}{BR_{\rm MC}(B^0 \to \phi K_L^0)}. \tag{8.5}$$

For the nonresonant $B^0 \to K_S^0 K_L^0 K_L^0$ sample, the Dalitz-plot distribution measured in $B^0 \to K_S^0 K_S^0 K_S^0$ decays is used to model the kinematics. The $B^0 \to K_S^0 K_S^0 K_S^0$ decay is



(a) (b)

Figure 8.14: Dalitz plots for $B^+ \to K^+ K^0_L K^0_L$ decays generated using (a) a flat phase-space model and (b) a model based on experimental results [1]. Here, s_{12} is the squared invariant mass of the $K^0_L K^0_L$ system, and s_{23} is the larger of the two squared invariant masses of the $K^0_L K^+$ combinations. A small distortion is present in the flat-phase-space distribution because of PHOTOS.

modeled by the following formula [2]:

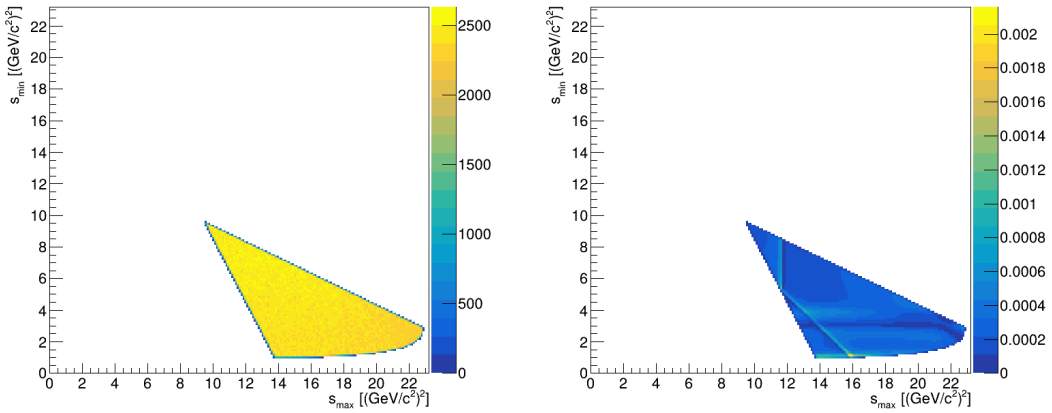
$$F_j^L\left(s_{\min}, s_{\max}, L\right) = R_j\left(m\right) X_L\left(|\overrightarrow{p}^*|\, r'\right) X_L\left(|\overrightarrow{q}|\, r\right) T_j\left(L, \overrightarrow{p}, \overrightarrow{q}\right), \tag{8.6}$$

where $s_{\min}$ and $s_{\max}$ are the smallest and largest squared invariant masses of $K^0_S K^0_S$, respectively. The correction factor is obtained from the bin-by-bin ratio of the Dalitz-plot distributions generated using a flat phase-space model and a model based on experimental results [2]. Figure 8.15 shows these Dalitz plots. The branching ratio correction factor is calculated using the PDG value [22]:

$$w = \frac{1}{3}\frac{BR_{\rm PDG}(B^0 \to K^0_S K^0_S K^0_S)}{BR^{\rm nonres}_{\rm MC}(B^0 \to K^0_S K^0_L K^0_L)}. \tag{8.7}$$

To correct $B \to K^* K^0_L K^0_L$ and $B \to X_s K^0_L K^0_L$, experimental results for $B \to K^* K^0_S K^0_S$ and $B \to X_s K^0_S K^0_S$ decays are needed. However, no such experimental measurements are available. Therefore, studies of $B \to K^* K^0_S K^0_S$ and $B \to X_s K^0_S K^0_S$ decays are performed.

To reconstruct $B \to K^* K^0_S K^0_S$ decay, the $K\pi$ and $B_{\rm tag}$ candidates are reconstructed using the method described in Section 4. The $K\pi$ candidate is combined with two K^0_S candidates to reconstruct $B_{\rm sig} \to K^* K^0_S K^0_S$. After the reconstruction, several cuts are applied. Table 8.3 shows the cut flow for $B \to K^* K^0_S K^0_S$ analysis. Figure 8.16 shows the $M_{\rm bc}$ distribution of the $B_{\rm sig}$ candidates after BCS. One event is observed in data in the region $M_{\rm bc} > 5.27$ GeV/c^2, with an expected background yield of 1.52 events. Therefore, an upper limit on the signal yield is determined. Using the Feldman–Cousins method, the



(a) (b)

Figure 8.15: Dalitz plots for $B^0 \to K^0_S K^0_L K^0_L$ decays generated using (a) a flat phase-space model and (b) a model based on experimental results [2]. Here, $s_{\min}$ and $s_{\max}$ are the smallest and largest squared invariant masses among the $K^0_L K^0_L$ and the two $K^0_S K^0_L$ combinations, respectively. A small distortion is present in the flat-phase-space distribution because of `PHOTOS`.

Table 8.3: Cut flow for the $B \to K^* K^0_S K^0_S$ analysis.

Selection
Generated
Skim
Reconstruction and preselections
$M_{bc} > 5.2$ GeV/c^2 for B_{sig}
$\lvert\Delta E\rvert < 0.05$ GeV for B_{sig}
$\lvert\Delta E\rvert < 0.2$ GeV for B_{tag}
$E_{\text{ecl}} < 1.3$ GeV
`sphericity` > 0.2
`R2` < 0.4 for B_{sig}
`cosTBTO` < 0.9 for B_{sig}
$0.6 < M_{X_s} < 1.0$ GeV/c^2
BCS

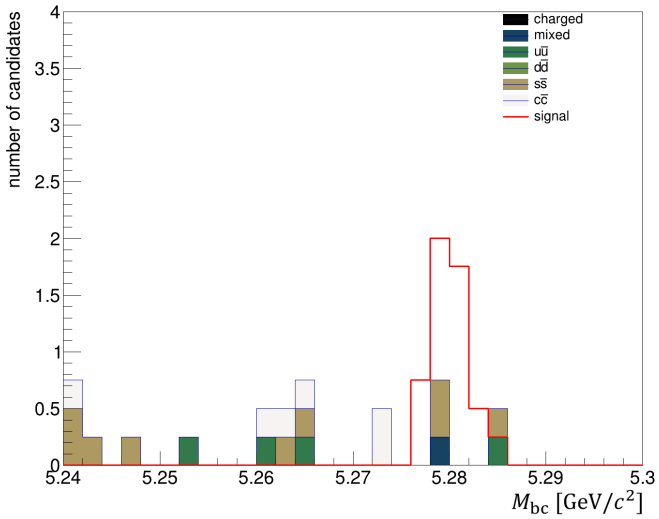


Figure 8.16: Distribution of $M_{\rm bc}$ for the $B_{\rm sig}$ candidates in the $B \to K^* K^0_S K^0_S$ analysis.

upper limit on the signal yield is 2.90 events at the 90% confidence level. The expected signal yield in MC is 3.89 events. The correction factor is therefore determined to be $2.90/3.89 = 0.74$.

To reconstruct $B \to X_s K^0_S K^0_S$ decay, the X_s and $B_{\rm tag}$ candidates are reconstructed using the method described in Section 4. The X_s candidate is combined with two K^0_S candidates to reconstruct $B_{\rm sig} \to X_s K^0_S K^0_S$. After the reconstruction, several cuts are applied. Table 8.4 shows the cut flow for $B \to X_s K^0_S K^0_S$ analysis. Figure 8.17 shows the $M_{\rm bc}$ of $B_{\rm sig}$ distribution after BCS. A fit is performed to extract the signal yield. Table 8.5 summarizes the fit model used in the $B \to X_s K^0_S K^0_S$ analysis. Figure 8.18 shows the fitting result. Figure 8.19 shows the comparison between the signal yields extracted from data and those expected from MC. The blue points show the signal yields extracted from data, and the red points show the expected yields from MC. The correction factors are defined as the ratios of the data yields to the corresponding MC yields.

8.11 $D \to K^0_L X$ decay from B meson

It was found that the data-MC discrepancy is reduced by applying a scale factor to $D \to K^0_L X$ decays from B mesons [56]. In this analysis, a scale factor of 1.3 ± 0.1 is applied.

Table 8.4: Cut flow for $B \to X_s K_S^0 K_S^0$ analysis.

Selection
Generated
Skim
Reconstruction and preselections
$M_{bc} > 5.2$ GeV/c^2 for $B_{\rm sig}$
$\lvert\Delta E\rvert < 0.05$ GeV for $B_{\rm sig}$
$\lvert\Delta E\rvert < 0.2$ GeV for $B_{\rm tag}$
$E_{\rm ecl} < 1.3$ GeV
`sphericity` > 0.2
`R2` < 0.4 for $B_{\rm sig}$
`cosTBTO` < 0.9 for $B_{\rm sig}$
$M_{X_s} > 1.0$ GeV/c^2
BCS

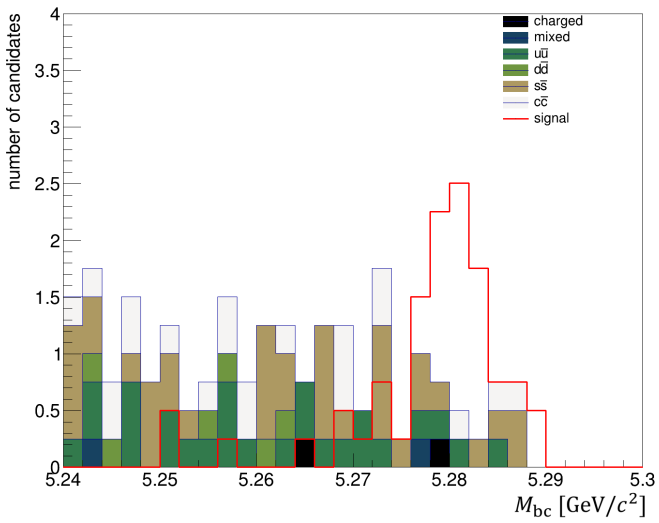


Figure 8.17: Distribution of $M_{\rm bc}$ for the $B_{\rm sig}$ candidates in the $B \to X_s K_S^0 K_S^0$ analysis.

Table 8.5: Fit model used in the $B \to X_s K^0_S K^0_S$ analysis.

Function	Parameter	
Crystal Ball	mean	fixed using $B \to K K^0_S K^0_S$ data
	width	fixed using $B \to K K^0_S K^0_S$ data
	alpha	fixed from MC
	n	fixed from MC
ARGUS	end point	fixed to the beam energy
	curvature	floated
	power	fixed to 0.5

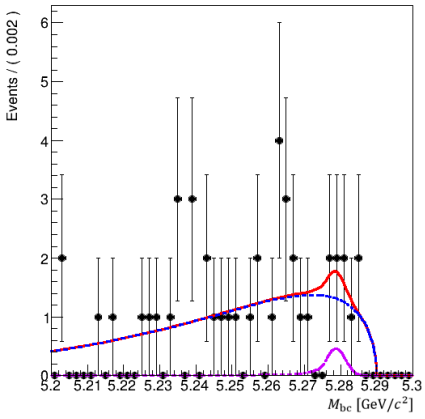


Figure 8.18: Fit results for the $B \to X_s K^0_S K^0_S$ analysis. The fit variable is $M_{\rm bc}$ of the $B_{\rm sig}$ candidate. The blue curve shows the background component, the purple curve shows the $B \to X_s K^0_S K^0_S$ signal component, and the red curve shows their sum.

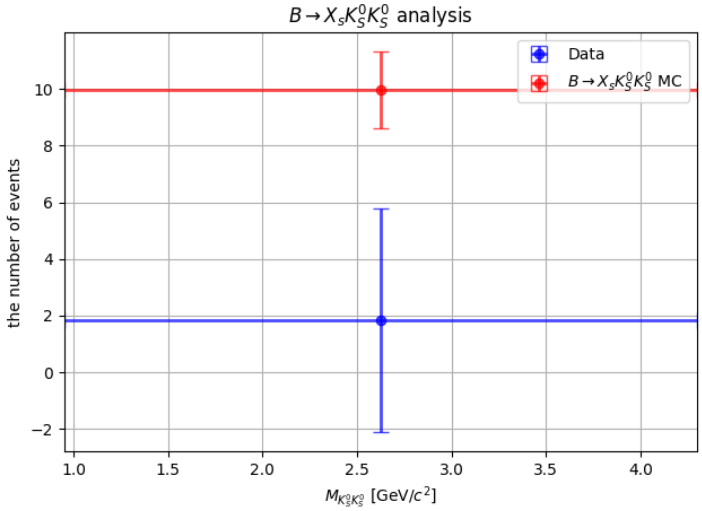


Figure 8.19: Comparison of the $M_{K_S^0 K_S^0}$ distributions in data and MC for $B \to X_s K_S^0 K_S^0$ decays. The blue points show the signal yields extracted from data, and the red points show the expected yields from MC.

Chapter 9

Systematic uncertainties

9.1 Tracking efficiency

The official relative uncertainty of $\pm 0.27\%$ is assigned to each track among the daughters of the X_s candidates. This uncertainty is determined using the tag-and-probe method with 1×3-prong $\tau^+\tau^-$ events in $e^+e^- \to \tau^+\tau^-$ collisions.

9.2 Hadron PID

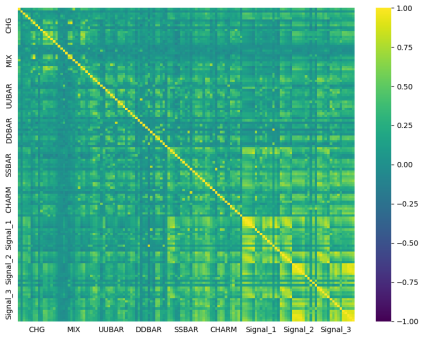
As shown in Appendix G, the PID correction factors and their uncertainties are provided in bins of particle momentum and polar angle. Because a large number of kinematic bins are used, treating the uncertainty in each PID-correction bin as an independent nuisance parameter would introduce a large number of nuisance parameters into the binned fit. Furthermore, variations of the correction factors in different PID-correction bins can affect the same fit bins, thereby inducing correlated variations in the predicted yields.

To obtain a compact set of independent systematic variations, principal-component analysis (PCA) is performed. A covariance matrix is constructed for the predicted yields across the fit bins and event categories. The covariance matrix is estimated using toy samples in which the PID correction factors are fluctuated according to their uncertainties. The covariance matrix is diagonalized as

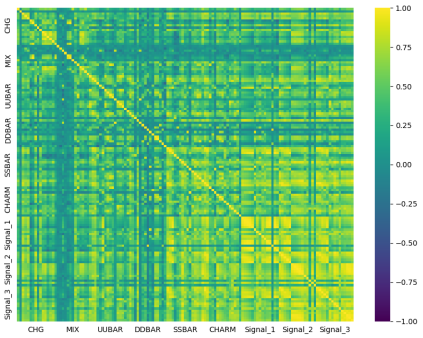
$$\Sigma = \sum_{i=1}^{p} \sigma_i^2 \mathbf{v}_i \mathbf{v}_i^T, \tag{9.1}$$

where Σ is the covariance matrix, p is its dimension, σ_i^2 is the ith eigenvalue, and $\mathbf{v}_i$ is the corresponding eigenvector. To reduce the number of nuisance parameters, the following approximation is used:

$$\Sigma \approx \sum_{i=1}^{t} \sigma_i^2 \mathbf{v}_i \mathbf{v}_i^T + \mathrm{diag}\left(\sum_{i=t+1}^{p} \sigma_i^2 \mathbf{v}_i \mathbf{v}_i^T \right), \tag{9.2}$$



(a) Correlation matrix among the fit bins and event categories induced by the `kaonID` uncertainties.



(b) Correlation matrix reconstructed using the leading 19 eigenvectors.

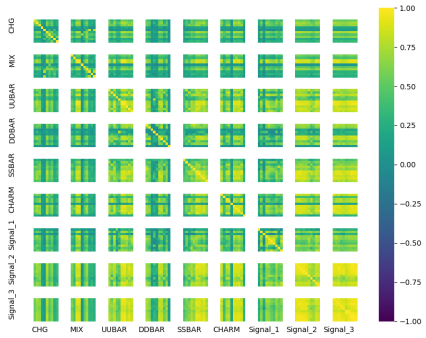
Figure 9.1: Correlation matrices associated with the `kaonID` systematic uncertainty. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading 19 eigenvectors.

where t is the number of retained components. The final diagonal term will be treated as a bin-by-bin uncorrelated uncertainty. The fraction of the total variance described by the retained components is quantified by the cumulative explained variance ratio:

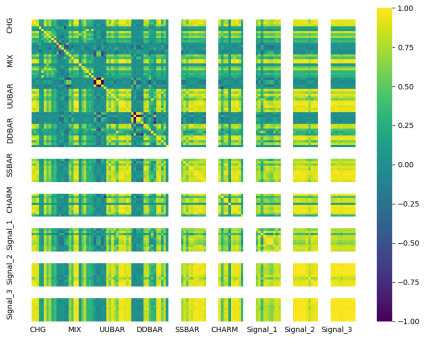
$$\frac{\sum_{i=1}^{t} \sigma_i^2}{\sum_{i=1}^{p} \sigma_i^2}. \tag{9.3}$$

A total of 500 toy samples are generated to estimate the covariance matrix associated with the `kaonID` uncertainties. For each toy sample, the correction factors for the kaon identification efficiency and the pion-to-kaon misidentification rate are fluctuated according to Gaussian distributions. Figure 9.1 shows the correlation matrices obtained using all eigenvectors and using only the leading 19 eigenvectors. The cumulative explained variance ratio for these 19 eigenvectors is 0.96. The leading 19 eigenvectors are retained and implemented as independent nuisance parameters in the fit.

The same procedure is applied to the `pionID` uncertainties. A total of 500 toy samples are generated by fluctuating the correction factors for the pion identification efficiency and the kaon-to-pion misidentification rate. Figure 9.2 shows the correlation matrices obtained using all eigenvectors and using only the leading 19 eigenvectors. The cumulative explained variance ratio for these 19 eigenvectors is 0.96. The leading 19 eigenvectors are retained and implemented as independent nuisance parameters in the fit.



(a) Correlation matrix among the fit bins and event categories induced by the `pionID` uncertainties. White regions correspond to bins with no pion candidates and therefore no yield variation induced by the `pionID` uncertainties.



(b) Correlation matrix reconstructed using the leading 19 eigenvectors.

Figure 9.2: Correlation matrices associated with the `pionID` systematic uncertainty. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading 19 eigenvectors.

9.3 K_S^0 efficiency

The official uncertainty associated with the K_S^0 efficiency correction factor, discussed in Section 8.4, is used. The same procedure described in Section 9.2 is used to reduce the number of nuisance parameters.

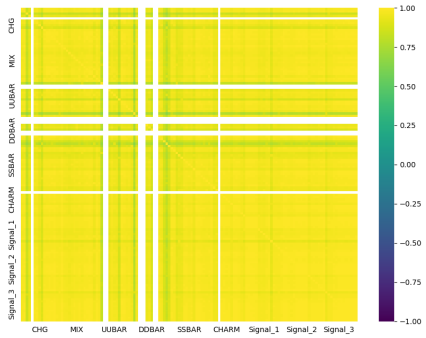
A total of 500 toy samples are generated to estimate the covariance matrix associated with the K_S^0 efficiency uncertainty. For each toy sample, the K_S^0 efficiency correction factors are varied according to Gaussian distributions defined by their uncertainties.

Figure 9.3 shows the correlation matrices obtained using all eigenvectors and using only the leading four eigenvectors. The cumulative explained variance ratio for these 4 eigenvectors is 0.97. The leading 4 eigenvectors are retained and implemented as independent nuisance parameters in the fit.

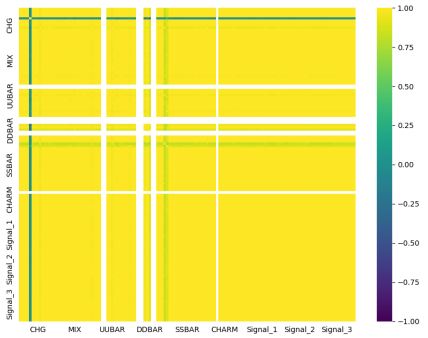
9.4 π^0 efficiency

The official uncertainty associated with the π^0 efficiency correction factor, discussed in Section 8.3, is used. The same procedure described in Section 9.2 is used to reduce the number of nuisance parameters.

A total of 500 toy samples are generated to estimate the covariance matrix associated with the π^0 efficiency uncertainty. For each toy sample, the π^0 efficiency correction factors are varied according to Gaussian distributions defined by their uncertainties.



(a) Correlation matrix among the fit bins and event categories induced by the K_S^0 efficiency uncertainties. White regions correspond to bins with no K_S^0 candidates and therefore no yield variation induced by the K_S^0 efficiency uncertainties.



(b) Correlation matrix reconstructed using the leading four eigenvectors.

Figure 9.3: Correlation matrices associated with the K_S^0 efficiency systematic uncertainty. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading four eigenvectors.

Figure 9.4 shows the correlation matrices obtained using all eigenvectors and using only the leading four eigenvectors. The cumulative explained variance ratio for these 7 eigenvectors is 0.96. The leading 7 eigenvectors are retained and implemented as independent nuisance parameters in the fit.

9.5 FastBDT efficiency

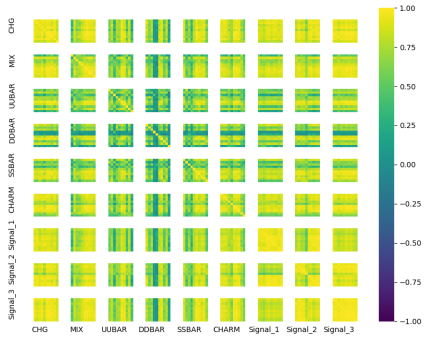
The $B \to X_s J/\psi$ control-sample study is described in Section 7. The uncertainties in the data-to-MC ratios of the FastBDT selection efficiency given in Eq. 7.2 are assigned as systematic uncertainties.

9.6 Form Factors for $B \to K^{(*)}\nu\bar{\nu}$

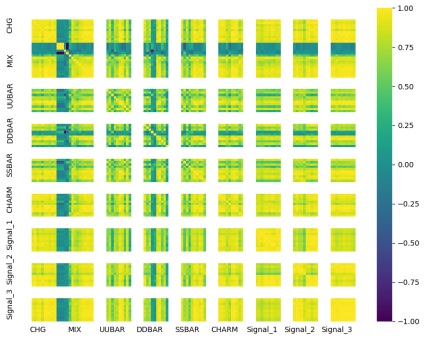
The decay amplitudes for $B \to K\nu\bar{\nu}$ and $B \to K^*\nu\bar{\nu}$ depend on the form factors given in Eqs. 1.5 and 1.9. These form factors have uncertainties arising from the parameters of the z expansion given in Eqs. 3.2 and 3.4.

The form-factor parameters are varied, and the resulting changes in the predicted event yields in the fit bins are evaluated. These yield variations are treated as systematic uncertainties.

Because the form-factor parameters are correlated [5, 8], PCA is performed to ob-



(a) Correlation matrix among the fit bins and event categories induced by the π^0 efficiency uncertainties. White regions correspond to bins with no π^0 candidates and therefore no yield variation induced by the π^0 efficiency uncertainties.



(b) Correlation matrix reconstructed using the leading 7 eigenvectors.

Figure 9.4: Correlation matrices associated with the π^0 efficiency systematic uncertainty.The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading 7 eigenvectors.

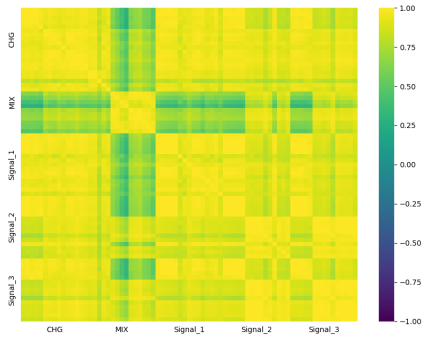
tain independent variation directions. The parameters are varied along the resulting eigenvectors.

9.7 FEI calibration for $B\bar{B}$

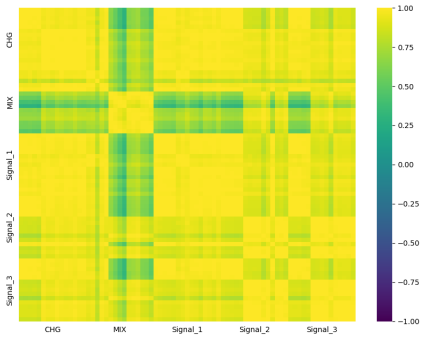
As discussed in Section 8.6, the official FEI calibration factors are applied. These calibration factors have associated uncertainties and correlations among the B_{tag} decay channels. The covariance matrix of the FEI calibration factors is diagonalized using PCA, and the resulting eigenvectors define independent variation directions.

A total of 500 toy samples are generated by varying the FEI calibration factors along these eigenvector directions according to their uncertainties. For each toy sample, the predicted event yields in the fit bins are recalculated. A covariance matrix of the predicted yields across the fit bins is then constructed from the toy samples.

The same procedure described in Section 9.2 is applied to this yield covariance matrix to reduce the number of nuisance parameters. Figure 9.5 shows the correlation matrices reconstructed using all eigenvectors and using only the leading three eigenvectors. The cumulative explained variance ratio for these three eigenvectors is 0.98. The leading three eigenvectors are retained and implemented as independent nuisance parameters in the fit.



(a) Correlation matrix among the fit bins and event categories induced by the FEI calibration uncertainties.



(b) Correlation matrix reconstructed using the leading three eigenvectors.

Figure 9.5: Correlation matrices associated with the FEI calibration systematic uncertainty. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading three eigenvectors.

9.8 Additional uncertainty in the FEI calibration

As discussed in Section 8.6, the official FEI efficiency calibration factors are applied. These factors are derived using $B \to X\ell\nu$ decays. As an additional study, a possible dependence of the FEI calibration on the signal-side decay topology and cut flow is investigated using $B \to K^{(*)}J/\psi$ decays.

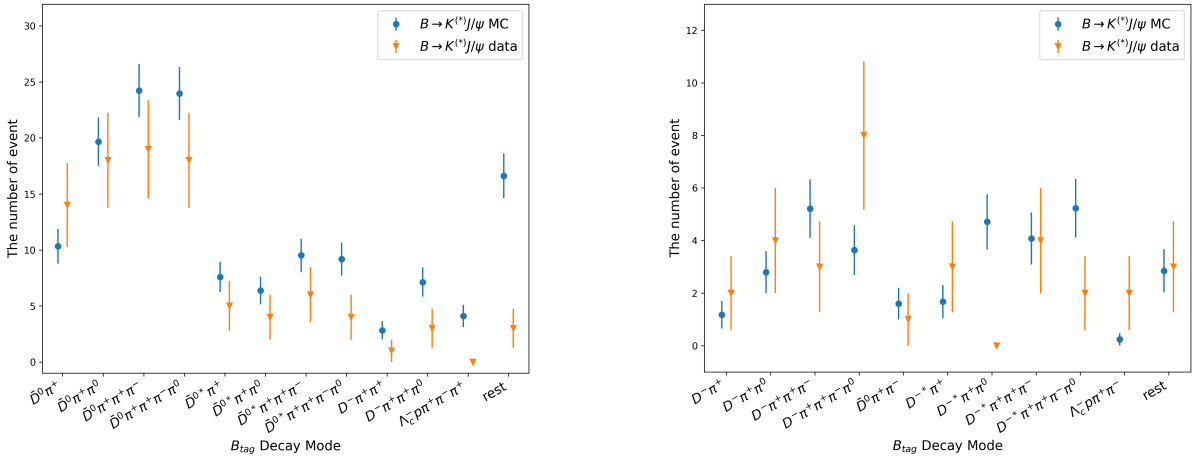
The control-sample analysis described in Section 7 is used. However, the X_s candidate is required to be reconstructed in either a K or $K\pi$ final state with $M_{X_s} < 1.0$ GeV/c^2. Figure 9.6 shows the event yields in data and MC for the different B_{tag} decay channels. Because of the limited statistical precision, the B_{tag} decay channels are grouped. For charged B_{tag} candidates, the decay channels are divided into three categories:

- Category 1: $\bar{D}^0\pi^+$, $\bar{D}^0\pi^+\pi^0$, $\bar{D}^0\pi^+\pi^+\pi^-$, and $\bar{D}^0\pi^+\pi^+\pi^-\pi^0$
- Category 2: $\bar{D}^{0*}\pi^+$, $\bar{D}^{0*}\pi^+\pi^0$, $\bar{D}^{0*}\pi^+\pi^+\pi^-$, and $\bar{D}^{0*}\pi^+\pi^+\pi^-\pi^0$
- Category 3: $D^-\pi^+\pi^+$, $D^-\pi^+\pi^+\pi^0$, $\Lambda_c^- p\pi^+\pi^-\pi^+$, and the remaining decay channels

The resulting FEI calibration factors are

$$\text{FEI calibration factor} = \begin{cases} 0.88 \pm 0.12 & \text{(Category 1)}, \\ 0.58 \pm 0.14 & \text{(Category 2)}, \\ 0.23 \pm 0.09 & \text{(Category 3)}. \end{cases} \tag{9.4}$$

For neutral B_{tag} candidates, all decay channels are combined because of the limited sample



(a) Charged B_{tag}

(b) Neutral B_{tag}

Figure 9.6: Comparison of the event yields in data and MC for $B \to K^{(*)}J/\psi$ decays. The horizontal axis indicates the B_{tag} decay channel.

size. The resulting FEI calibration factor is

$$\text{FEI calibration factor} = 0.96 \pm 0.19. \tag{9.5}$$

Figure 9.7 compares two signal templates. The blue histogram is obtained using the official FEI calibration factors for all B_{tag} candidates. For the red histogram, the alternative calibration factors given in Eqs. 9.4 and 9.5 are applied only to correctly reconstructed B_{tag} candidates satisfying $\texttt{isSignal}(B_{\text{tag}}) = 1$. The official FEI calibration factors are retained for the remaining candidates.

In this analysis, the systematic uncertainty is evaluated by varying the FEI calibration factors applied to signal MC events satisfying $\texttt{isSignal}(B_{\text{tag}}) = 1$. For each toy sample, the alternative FEI calibration factor for each B_{tag} category is first varied according to a Gaussian distribution defined by its central value and statistical uncertainty in Eqs. 9.4 and 9.5. The FEI calibration factor used in the toy sample is then varied around the official calibration factor. The width of this variation is given by the difference between the official factor and the fluctuated alternative factor. A common standard-Gaussian random variable is used for all B_{tag} categories within each toy sample, while the alternative calibration factors are varied independently according to their statistical uncertainties.

A total of 500 toy samples are generated using this procedure. For each toy sample, the predicted yields in the fit bins and event categories are recalculated, and the covariance matrix of the predicted yields is constructed. PCA is applied to this covariance matrix to reduce the number of nuisance parameters while preserving the correlations among the predicted yields. The leading three eigenvectors and their corresponding eigenvalues are retained as independent nuisance parameters in the fit. The cumulative explained variance ratio for these three eigenvectors is 1.00. Figure 9.8 shows the correlation matrices associated with this secondary FEI calibration uncertainty.

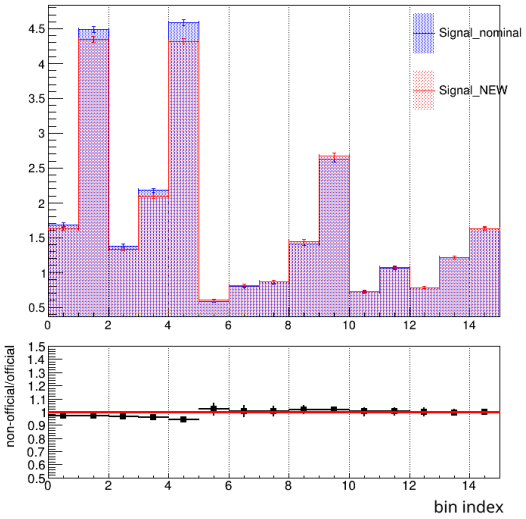
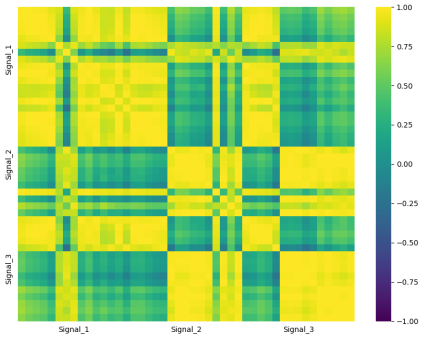
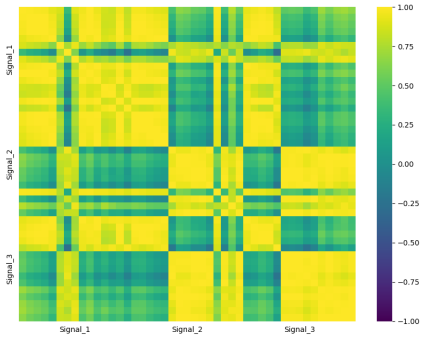


Figure 9.7: Comparison of the nominal and alternative signal templates. The blue histogram is obtained using the official FEI calibration factors. For the red histogram, the alternative calibration factors are applied to correctly reconstructed B_{tag} candidates with $\texttt{isSignal}(B_{\text{tag}}) = 1$, while the official factors are retained for all other candidates.



(a) Correlation matrix among the fit bins and event categories induced by the secondary FEI calibration uncertainty.



(b) Correlation matrix reconstructed using the leading three eigenvectors.

Figure 9.8: Correlation matrices associated with the secondary uncertainty in the FEI efficiency correction. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading three eigenvectors.

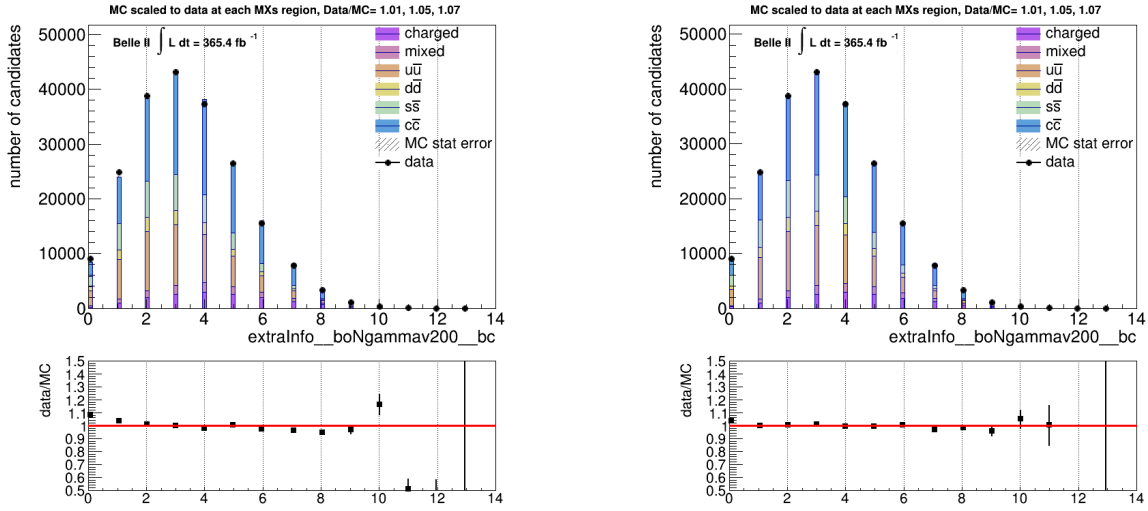


Figure 9.9: Distribution of the number of remaining photon candidates in the M_{bc} sideband. The left plot shows the distribution obtained with the nominal photon multiplicity correction, while the right plot shows the distribution obtained with the $+1\sigma$ systematic variation. The data–MC agreement is improved by the variation.

9.9 $q\bar{q}$ efficiency

The $q\bar{q}$ background efficiency is corrected using correction factors derived from the off-resonance sample described in Section 6.1. The uncertainties in the data-to-MC ratios given in Eq. 6.2 are assigned as systematic uncertainties.

9.10 Photon multiplicity

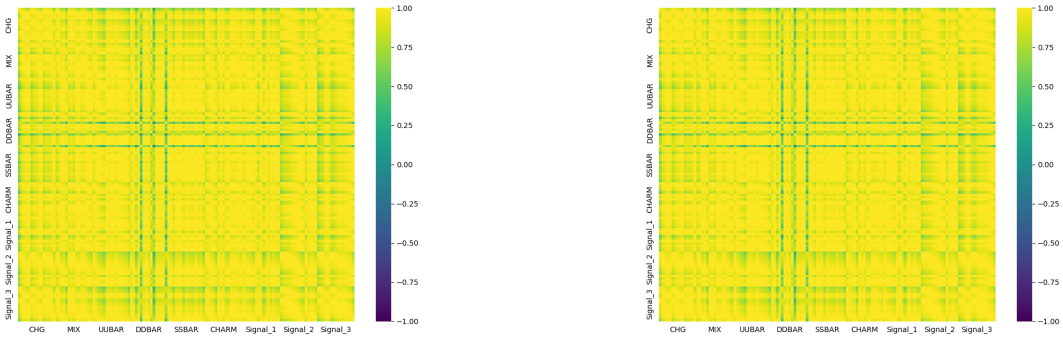
As discussed in Section 8.2, the neutral-tag wrong-charge sideband is used to estimate the uncertainty associated with the photon multiplicity correction.

Figure 8.3 shows the data-to-MC ratios as a function of the number of remaining photon candidates. These data-to-MC ratios are used to define systematic variations of the photon multiplicity correction factors.

As a validation of this systematic uncertainty, its effect is examined in the M_{bc} sideband. Figure 9.9 shows the distribution of the number of remaining photon candidates. Applying the $+1\sigma$ systematic variation improves the data–MC agreement.

The same procedure described in Section 9.2 is used to reduce the number of nuisance parameters.

A total of 500 toy samples are generated to estimate the covariance matrix associated with the uncertainties in the photon multiplicity correction factors. For each toy sample, the photon multiplicity correction factors are varied according to Gaussian distributions



(a) Correlation matrix among the fit bins and event categories induced by the uncertainties in the photon multiplicity correction factors.

(b) Correlation matrix reconstructed using the leading two eigenvectors.

Figure 9.10: Correlation matrices associated with the systematic uncertainty in the photon multiplicity correction factors. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading two eigenvectors.

defined by their uncertainties.

Figure 9.10 shows the correlation matrices obtained using all eigenvectors and using only the leading two eigenvectors. The cumulative explained variance ratio for these two eigenvectors is 1.00. The leading two eigenvectors are retained and implemented as independent nuisance parameters in the fit.

9.11 Fraction of decay modes

Because the branching fractions of $B \to K\nu\bar{\nu}$, $B \to K^*\nu\bar{\nu}$, and the remaining $B \to X_s\nu\bar{\nu}$ have uncertainties, the relative fractions of these decay modes are also uncertain. The branching fractions are [6, 51]

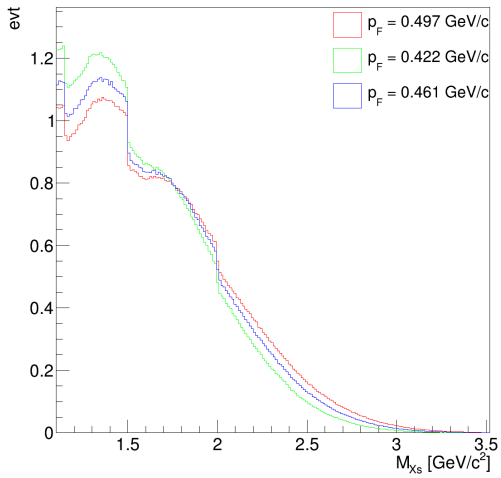
$$BR(\mathrm{B}^+ \to K^+\nu\bar{\nu})_{SM} = (5.044 \pm 0.402) \times 10^{-6}, \tag{9.6}$$

$$BR(\mathrm{B}^0 \to K^{*0}\nu\bar{\nu})_{SM} = (9.19 \pm 0.86 \pm 0.50) \times 10^{-6}. \tag{9.7}$$

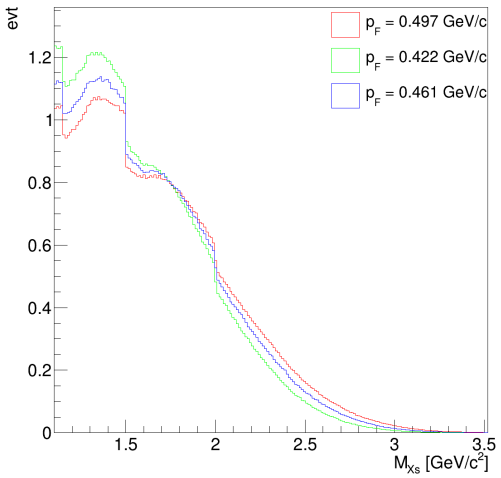
The first uncertainty in $BR(B^0 \to K^{*0}\nu\bar{\nu})_{\mathrm{SM}}$ arises from the form factors. Because the form-factor uncertainty is already included in Section 9.6, only the second uncertainty is considered here.

In addition, the uncertainties in the $B \to K$ form factors result in a relative uncertainty of $^{+0.056}_{-0.055}$ in the $B \to K\nu\bar{\nu}$ branching fraction. This relative uncertainty is calculated using the correlation matrix given in Ref. [5]. After excluding the form-factor contribution, the remaining relative uncertainty is ± 0.057.

Using these uncertainties, the uncertainties in the relative fractions of the $B \to K\nu\bar{\nu}$, $B \to K^*\nu\bar{\nu}$, and the remaining $B \to X_s\nu\bar{\nu}$ decay modes are evaluated.



(a) M_{X_s} distributions for the charged signal sample obtained with different p_F values. The discontinuities arise from the fragmentation correction described in Section 8.8.



(b) M_{X_s} distributions for the neutral signal sample obtained with different p_F values. The discontinuities arise from the fragmentation correction described in Section 8.8.

Figure 9.11

9.12 Fermi motion momentum

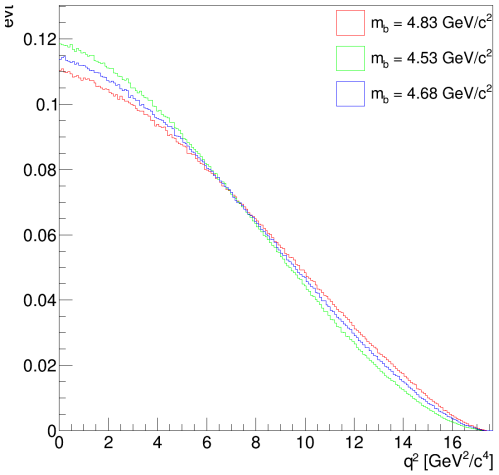
The Fermi motion momentum p_F in Eq. 3.8 is used to produce the signal MC samples. To estimate the uncertainty associated with the Fermi motion momentum, signal samples with different p_F values are produced without detector simulation.

Figure 9.11 shows the M_{X_s} distributions obtained with different p_F values. The discontinuities arise from the fragmentation correction described in Section 8.8. The weights are obtained from the ratios of the M_{X_s} distributions generated with varied and nominal p_F values. These weights are applied to the nominal signal MC sample, and the resulting changes in the signal templates are assigned as the systematic uncertainty associated with the Fermi motion momentum.

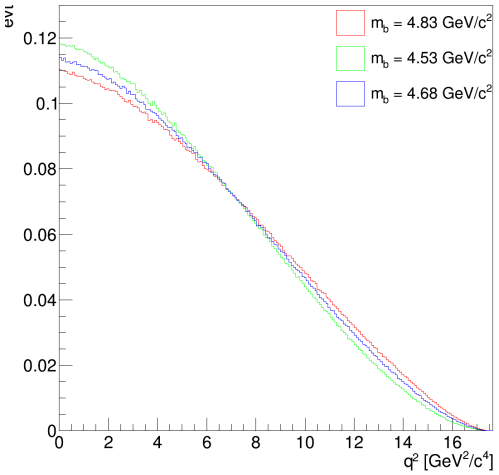
9.13 $K^* - X_s$ transition

As discussed in Section 3.2.2, the remaining $B \to X_s\nu\bar{\nu}$ sample, excluding $B \to K\nu\bar{\nu}$ and $B \to K^*\nu\bar{\nu}$, is generated with the requirement $M_{X_s} > 1.1$ GeV/c^2.

To estimate the systematic uncertainty associated with this mass threshold, the remaining $B \to X_s\nu\bar{\nu}$ samples are generated with varied threshold values of $M_{X_s} = 1.1 \pm 0.1$ GeV/c^2. The variation of 0.1 GeV/c^2 is motivated by the M_{X_s} distribution in the $B \to X_s\gamma$ study [47]. The resulting changes in the event yields in the fit bins are assigned as the systematic uncertainty.



(a) q^2 distributions for the charged signal sample obtained with different m_b values.



(b) q^2 distributions for the neutral signal sample obtained with different m_b values.

Figure 9.12: q^2 distributions for the charged and neutral signal samples generated with different b-quark mass values.

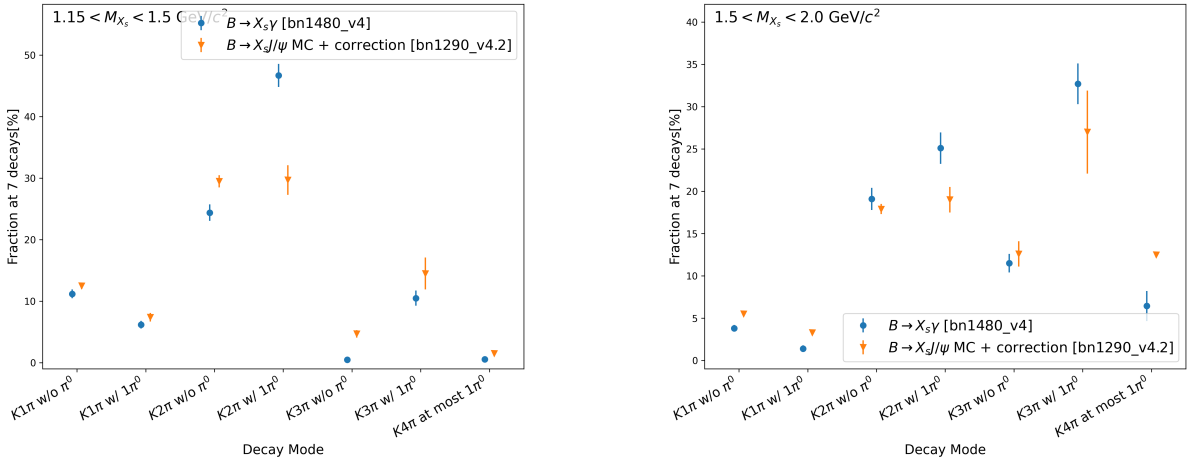
Non-$K^{(*)}$ components may contribute below 1.0 GeV/c^2. To quantify this effect, the impact of the S-wave component in $K\pi\nu\bar{\nu}$ decays is examined. As discussed in Section J, the S-wave component does not change the final result at the first decimal place. Therefore, no additional systematic uncertainty associated with non-$K^{(*)}$ components below 1.0 GeV/c^2 is included.

9.14 b-quark mass

The mass of the b quark is used as a parameter in the decay amplitude given in Eq. 1.14. The value of this parameter varies among theoretical calculations [3, 4]. Conservatively, the following value is used in this analysis:

$$m_b = 4.68 \pm 0.15 \text{ GeV}/c^2. \tag{9.8}$$

To estimate the systematic uncertainty associated with the b-quark mass, signal samples with varied b-quark mass values are generated without detector simulation. Figure 9.12 shows the q^2 distributions obtained with different m_b values. Weights are derived from the ratios of the q^2 distributions generated with varied and nominal m_b values. These weights are applied to the nominal signal MC sample, and the resulting changes in the signal templates are assigned as the systematic uncertainty associated with the b-quark mass.



(a) $1.15 < M_{X_s} < 1.5$ GeV/c^2 (b) $1.5 < M_{X_s} < 2.0$ GeV/c^2

Figure 9.13: Comparison of the fragmentations in $B \to X_s\gamma$ and $B \to X_s J/\psi$ decays.

9.15 MC statistics

Because of the finite sizes of the MC samples, the histogram templates have statistical uncertainties. These uncertainties are included as systematic uncertainties.

9.16 Fragmentation of X_s

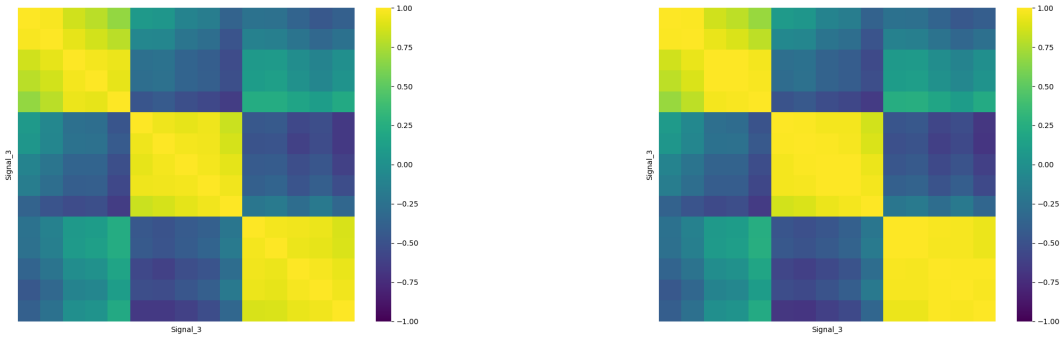
As discussed in Section 8.8, the correction for the fragmentation of X_s is derived using $B \to X_s\gamma$ decays. The systematic uncertainty associated with the fragmentation correction is estimated using two methods. The first method uses the difference between the fragmentations observed in $B \to X_s\gamma$ and $B \to X_s J/\psi$ decays. The second method is based on fluctuations of PYTHIA parameters. Both methods are used because the previous $B \to X_s J/\psi$ study does not cover all X_s decay modes and M_{X_s} regions.

Figure 9.13 compares the fragmentations in $B \to X_s\gamma$ [61, 62] and $B \to X_s J/\psi$ decays [68, 69]. Here, the fragmentation in $B \to X_s J/\psi$ decays is obtained from the MC sample after applying the overall correction factor. This correction factor is an overall factor and does not depend on M_{X_s}. In general, the fragmentations in $B \to X_s\gamma$ and $B \to X_s J/\psi$ decays are in good agreement. However, some differences are observed for several modes, such as the $K2\pi$ mode with one π^0. In the regions $1.15 < M_{X_s} < 1.5$ GeV/c^2 and $1.5 < M_{X_s} < 2.0$ GeV/c^2, the differences for the seven decay modes shown in Figure 9.13 are assigned as systematic uncertainties.

For the remaining decay categories and M_{X_s} regions, the systematic uncertainty in the fragmentation is estimated by fluctuating PYTHIA parameters. This method has been used in other $B \to X_s\gamma$ studies [61, 47]. Table 9.1 summarizes the varied PYTHIA parameters and their tested values. The tested values are chosen based on the fragmentation in the

Table 9.1: Summary of the fluctuated PYTHIA parameters. Among the tested values, combinations are selected such that the fragmentations of the seven decay categories shown in Figure 9.13 are within 2σ of the $B \to X_s\gamma$ result. Here, σ is defined as the difference between the $B \to X_s\gamma$ and $B \to X_s J/\psi$ studies.

Parameter name	Description	Tested values
`StringFlav:probStoUD`	The suppression of s quark pair production in the field compared with u or d pair production	$[0.0, 0.3, 0.6]$
`StringFlav:etaSup`	Extra suppression factor for η production in fragmentation	$[0.4, 0.7, 1.0]$
`StringFlav:mesonUDvector`	The relative production ratio vector/pseudoscalar for light mesons	$[2.0, 2.5, 3.0]$
`StringFlav:mesonSvector`	The relative production ratio vector/pseudoscalar for strange mesons	$[2.0, 2.5, 3.0]$
`StringFlav:mesonSL1S1J0`	The relative scalar production ratio (L=1,S=1,J=0)/pseudoscalar for strange mesons	$[2.0, 2.5, 3.0]$
`StringFlav:mesonSL1S1J2`	The relative tensor production ratio (L=1,S=1,J=2)/pseudoscalar for strange mesons	$[2.3, 2.8, 3.0]$



(a) Correlation matrix among the fit bins and event categories induced by the fragmentation uncertainties.

(b) Correlation matrix reconstructed using the leading four eigenvectors.

Figure 9.14: Correlation matrices associated with the systematic uncertainty in the fragmentation. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading four eigenvectors.

$B \to X_s\gamma$ study. Among the parameter combinations listed in Table 9.1, combinations are selected such that the fragmentations of the seven decay categories shown in Figure 9.13 are within 2σ of the $B \to X_s\gamma$ result. Here, σ is defined as the difference between the $B \to X_s\gamma$ and $B \to X_sJ/\psi$ studies. The systematic uncertainty is assigned as the maximum deviation from the nominal value. Table 9.2 shows the nominal fragmentation fractions used in this analysis and the maximally deviated fractions obtained from the `PYTHIA` parameter fluctuations. An entry is left empty when the systematic uncertainty is estimated from the difference between the $B \to X_s\gamma$ and $B \to X_sJ/\psi$ studies.

The same procedure described in Section 9.2 is used to reduce the number of nuisance parameters.

A total of 500 toy samples are generated to estimate the covariance matrix associated with the fragmentation uncertainties. For each toy sample, the fragmentation fractions are varied according to Gaussian distributions defined by their uncertainties.

Figure 9.14 shows the correlation matrices obtained using all eigenvectors and using only the leading four eigenvectors. The cumulative explained variance ratio for these four eigenvectors is 0.984. The leading four eigenvectors are retained and implemented as independent nuisance parameters in the fit.

Resonant contributions are not well modeled in `PYTHIA`. To check the impact of this mismodelling, the reconstruction efficiency is studied using the $K_1(1270)$ resonance. As described in Appendix I, the efficiencies are calculated as functions of M_{X_s} for the samples generated with `PYTHIA` and with the $K_1(1270)$ resonance. The efficiency differences between these two samples are smaller than the differences among decay modes with different multiplicities. Consequently, among the systematic uncertainties associated with

Table 9.2: Summary of the results obtained by fluctuating the PYTHIA parameters. The fractions of the nine decay categories are shown for the nominal MC sample and for the maximally deviated PYTHIA parameter combination. The fractions of missing decay modes are also shown. An entry is left empty when the systematic uncertainty is estimated from the difference between the $B \to X_s\gamma$ and $B \to X_s J/\psi$ studies.

	nominal and maximally deviated fraction in each M_{X_s} region (GeV)									
	[0.0,1.15]		[1.15,1.5]		[1.5,2.0]		[2.0,2.4]		[2.4,∞)	
decay category	nom	max	nom	max	nom	max	nom	max	nom	max
$K1\pi$ w/o π^0	39%	17%	11%	-	3.8%	-	1.7%	2.6%	9.9%	1.1%
$K1\pi$ w/ $1\pi^0$	19%	8.4%	6.2%	-	1.4%	-	0.5%	1.3%	5.0%	0.5%
$K2\pi$ w/o π^0	17%	33%	24%	-	19%	-	9.0%	10%	11%	5.6%
$K2\pi$ w $1\pi^0$	17%	34%	47%	-	25%	-	8.0%	13%	13%	7.6%
$K3\pi$ w/o π^0	0.4%	0.1%	0.5%	-	12%	-	17%	9.3%	6.6%	9.8%
$K3\pi$ w $1\pi^0$	3.0%	0.2%	11%	-	33%	-	27%	24%	16%	22%
$K4\pi$ at most $1\pi^0$	0.0%	0.0%	0.5%	-	6.4%	-	10%	21%	16%	29%
including $2\pi^0$	4.4%	8.4%	4.7%	7.8%	16%	14%	22%	20%	17%	26%
including $3K$	0.0%	0.0%	0.0%	0.0%	2.2%	0.0%	5.4%	0.0%	5.2%	0.0%
missing modes	7.6%	1.6%	8.4%	2.4%	16%	5.6%	35%	23%	57%	47%

PYTHIA mismodelling, only the uncertainty from the fragmentation is considered.

9.17 Background normalization

As discussed in Section 6, data–MC discrepancies are observed in the event yields in the $M_{\rm bc}$ sideband and the FastBDT sideband. The data-to-MC ratios in the $M_{\rm bc}$ sideband region are

$$\frac{\text{Data}}{\text{MC}} = \begin{cases} 1.12 \pm 0.03 & (0.0 < M_{X_s}^{\rm reco} < 0.6 \text{ GeV}/c^2), \\ 1.04 \pm 0.05 & (0.6 \leq M_{X_s}^{\rm reco} < 1.0 \text{ GeV}/c^2), \\ 1.10 \pm 0.03 & (1.0 \leq M_{X_s}^{\rm reco} < 2.0 \text{ GeV}/c^2), \end{cases}$$

where only the statistical uncertainties of data and MC are included. The data-to-MC ratios in the FastBDT sideband region are

$$\frac{\text{Data}}{\text{MC}} = \begin{cases} 1.11 \pm 0.06 & (0.0 < M_{X_s}^{\rm reco} < 0.6 \text{ GeV}/c^2), \\ 1.02 \pm 0.07 & (0.6 \leq M_{X_s}^{\rm reco} < 1.0 \text{ GeV}/c^2), \\ 1.10 \pm 0.04 & (1.0 \leq M_{X_s}^{\rm reco} < 2.0 \text{ GeV}/c^2), \end{cases}$$

where only the statistical uncertainties of data and MC are included.

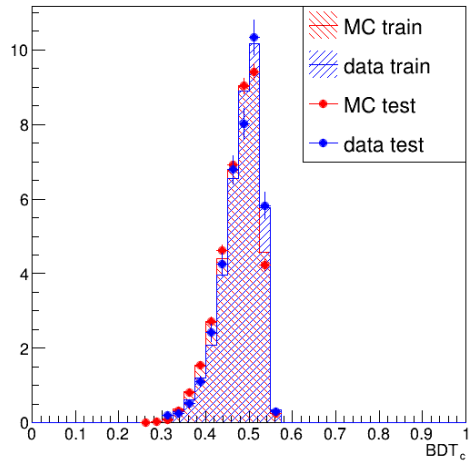
Conservatively, relative uncertainties of $\pm 20\%$ are assigned to the overall background normalization in each $M_{X_s}^{\rm reco}$ region. These uncertainties are applied separately to each background component: charged $B\bar{B}$, mixed $B\bar{B}$, $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ templates.

9.18 $q\bar{q}$ background shape

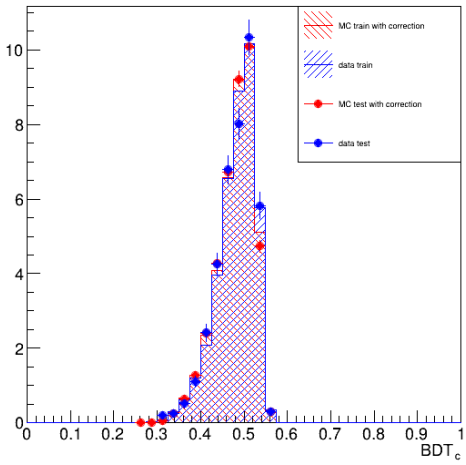
The FastBDT output is used as the fit variable. There may be discrepancies between data and MC in the FastBDT output. In this analysis, these discrepancies are corrected by training an additional FastBDT, denoted as BDTc [70].

BDTc is trained to distinguish off-resonance data from off-resonance MC. If the MC perfectly models the data, the BDTc output distributions for data and MC should be the same. However, if there is a data–MC discrepancy, the BDTc output distributions are different.

Figure 9.15 shows the BDTc output distribution for the off-resonance sample. For each event, a weight of $\sqrt{\frac{p}{1-p}}$ is applied, where p is the BDTc output. If an event is data-like, the BDTc output is close to 1, and therefore $\sqrt{\frac{p}{1-p}}$ becomes large. On the other hand, if an event is MC-like, the BDTc output is close to 0, and the weight becomes small. Consequently, this weight makes the MC distribution more data-like.



(a) BDTc output distribution for the off-resonance sample before the correction



(b) BDTc output distribution for the off-resonance sample after the correction

Figure 9.15: BDTc output distributions for the off-resonance sample before and after the correction. A weight of $\sqrt{\frac{p}{1-p}}$ is applied for the correction, where p is the BDTc output. After applying the correction, a normalization factor is applied to preserve the total number of events.

After applying the correction, a normalization factor is applied to preserve the total number of events. Figure 9.15 shows the BDTc output distribution after the correction. The discrepancy between data and MC is reduced after the correction.

Figure 9.16 shows the distributions of selected variables for the off-resonance sample. The distributions are slightly modified by the BDTc correction. The FastBDT distribution after the BDTc correction is used to evaluate the systematic uncertainty. One nuisance parameter is added for this uncertainty.

9.19 $B\bar{B}$ counting

The number of $\Upsilon(4S)$ events is $N_{\Upsilon(4S)} = (387.1 \pm 5.6) \times 10^6$. Therefore, a relative uncertainty of 1.5% is assigned due to the uncertainty in the number of $B\bar{B}$ pairs.

9.20 Branching ratios of B meson decays

The branching ratios of B meson decays are determined by the decay files used in BASF2. These values have uncertainties.

Figure 9.17 shows the decay modes of B mesons in the charged and mixed $B\bar{B}$ background samples after all selection requirements are applied.

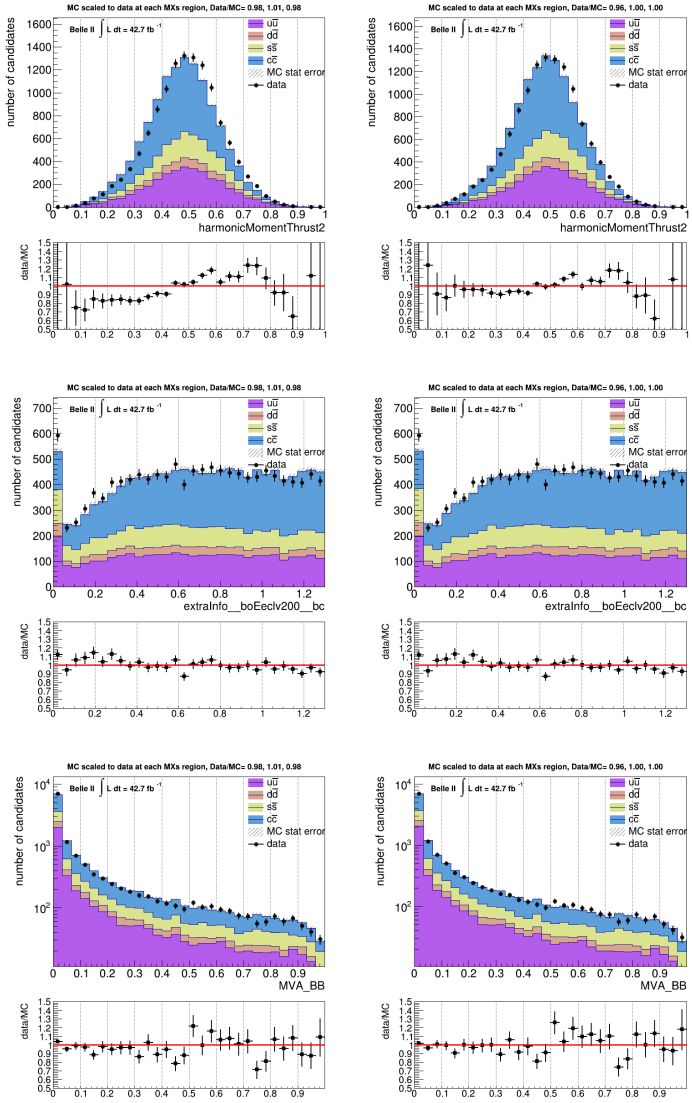
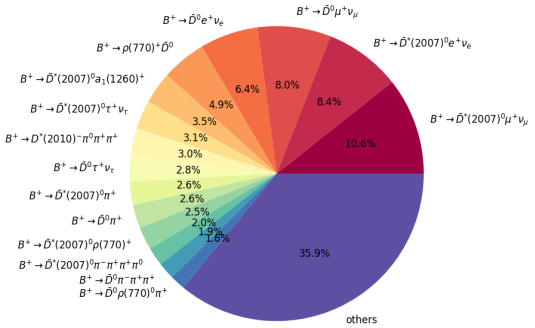
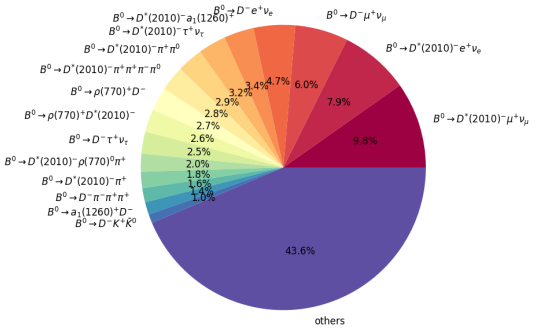


Figure 9.16: Distributions of (top) `harmonicMomentThrust2`, (middle) E_{ECL}, and (bottom) the FastBDT output. The definition of `harmonicMomentThrust2` is given in Appendix A. The plots are obtained from the off-resonance sample. The left plots show the distributions before the BDTc correction, while the right plots show the distributions after the BDTc correction.



(a) Decay modes of the charged $B\bar{B}$ background



(b) Decay modes of the mixed $B\bar{B}$ background

Figure 9.17: Remaining components of the (a) charged and (b) mixed $B\bar{B}$ backgrounds after all selection requirements are applied. The numbers in the pie charts are given in arbitrary units.

The same procedure described in Section 9.2 is used to reduce the number of nuisance parameters.

A total of 500 toy samples are generated to estimate the covariance matrix associated with the uncertainties in the B meson branching ratios. For each toy sample, the branching ratios are varied according to Gaussian distributions defined by their uncertainties from the PDG. Because this procedure is performed manually, only 70.7% and 60.1% of the B meson decays are varied for the charged and mixed backgrounds, respectively. In addition, $B \to K K_L^0 K_L^0$ decays are considered separately to check their impact.

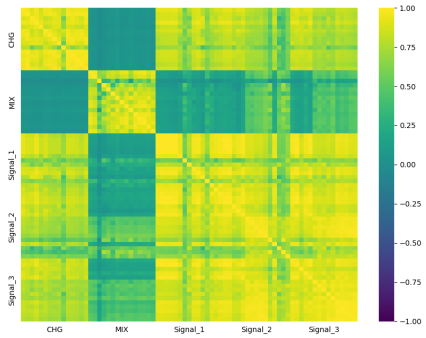
Figure 9.18 shows the correlation matrices obtained using all eigenvectors and using only the leading eight eigenvectors. The cumulative explained variance ratio for these eight eigenvectors is 0.974. The leading eight eigenvectors are retained and implemented as independent nuisance parameters in the fit.

9.21 Mismodelling of $B \to X_s n\bar{n}$ decays

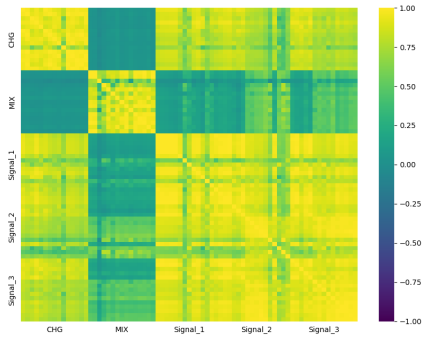
As discussed in Section 8.9, the $M_{n\bar{n}}$ distribution is modeled in this analysis. To cover possible mismodelling, a relative uncertainty of $\pm 100\%$ is assigned to the branching ratio of $B \to X_s n\bar{n}$ decays.

9.22 Mismodelling of $B \to X_s K_L^0 K_L^0$ decays

As discussed in Section 8.10, the Dalitz-plot distribution is modeled in this analysis. To cover possible mismodelling, a relative uncertainty of $\pm 100\%$ is assigned to the branching ratio of $B \to X_s K_L^0 K_L^0$ decays.



(a) Correlation matrix among the fit bins and event categories induced by the uncertainties in the branching ratios.



(b) Correlation matrix reconstructed using the leading eight eigenvectors.

Figure 9.18: Correlation matrices associated with the systematic uncertainties in the B meson branching ratios. The left plot is obtained using all eigenvectors, while the right plot is reconstructed using the leading eight eigenvectors.

9.23 $D \to K_L^0 X$ decay from B meson

As discussed in Section 8.11, a scale factor of 1.3 ± 0.1 is applied to $D \to K_L^0 X$ decays from B mesons. The uncertainty in this scale factor is adopted as a systematic uncertainty.

9.24 K_L^0 ECL efficiency

In the $B^+ \to K^+ \nu\bar{\nu}$ analysis [56], the K_L^0 reconstruction efficiency as the ECL cluster had a significant impact on the result for the inclusive-tag analysis. For the hadronic-tag analysis, however, this effect is not dominant.

In this analysis, ECL clusters are investigated. Only $\mathcal{O}(0.1\%)$ of ECL clusters originate from K_L^0. Therefore, no correction is applied. Instead, a relative uncertainty of $\pm 17\%$ is assigned to events containing ECL clusters originating from K_L^0. This uncertainty is motivated by the hadronic-tag $B^+ \to K^+ \nu\bar{\nu}$ analysis [56].

9.25 Uncorrelated uncertainties

As discussed in Section 9 and Eq. 9.2, PCA is performed for some systematic uncertainties, and the dominant eigenvectors are retained. The remaining component of the covariance matrix is approximated by a diagonal matrix. The diagonal elements are adopted as uncorrelated bin-by-bin uncertainties. To reduce the number of nuisance parameters, these uncorrelated uncertainties are added in quadrature to the MC statistical

uncertainties.

9.26 Summary

Table 9.3 summarizes the systematic uncertainties and the statistical uncertainty from the MC samples. Here, σ_{μ_1}, σ_{μ_2}, and σ_{μ_3} denote the uncertainties in μ_1, μ_2, and μ_3, respectively. To evaluate the impact of each systematic source on σ_μ, a fit to the Asimov dataset with $\mu = 1$ is performed. The asymmetric `MINOS` errors from the fit are taken as the uncertainties in μ. To estimate the contribution from a specific nuisance parameter, the fit to the Asimov dataset is repeated with that nuisance parameter removed. The uncertainty from the specific nuisance parameter is then calculated as

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{without syst}} + \sigma^2_{\text{syst}}}, \tag{9.9}$$

where σ_{total} is the uncertainty obtained with all nuisance parameters included, $\sigma_{\text{without syst}}$ is the uncertainty obtained with the specific nuisance parameter removed, and σ_{syst} is the contribution from that nuisance parameter.

In the table, the asymmetric `MINOS` errors are first symmetrized by taking the average of the absolute values of the positive and negative errors. The contribution from each systematic source is then calculated using the symmetrized uncertainties.

The statistical uncertainty is estimated by fixing all nuisance parameters. The resulting statistical uncertainty is also shown in Table 9.3.

9.27 Fit bias

As discussed in Sections 10 and 11.2, a bias associated with the fitter is observed. This effect is included as a systematic uncertainty. However, to quantify its impact precisely, the uncertainty is assigned only after unblinding the signal region.

Table 9.3: Summary of the systematic uncertainties and statistical uncertainty. The values are obtained using the Asimov dataset.

Source	Impact on σ_{μ_1}	Impact on σ_{μ_2}	Impact on σ_{μ_3}
Background normalization	1.17	2.22	3.96
MC statistics	0.87	1.55	3.47
FastBDT efficiency	0.09	0.34	1.27
$K^* - X_s$ transition	0.01	0.02	1.17
BR of main B meson decays	0.28	0.65	1.05
Photon multiplicity	0.08	0.24	0.95
$q\bar{q}$ efficiency	0.20	0.25	0.93
Additional FEI uncertainty	0.18	0.34	0.55
$q\bar{q}$ shape	0.25	0.26	0.52
Fragmentation of X_s	0.01	0.05	0.44
FEI calibration for $B\bar{B}$	0.19	0.28	0.42
Fermi motion momentum	0.00	0.00	0.39
Mismodelling of $B \to X_s K_L^0 K_L^0$	0.02	0.41	0.35
$B \to K^*$ form factor	0.01	0.44	0.32
$B\bar{B}$ counting	0.05	0.12	0.30
Fraction of decay modes	0.14	0.24	0.25
b-quark mass	0.00	0.00	0.19
$B \to K$ form factor	0.15	0.00	0.14
Pion ID	0.01	0.09	0.14
BR of $D \to K_L^0 X$ from B meson	0.02	0.12	0.11
Tracking efficiency	0.01	0.02	0.09
π^0 efficiency	0.01	0.04	0.08
Mismodelling of $B \to X_s n\bar{n}$	0.15	0.52	0.04
K_L^0 ECL efficiency	0.01	0.02	0.04
K_S^0 efficiency	0.01	0.03	0.04
Kaon ID	0.06	0.02	0.03
Statistical uncertainty	1.65	2.86	5.43

Chapter 10

Fit quality check

In this section, the fitter bias is investigated using toy MC studies and linearity tests.

10.1 Toy MC study

To check the quality of the fitter, toy MC studies are performed. Toy MC samples are generated based on the signal and background histogram templates expected from simulation. A total of 10000 toy MC experiments are performed. In each experiment, the numbers of signal and background events are sampled from Poisson distributions whose means are fixed to the expected numbers of events from simulation. The nuisance parameters associated with the systematic uncertainties are sampled from Gaussian distributions.

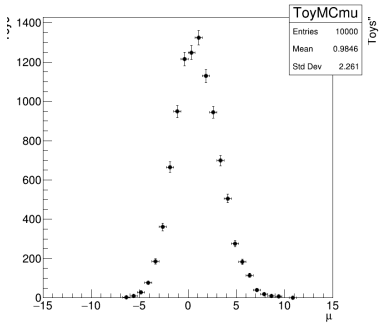
Because the number of bins is small, a bias can arise in the pull distribution when a symmetric error is used. Therefore, the pull is calculated using the asymmetric `MINOS` errors [71]:

$$\text{pull} = \begin{cases} \frac{(\text{true value of } \mu) - (\text{fit result of } \mu)}{(\text{positive MINOS error of } \mu)} & \text{if (fit result)} \leq \text{(true value)} \\ \frac{(\text{fit result of } \mu) - (\text{true value of } \mu)}{(\text{negative MINOS error of } \mu)} & \text{otherwise,} \end{cases} \tag{10.1}$$

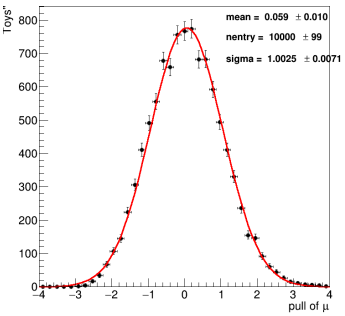
Figures 10.1–10.3 show the results of the toy MC studies for the three signal-strength parameters: μ_1, μ_2, and μ_3.

To quantify the agreement between the pull distributions and the standard normal distribution, the pull distributions are fitted with Gaussian functions. The means of the pull distributions are

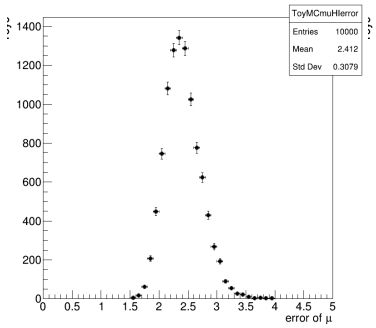
$$\text{mean} = \begin{cases} 0.059 \pm 0.010 & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 0.020 \pm 0.010 & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 0.027 \pm 0.010 & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases} \tag{10.2}$$



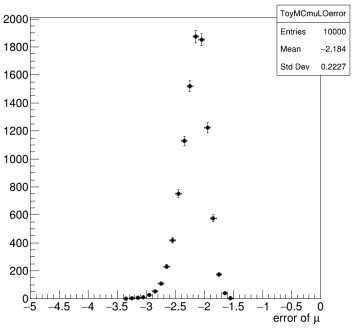
(a) Distribution of μ_1



(b) Pull of μ_1

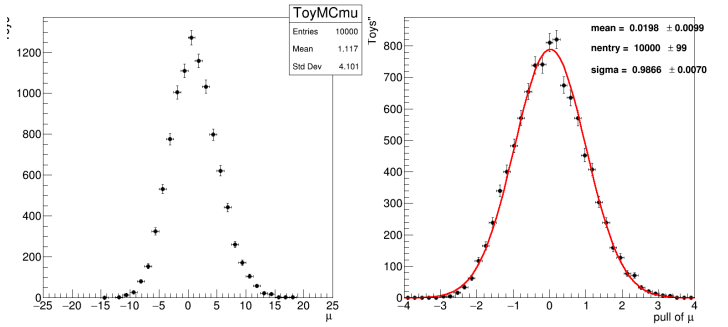


(c) Positive error of μ_1

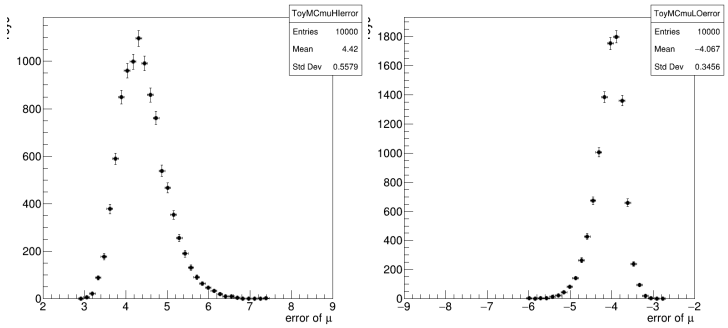


(d) Negative error of μ_1

Figure 10.1: Results of 10000 toy MC experiments for $0.0 < M_{X_s} < 0.6$ GeV/c^2.

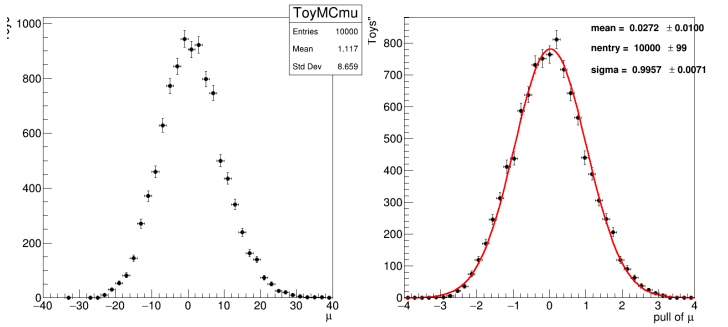


(a) Distribution of μ_2 (b) Pull of μ_2

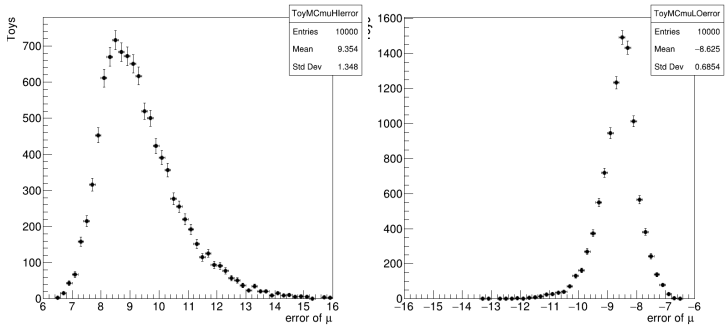


(c) Positive error of μ_2 (d) Negative error of μ_2

Figure 10.2: Results of 10000 toy MC experiments for $0.6 < M_{X_s} < 1.0$ GeV/c^2.

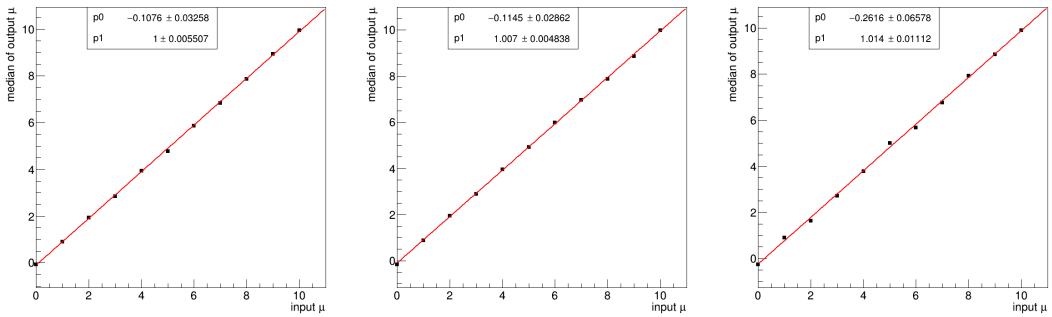


(a) Distribution of μ_3 (b) Pull of μ_3



(c) Positive error of μ_3 (d) Negative error of μ_3

Figure 10.3: Results of 10000 toy MC experiments for 1.0 GeV/$c^2 < M_{X_s}$.



(a) $0.0 < M_{X_s}^{\text{true}} < 0.6$ GeV (b) $0.6 < M_{X_s}^{\text{true}} < 1.0$ GeV (c) $1.0 \text{ GeV} < M_{X_s}^{\text{true}}$

Figure 10.4: Results of the linearity tests for the three signal-strength parameters. Each point shows the median value of the fitted μ distribution.

The widths of the pull distributions are

$$\text{width} = \begin{cases} 1.003 \pm 0.007 & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 0.987 \pm 0.007 & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 0.996 \pm 0.007 & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases} \tag{10.3}$$

In general, the pull distributions show good agreement with the standard normal distribution.

10.2 Linearity test

It is also important to check the linearity of the fitter. To correctly interpret possible effects from new physics, the fitted signal strength should be proportional to the injected signal strength.

Linearity tests are performed by varying the injected value of μ from 0 to 10. In total, 11 different values of μ are tested. For each injected value of μ, 10000 toy MC experiments are performed. Therefore, a total of $11 \times 10000 = 110000$ toy MC experiments are performed. For each injected value of μ, the median value of the μ distribution is obtained. Figure 10.4 shows the median fitted values for each injected value of μ_1, μ_2, and μ_3. To quantify the linearity of the fitter, the results of the linearity tests are fitted with first-order polynomials. The fit results are

$$y = \begin{cases} 1.00 \cdot x - 0.11 & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 1.01 \cdot x - 0.11 & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 1.01 \cdot x - 0.26 & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases} \tag{10.4}$$

The slopes are in good agreement with unity, while the intercepts show some deviations from zero.

10.3 Summary

A small fitter bias is observed. To account for this bias, additional post-unblinding systematic uncertainties are assigned. These post-unblinding systematic uncertainties are discussed in Section 11.2.

Chapter 11

Results

11.1 Results of MC study

Using the Asimov dataset with $\mu = 1$, the expected upper limits are

$$UL(\mu) = \begin{cases} 4.80 & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 8.16 & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 16.75 & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases} \tag{11.1}$$

These limits can be converted into branching-ratio upper limits as

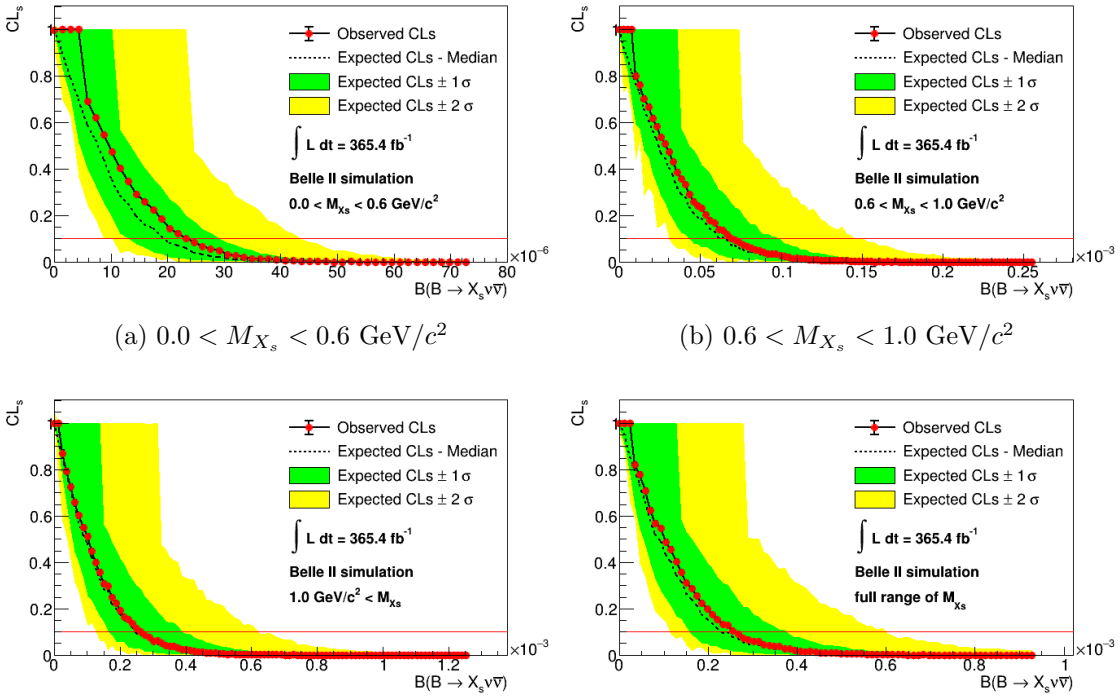
$$UL(B \to X_s \nu\bar{\nu}) = BR(B \to X_s \nu\bar{\nu})_{SM} \times UL(\mu), \tag{11.2}$$

where $UL(B \to X_s \nu\bar{\nu})$ is the branching-ratio upper limit, $BR(B \to X_s \nu\bar{\nu})_{\rm SM}$ is the SM prediction for the branching ratio, and $UL(\mu)$ is the upper limit on μ in each M_{X_s} region. The value of $BR(B \to X_s \nu\bar{\nu})_{\rm SM}$ in each M_{X_s} region is calculated from the signal MC sample based on the Standard Model.

In the MC, the expected branching ratios are

$$BR(B \to X_s \nu\bar{\nu})_{SM} = \frac{N_{B\to X_s \nu\bar{\nu}}}{N_B} = \begin{cases} 4.85 \times 10^{-6} & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 8.50 \times 10^{-6} & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 15.67 \times 10^{-6} & (1.0 \text{ GeV}/c^2 < M_{X_s}), \end{cases} \tag{11.3}$$

where $N_{B\to X_s \nu\bar{\nu}}$ is the number of B mesons that decay into $X_s \nu\bar{\nu}$ in a given M_{X_s} region, and N_B is the total number of B mesons. The corresponding branching-ratio upper limits



(a) $0.0 < M_{X_s} < 0.6$ GeV/c^2

(b) $0.6 < M_{X_s} < 1.0$ GeV/c^2

(c) 1.0 GeV/$c^2 < M_{X_s}$

(d) Full M_{X_s} range

Figure 11.1: CL_s values as functions of the tested branching ratio, obtained using the Asimov dataset.

are

$$UL(B \to X_s \nu\bar{\nu}) = \begin{cases} 2.33 \times 10^{-5} & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 6.93 \times 10^{-5} & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 26.25 \times 10^{-5} & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases} \tag{11.4}$$

Figure 11.1 shows the results of the CL_s method using the Asimov dataset.

11.2 Post-unblinding results

Figure 11.2 shows the bin-index distribution before the fit. The chi squared value for this distribution is

$$\chi^2_{15} = 54.3,$$

where the number of degrees of freedom is 15 because there are 15 bins. The p-value for this chi squared value is

$$\text{p-value} = \mathcal{P}(\chi^2_{15} > 54.3) < 0.01. \tag{11.5}$$

This low p-value mainly originates from the $0.6 < M_{X_s}^{\text{reco}} < 2.0$ GeV region, as shown in Figure 11.2.

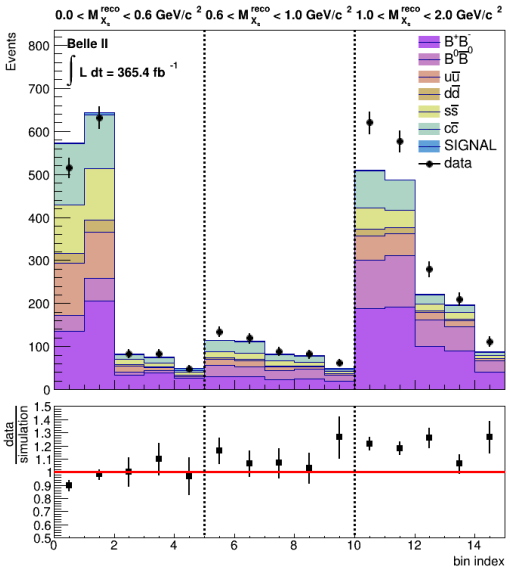


Figure 11.2: The bin-index distribution before the fit.

Figure 11.3 shows the bin-index distribution after the fit. The chi squared value for this distribution is

$$\chi^2_{12} = 5.4, \tag{11.6}$$

where the number of degrees of freedom is 12 because three free parameters are used for 15 bins. The p-value for this chi squared value is

$$\text{p-value} = \mathcal{P}(\chi^2_{12} > 5.4) = 0.94. \tag{11.7}$$

The fit result is

$$\mu = \begin{cases} 0.5^{+1.7}_{-1.6}(\text{stat})^{+1.7}_{-1.4}(\text{syst}) = 0.5^{+2.4}_{-2.1} & (0.0 < M_{X_s} < 0.6\ \text{GeV}/c^2), \\ 4.1^{+3.1}_{-3.0}(\text{stat})^{+3.6}_{-3.0}(\text{syst}) = 4.1^{+4.8}_{-4.2} & (0.6 < M_{X_s} < 1.0\ \text{GeV}/c^2), \\ 3.2^{+5.9}_{-5.6}(\text{stat})^{+8.1}_{-7.0}(\text{syst}) = 3.2^{+10.0}_{-9.0} & (1.0\ \text{GeV}/c^2 < M_{X_s}), \end{cases} \tag{11.8}$$

where the fitter bias is not considered.

However, as discussed in Section 10, a bias associated with the fitter is observed. To include this effect as a post-unblinding systematic uncertainty, a toy MC study is performed. A total of 10000 toy MC samples are generated with the μ values obtained in Eq. 11.8. The nuisance parameters are varied according to Gaussian distributions. The pull value is then calculated using Eq. 10.1. The systematic uncertainty from the fitter bias is calculated as follows:

$$\sigma_{\text{syst mean}} = \sigma_{\text{total}} \times \mu_{\text{pull}}, \tag{11.9}$$

$$\sigma_{\text{syst width}} = \sigma_{\text{total}} \times (\sigma_{\text{pull}} - 1), \tag{11.10}$$

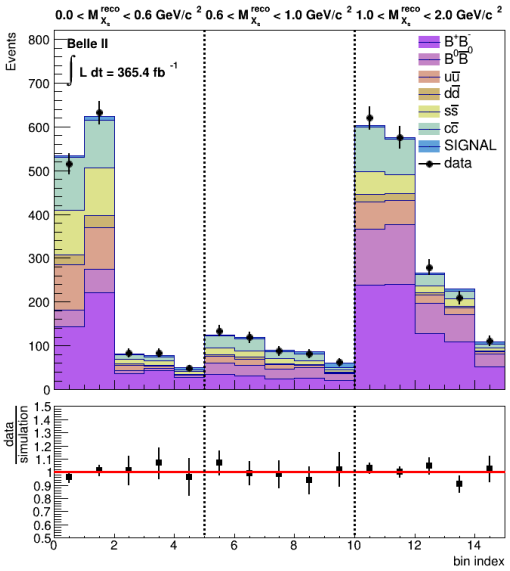


Figure 11.3: The bin-index distribution after the fit. The three signal strengths are set as free parameters.

where $\sigma_{\text{syst mean}}$ is the systematic uncertainty from the mean of the pull distribution, μ_{pull} is the mean of the pull distribution, $\sigma_{\text{syst width}}$ is the systematic uncertainty from the width of the pull distribution, σ_{pull} is the width of the pull distribution, and σ_{total} is the total uncertainty from the fit. Here, σ_{total} is obtained from the fit result.

After including this systematic uncertainty, the fit result is

$$\mu = \begin{cases} 0.5^{+1.7}_{-1.6}(\text{stat})^{+1.7}_{-1.5}(\text{syst}) = 0.5^{+2.4}_{-2.1} & (0.0 < M_{X_s} < 0.6\ \text{GeV}/c^2), \\ 4.1^{+3.1}_{-3.0}(\text{stat})^{+3.6}_{-3.0}(\text{syst}) = 4.1^{+4.8}_{-4.2} & (0.6 < M_{X_s} < 1.0\ \text{GeV}/c^2), \\ 3.2^{+5.9}_{-5.6}(\text{stat})^{+8.3}_{-7.0}(\text{syst}) = 4.6^{+10.1}_{-9.0} & (1.0\ \text{GeV}/c^2 < M_{X_s}), \end{cases} \tag{11.11}$$

When converted into branching ratios, the results become

$$BR(B \to X_s \nu\bar{\nu}) = \begin{cases} [0.3^{+0.8}_{-0.8}(\text{stat})^{+0.8}_{-0.7}(\text{syst})] \times 10^{-5} & (0.0 < M_{X_s} < 0.6\ \text{GeV}/c^2), \\ [3.5^{+2.7}_{-2.5}(\text{stat})^{+3.1}_{-2.6}(\text{syst})] \times 10^{-5} & (0.6 < M_{X_s} < 1.0\ \text{GeV}/c^2), \\ [5.1^{+9.2}_{-8.8}(\text{stat})^{+12.9}_{-11.0}(\text{syst})] \times 10^{-5} & (1.0\ \text{GeV}/c^2 < M_{X_s}). \end{cases} \tag{11.12}$$

When the three mass regions are combined, the signal strength and branching ratio are obtained as

$$\mu = 3.0^{+2.9}_{-2.8}(\text{stat})^{+4.4}_{-3.7}(\text{syst}) = 3.0^{+5.2}_{-4.7}, \tag{11.13}$$

and

$$BR(B \to X_s \nu\bar{\nu}) = [8.8^{+8.5}_{-8.2}(\text{stat})^{+12.6}_{-10.7}(\text{syst})] \times 10^{-5}. \tag{11.14}$$

Table 11.1 shows the impact of each systematic uncertainty on σ_μ. These values are determined using Eq. 9.9. Table 11.2 shows the impact on the branching ratios for each systematic uncertainty.

Because no evidence for a signal is observed, upper limits are calculated. For each M_{X_s} region, the upper limits are

$$UL(\mu) = \begin{cases} 4.5 & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 11.0 & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 19.7 & (1.0 \text{ GeV}/c^2 < M_{X_s}), \end{cases} \tag{11.15}$$

at the 90% confidence level, including the fitter-bias systematic uncertainty. When converted into branching-ratio upper limits, these become

$$UL(B \to X_s \nu\bar{\nu}) = \begin{cases} 2.2 \times 10^{-5} & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ 9.3 \times 10^{-5} & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2), \\ 30.9 \times 10^{-5} & (1.0 \text{ GeV}/c^2 < M_{X_s}), \end{cases} \tag{11.16}$$

When the three mass regions are combined, the upper limits are

$$UL(\mu) = 11.3, \tag{11.17}$$

and

$$UL(B \to X_s \nu\bar{\nu}) = 32.7 \times 10^{-5}, \tag{11.18}$$

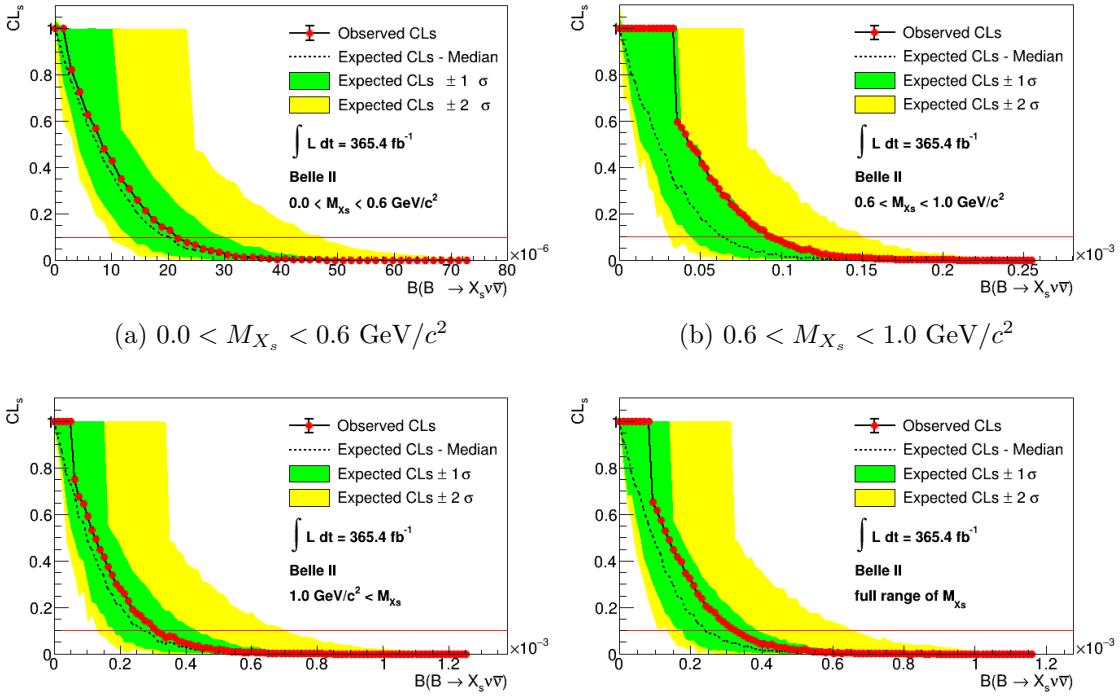
at the 90% confidence level, including the fitter-bias systematic uncertainty. Figure 11.4 shows the results of the CL_s method.

Table 11.1: Summary of the systematic and statistical uncertainties for the three signal-strength parameters. The values are obtained from data.

Source	Impact on σ_{μ_1}	Impact on σ_{μ_2}	Impact on σ_{μ_3}
Background normalization	1.17	2.34	4.19
MC statistics	0.90	1.78	4.04
FastBDT efficiency	0.10	0.49	1.44
$K^* - X_s$ transition	0.01	0.04	1.40
BR of main B meson decays	0.25	0.72	1.30
$q\bar{q}$ efficiency	0.19	0.25	1.16
Fitter bias	0.31	0.07	1.08
Photon multiplicity correction	0.05	0.29	1.02
Additional FEI calibration factor	0.17	0.46	0.63
FEI calibration for $B\bar{B}$	0.15	0.33	0.51
Fragmentation	0.02	0.08	0.46
$q\bar{q}$ shape	0.22	0.27	0.46
Fermi motion momentum	0.00	0.00	0.44
$B\overline{B}$ counting	0.04	0.16	0.38
$B \to K^*$ form factor	0.03	0.59	0.36
BR of $B \to X_s K_L^0 K_L^0$	0.01	0.39	0.35
Fraction of decay modes	0.13	0.32	0.28
b-quark mass	0.00	0.00	0.22
BR of $B \to X_s n\bar{n}$ decay	0.13	0.49	0.16
$B \to K$ form factor	0.13	0.00	0.16
Pion ID	0.01	0.09	0.15
BR of $D \to K_L^0 X$ from B meson	0.01	0.11	0.12
Tracking efficiency	0.01	0.03	0.10
π^0 reconstruction	0.01	0.04	0.09
K_S^0 reconstruction	0.02	0.04	0.05
K_L^0 ECL efficiency	0.01	0.02	0.05
Kaon ID	0.06	0.02	0.04
Statistical uncertainty	1.62	3.04	5.75

Table 11.2: Summary of the systematic and statistical uncertainties in the full range and in each mass region. The values are obtained from data.

	Impact on σ_{BR} $[10^{-5}]$ in $M_{X_s}^{\mathrm{reco}}$ regions $[\mathrm{GeV}/c^2]$			
Source	Full range	$[0, 0.6]$	$[0.6, 1.0]$	$[1.0, M_{X_s}^{\mathrm{max}}]$
MC statistics	6.0	0.4	1.5	6.3
Background normalization	5.7	0.6	2.0	6.6
BR of main B meson decays	2.3	0.1	0.6	2.0
$K^* - X_s$ transition	2.1	0.0	0.0	2.2
FastBDT efficiency	2.1	0.0	0.4	2.3
Photon multiplicity correction	1.8	0.0	0.2	1.6
$q\bar{q}$ efficiency	1.8	0.1	0.2	1.8
Fitter bias	1.6	0.2	0.1	1.7
FEI calibration for $B\bar{B}$	1.1	0.1	0.3	0.8
Additional FEI calibration factor	1.0	0.1	0.4	1.0
$q\bar{q}$ shape	1.0	0.1	0.2	0.7
BR of $B \to X_s K_L^0 K_L^0$	0.9	0.0	0.3	0.6
$B\overline{B}$ counting	0.7	0.0	0.1	0.6
Fermi motion momentum	0.7	0.0	0.0	0.7
Fragmentation	0.6	0.0	0.1	0.7
$B \to K^*$ form factor	0.6	0.0	0.5	0.6
BR of $B \to X_s n\bar{n}$ decay	0.5	0.1	0.4	0.2
Fraction of decay modes	0.4	0.1	0.3	0.4
b-quark mass	0.3	0.0	0.0	0.3
BR of $D \to K_L^0 X$ from B meson	0.3	0.0	0.1	0.2
Pion ID	0.2	0.0	0.1	0.2
$B \to K$ form factor	0.2	0.1	0.0	0.2
Tracking efficiency	0.2	0.0	0.0	0.1
π^0 reconstruction	0.1	0.0	0.0	0.1
Kaon ID	0.1	0.0	0.0	0.1
K_S^0 reconstruction	0.1	0.0	0.0	0.1
K_L^0 ECL efficiency	0.1	0.0	0.0	0.1
Total systematic sources	11.7	0.8	2.8	12.0
Statistical uncertainty	8.3	0.8	2.6	9.0



(a) $0.0 < M_{X_s} < 0.6$ GeV/c^2

(b) $0.6 < M_{X_s} < 1.0$ GeV/c^2

(c) 1.0 GeV/$c^2 < M_{X_s}$

(d) Full M_{X_s} range

Figure 11.4: CL_s values as functions of the tested branching ratio, obtained using data.

Appendix A

FastBDT inputs

Table A.1: List of input variables used for the FastBDT. Unless otherwise specified, the ROE denotes the rest of the event with respect to B_{tag}. The reconstruction of the D-meson candidates is described in Section 4.6.3. A total of 32 variables are used.

FastBDT variable	Description
`Bsig_cosTBTO`	Cosine of the angle between the thrust axis of the X_s candidate and that of the ROE.
`Bsig_KSFWVariables_hso01`, `Bsig_KSFWVariables_hso04`	KSFW variables [72] of the X_s candidate.
`Bsig_thrustBm`	Magnitude of the thrust of the X_s candidate.
`Bsig_useCMSFrame_p`	Momentum of the X_s candidate in the center-of-mass frame.
`Btag_CleoConeCS_1`, `Btag_CleoConeCS_2`, `Btag_CleoConeCS_3`	Cleo-cone variables [73] used for continuum suppression.
`Btag_cosTBTO`	Cosine of the angle between the thrust axis of the B_{tag} candidate and that of the ROE.
`Btag_KSFWVariables_hoo1`, `Btag_KSFWVariables_hoo2`, `Btag_KSFWVariables_hoo3`, `Btag_KSFWVariables_hoo4`, `Btag_KSFWVariables_hso02`, `Btag_KSFWVariables_hso24`	KSFW variables of the B_{tag} candidate.
`Btag_useCMSFrame_theta`	Polar angle of the B_{tag} candidate in the center-of-mass frame.

Continued on the next page

Table A.1 continued from the previous page

FastBDT variable	Description
`extraInfo__boEeclv200`	Extra energy deposited in the ECL.
`extraInfo__boNgammav200__bc`	Number of photon candidates in the ROE.
`foxWolframR3`, `foxWolframR4`	Ratios of the third- and fourth-order Fox–Wolfram moments to the zeroth-order moment, respectively [66].
`harmonicMomentThrust1`, `harmonicMomentThrust2`	Harmonic moments calculated with respect to the thrust axis [66].
`missingMomentumOfEvent`	Magnitude of the missing momentum in the laboratory frame.
`missingMomentumOfEvent_theta`	Polar angle of the missing momentum in the laboratory frame.
`nRemainingTracksInEvent`	Number of tracks in the ROE before applying additional track selections.
`roePTheta__bocleanMask__bc`	Polar angle of the ROE momentum. Here, the ROE is defined with respect to the B_{tag} and X_s candidates.
`Bsig_daughter_0_extraInfo_ Dc_pValue_std_noDCS`	Standard deviation of the vertex-fit p-values for $D^{\pm}$ in the first category.
`Bsig_daughter_0_extraInfo_ Dcsimpleveto_M_noDCS`	Mass of the $D^{\pm}$ candidate with the highest vertex-fit p-value in the first category.
`Bsig_daughter_0_extraInfo_ D0_pValue_std`	Standard deviation of the vertex-fit p-values for D^0 in the second category.
`Bsig_daughter_0_extraInfo_ D0simpleveto_M`	Mass of the D^0 candidate with the highest vertex-fit p-value in the second category.
`Bsig_daughter_0_extraInfo_ D0_pValue_std_yespizero`	Standard deviation of the vertex-fit p-values for D^0 in the fourth category.
`Bsig_daughter_0_extraInfo_ D0simpleveto_M_yespizero`	Mass of the D^0 candidate with the highest vertex-fit p-value in the fourth category.

Appendix B

Efficiency as a function of M_{X_S}

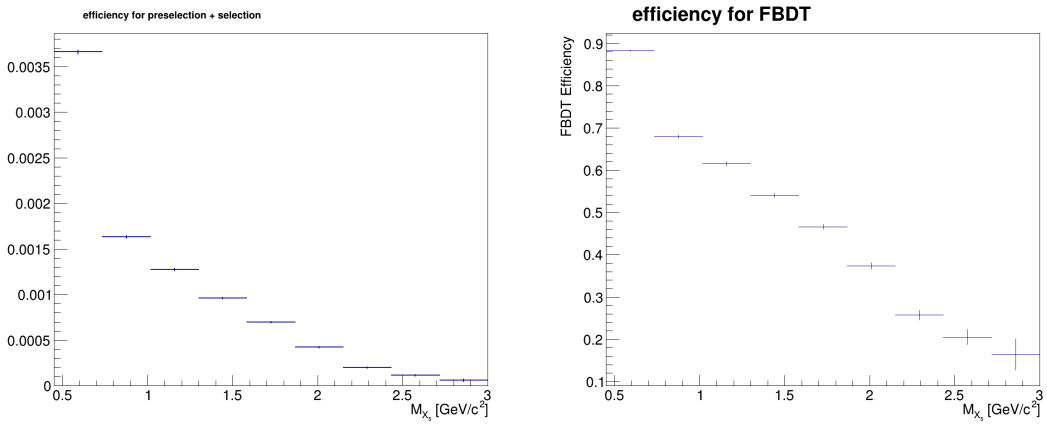


Figure B.1: Signal efficiency as a function of the true X_s mass, M_{X_s}. The left plot shows the combined efficiency of the preselection and selection requirements, and the right plot shows the efficiency of the FastBDT output requirement. The efficiency decreases with increasing M_{X_s}.

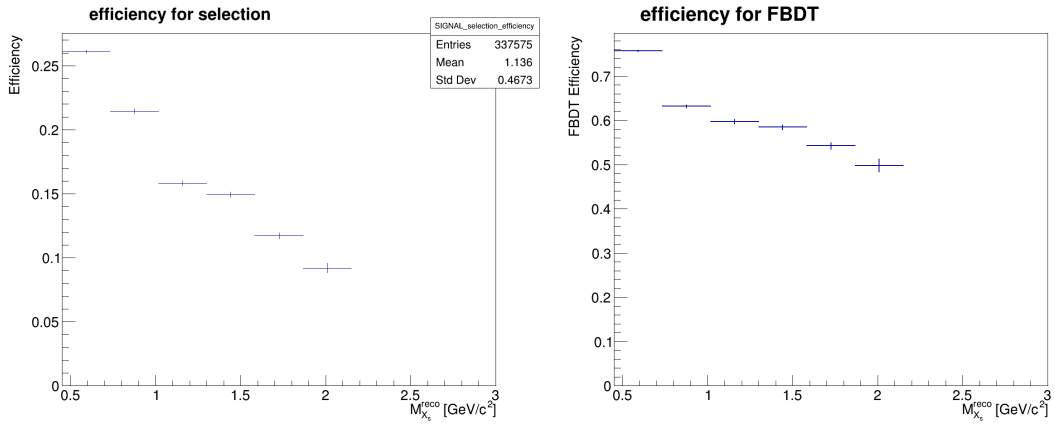


Figure B.2: Signal efficiency as a function of the reconstructed X_s mass, $M_{X_s}^{\text{reco}}$. The left plot shows the efficiency of the selection requirements, and the right plot shows the efficiency of the FastBDT output requirement. The efficiency decreases with increasing $M_{X_s}^{\text{reco}}$.

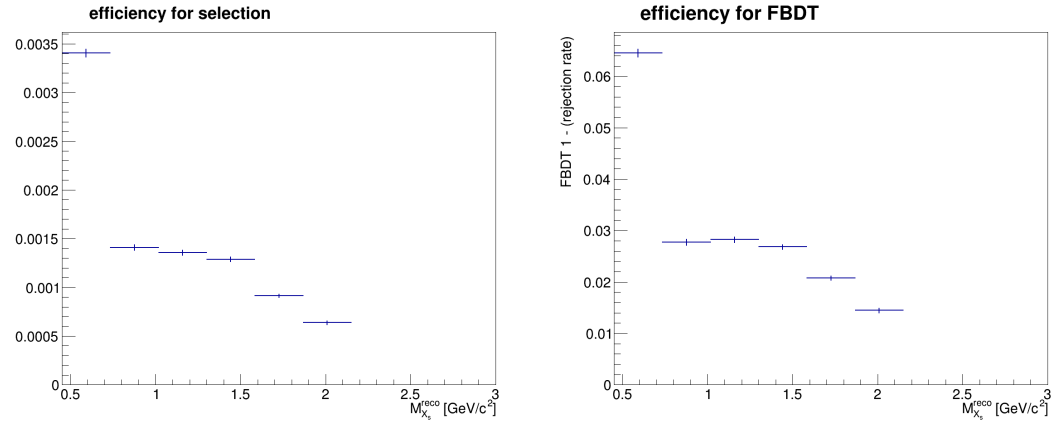


Figure B.3: Background efficiency as a function of the reconstructed X_s mass, $M_{X_s}^{\text{reco}}$. The left plot shows the efficiency of the selection requirements, and the right plot shows the efficiency of the FastBDT output requirement. The background efficiency decreases with increasing $M_{X_s}^{\text{reco}}$.

Appendix C

FastBDT input variable distributions

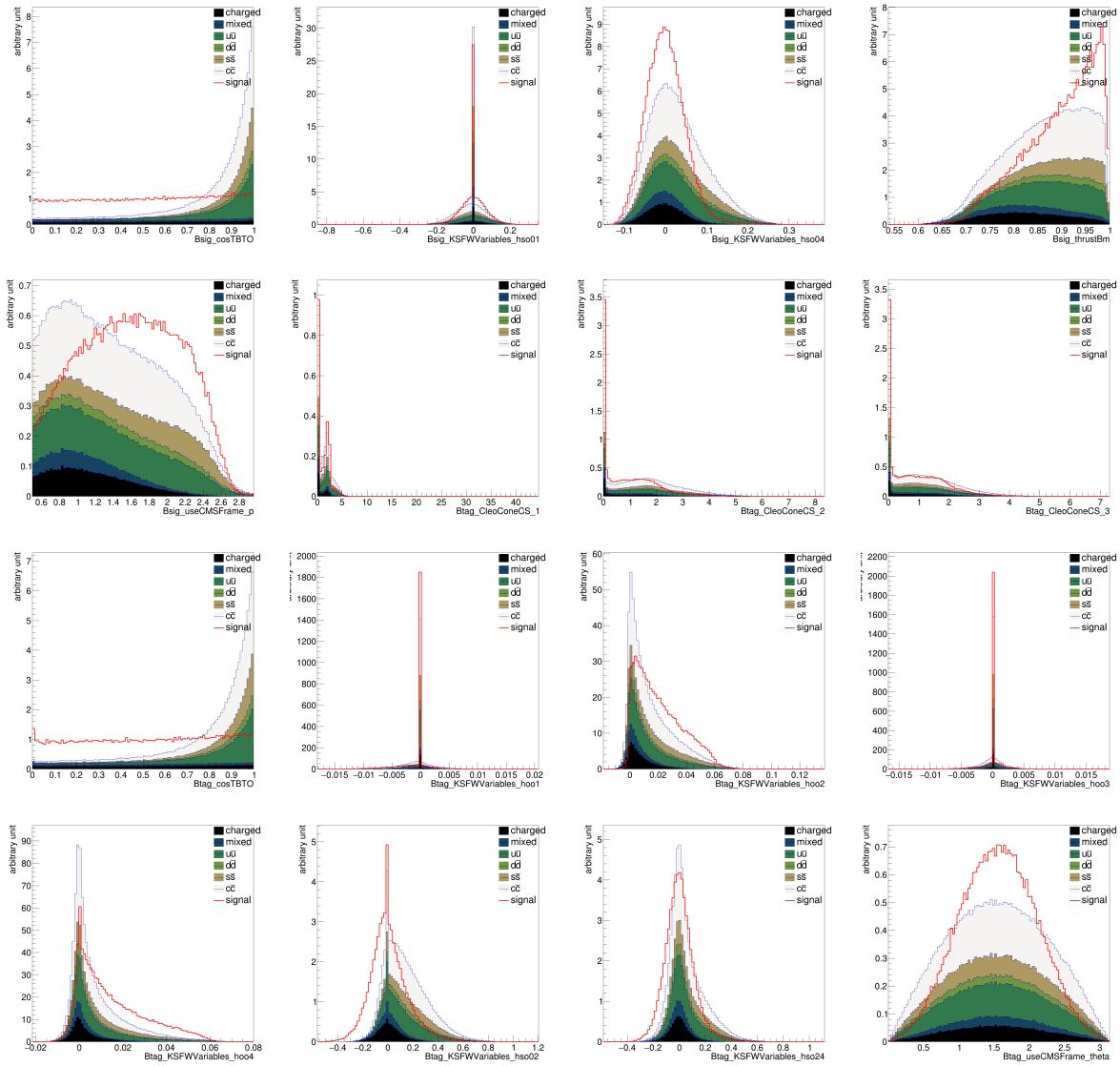


Figure C.1: FastBDT input variables.

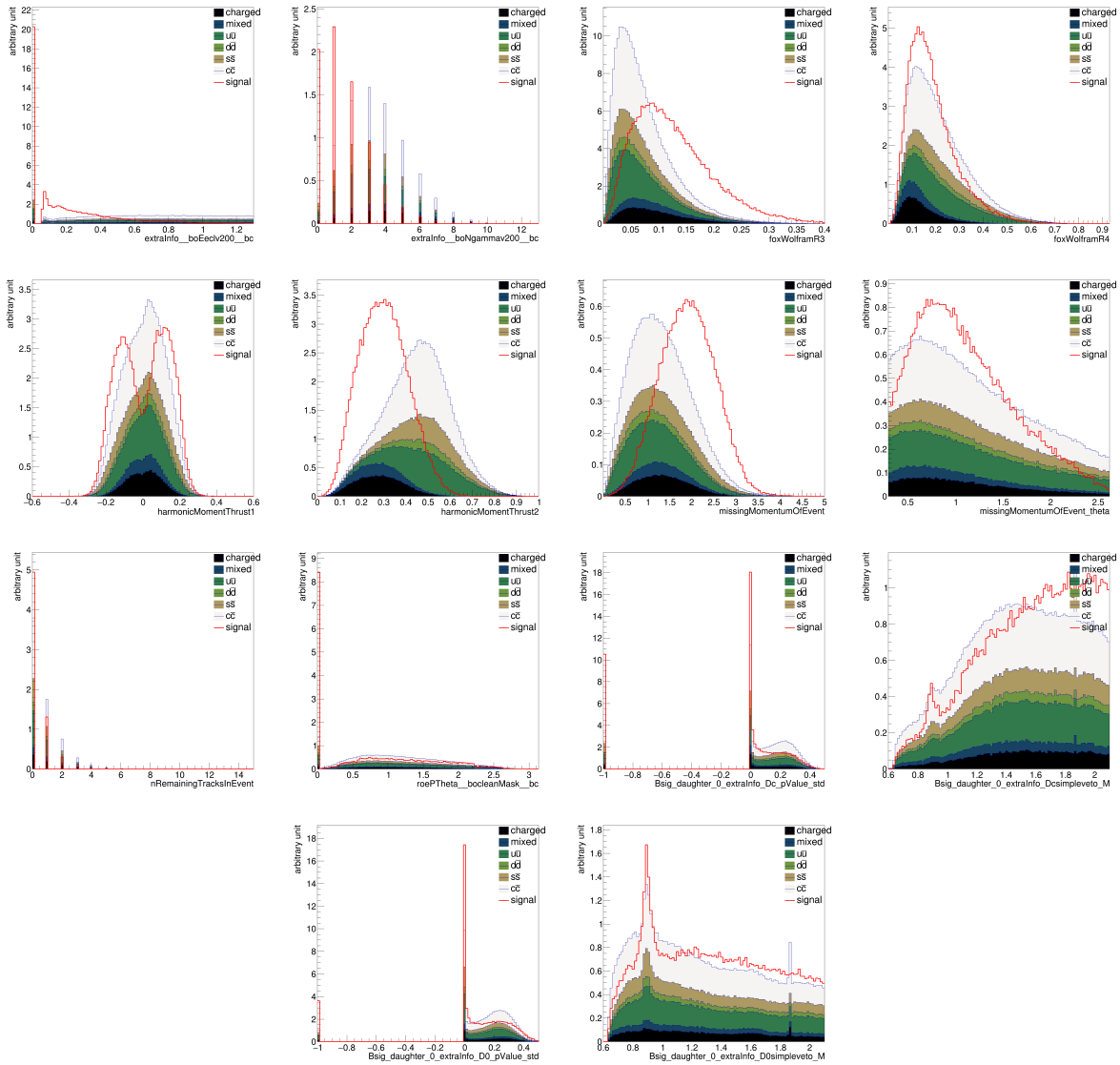


Figure C.2: FastBDT input variables.

Appendix D

Off-resonance variable distributions

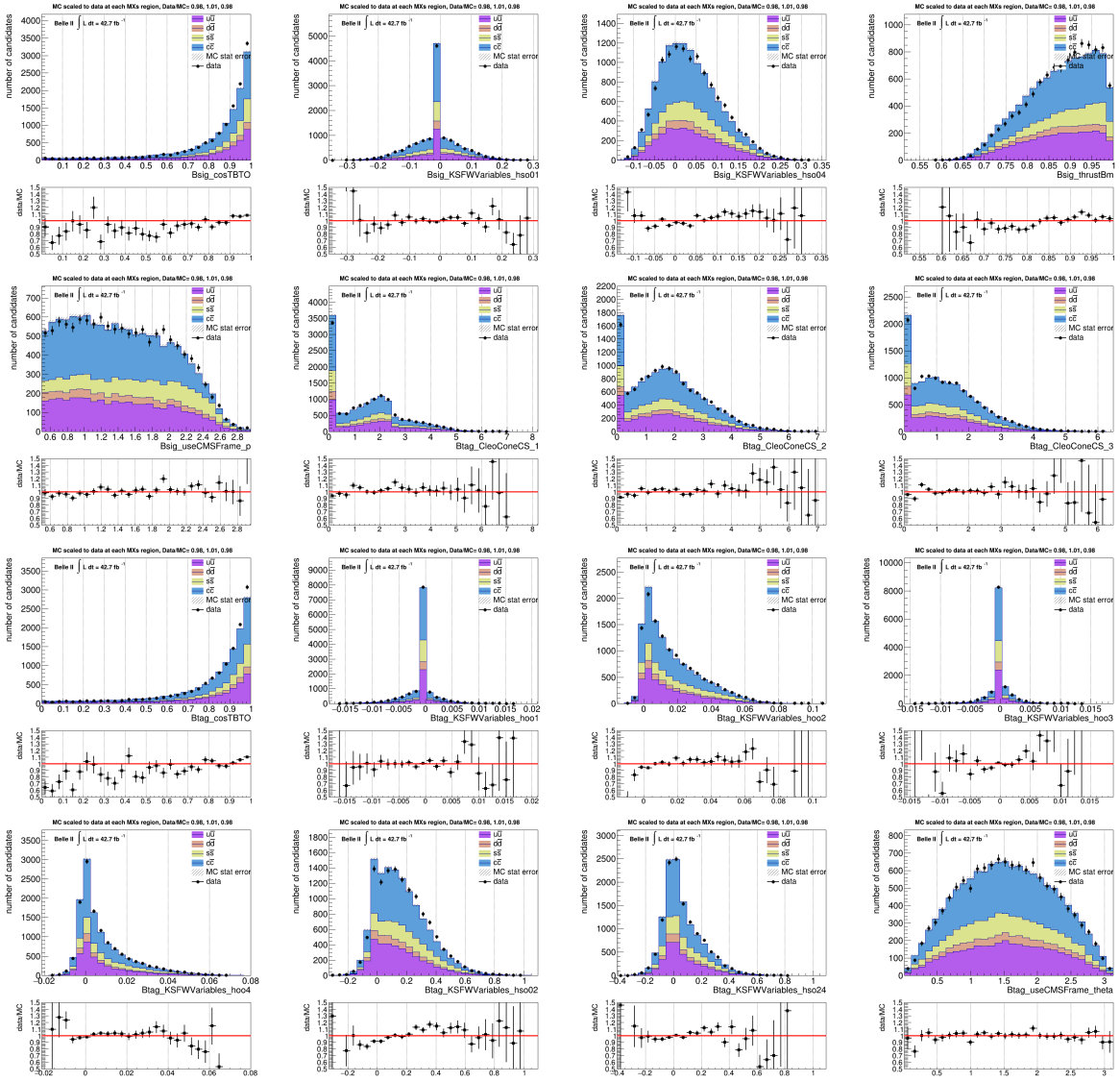


Figure D.1: FastBDT input variables.

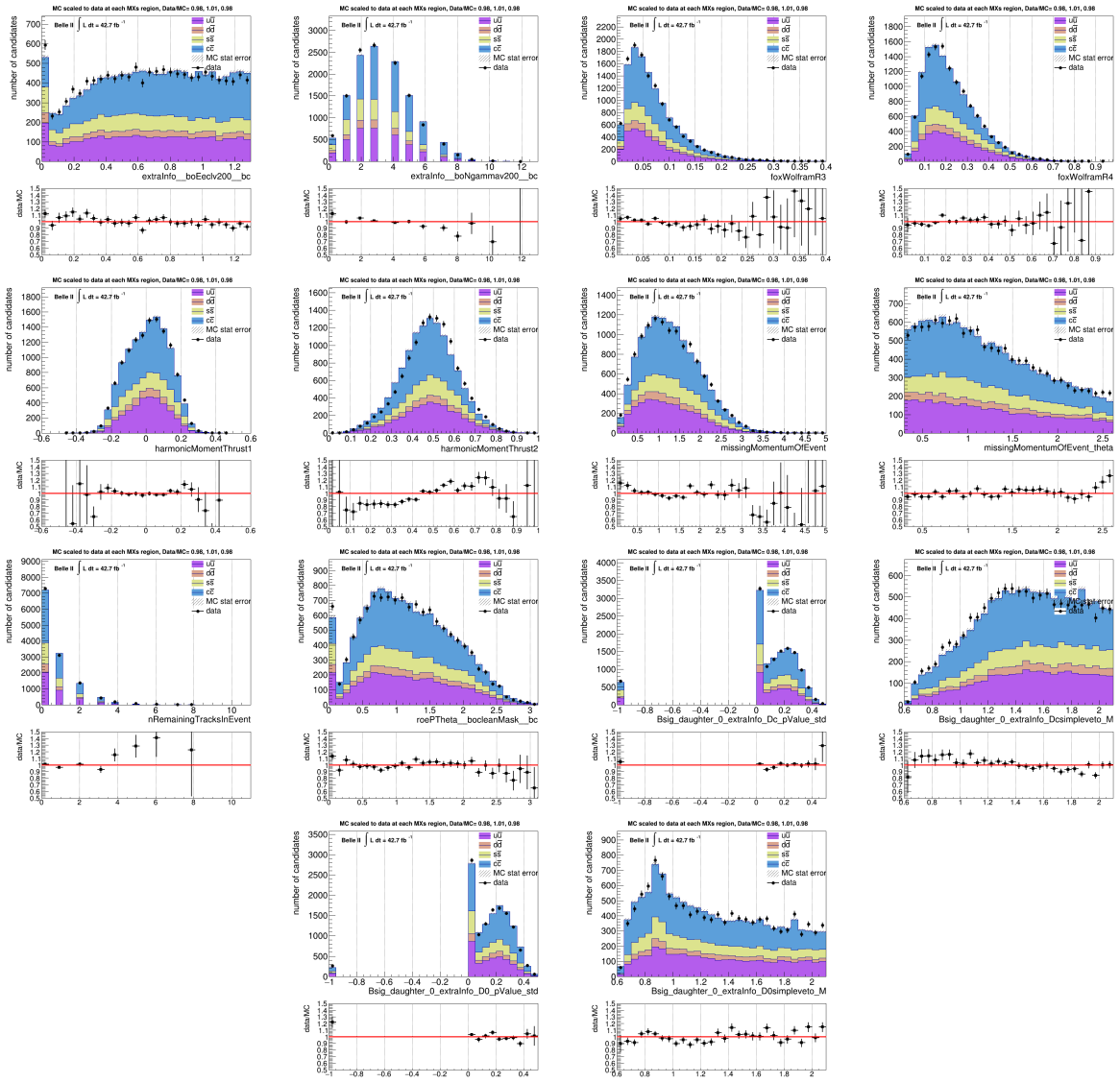


Figure D.2: FastBDT input variables.

Appendix E

M_{bc} Sideband variable distributions

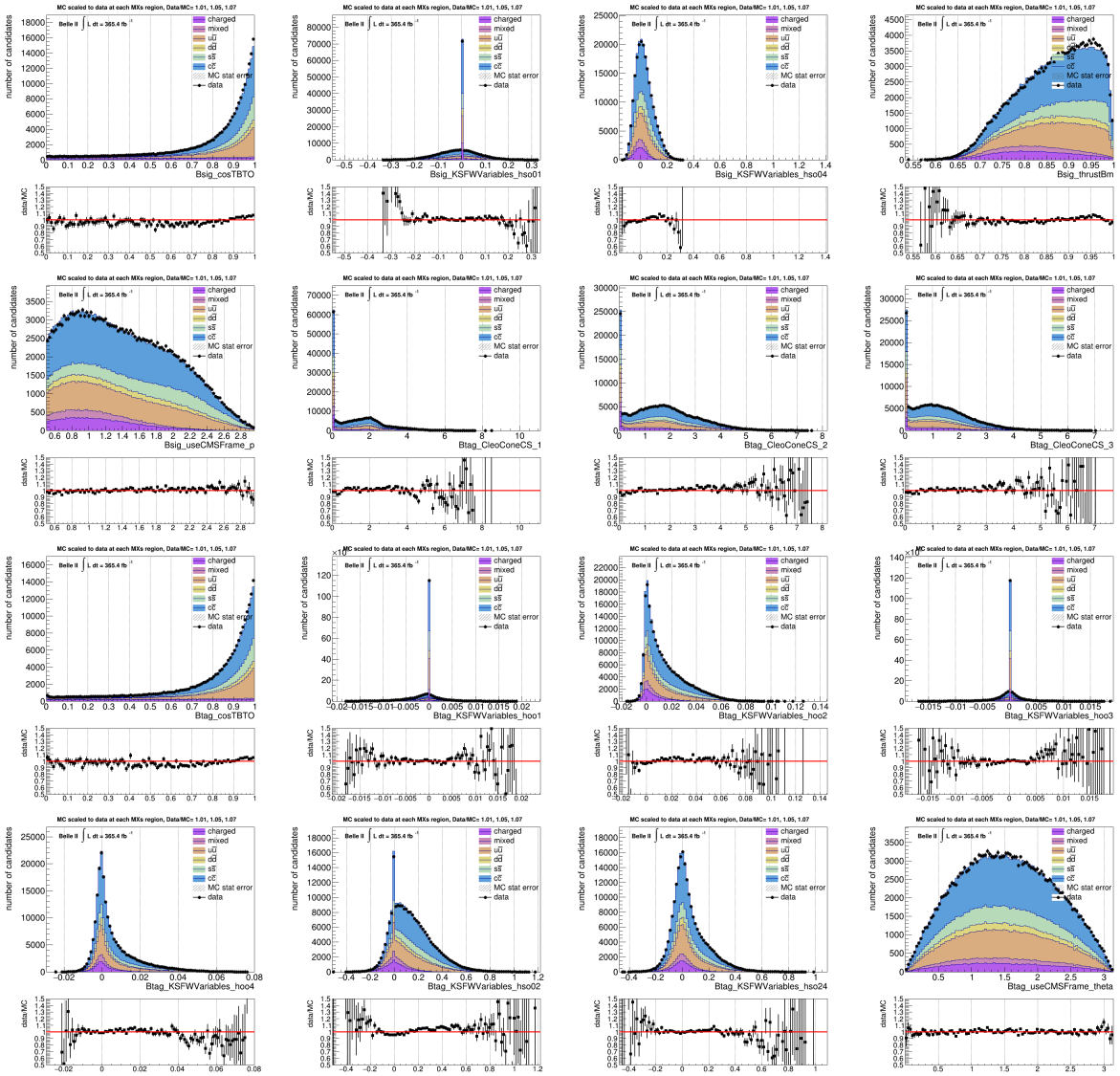


Figure E.1: FastBDT input variables.

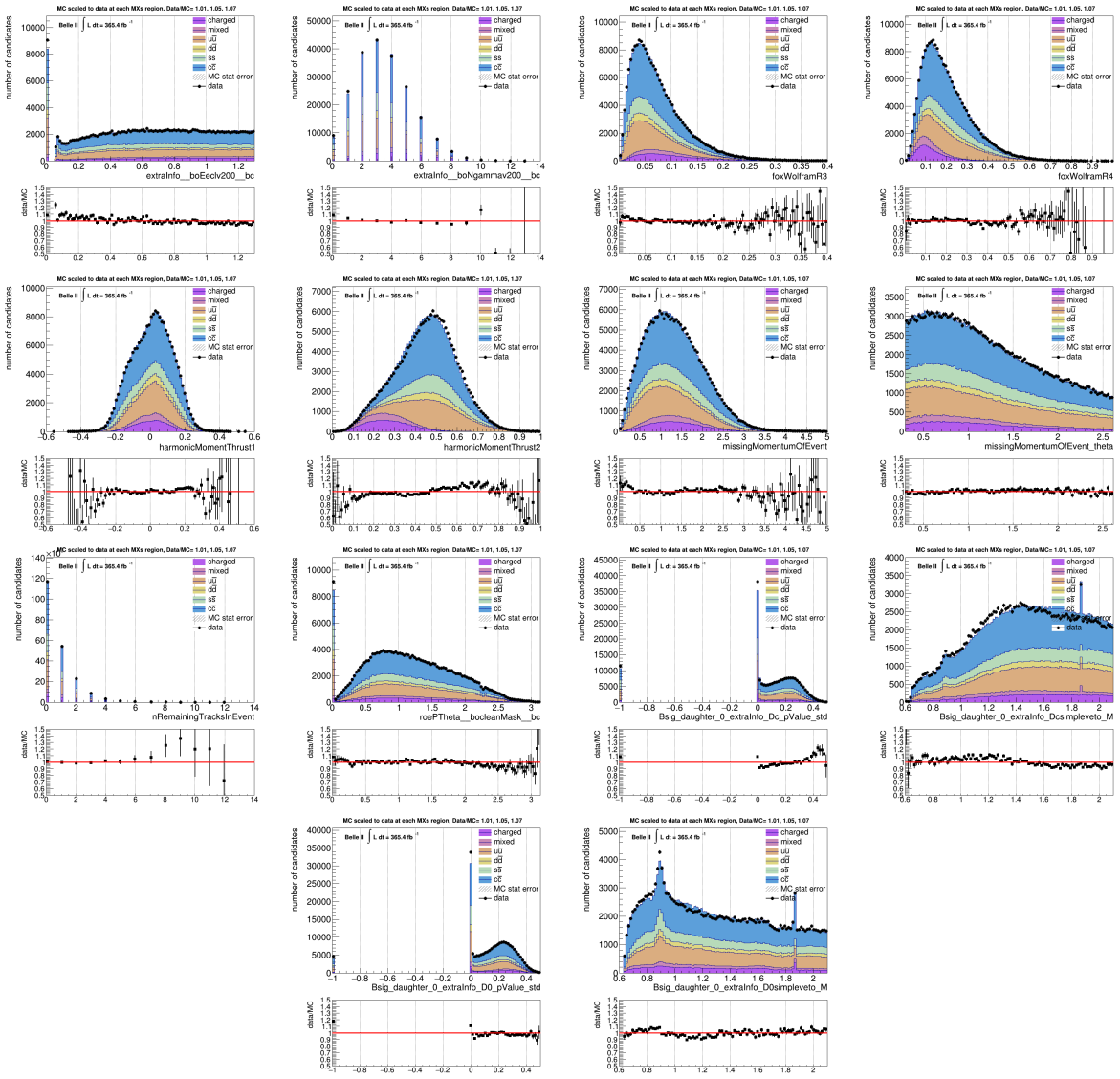


Figure E.2: FastBDT input variables.

Appendix F

$B \to X_s J/\psi$ variable distributions

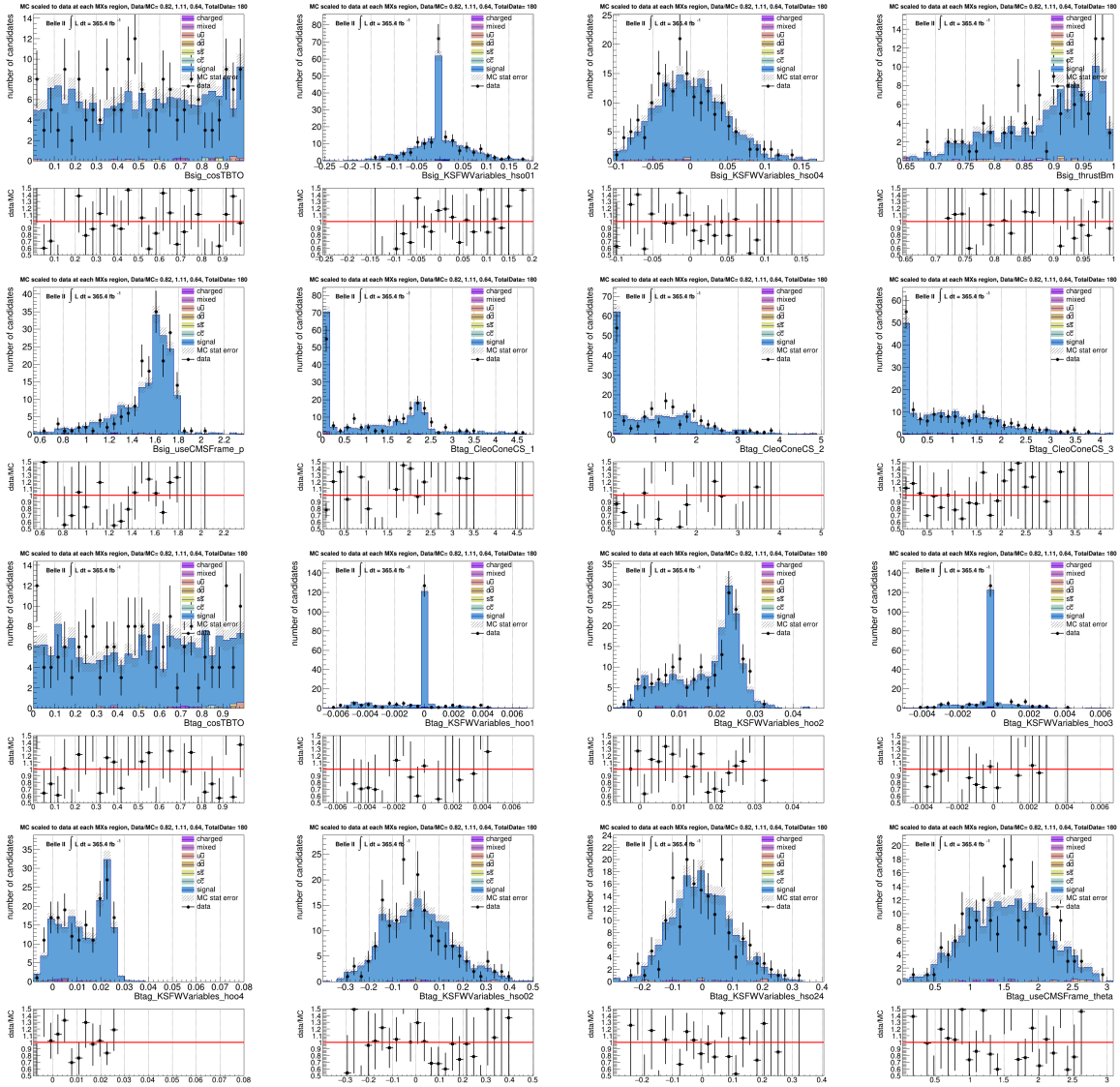


Figure F.1: FastBDT input variables.

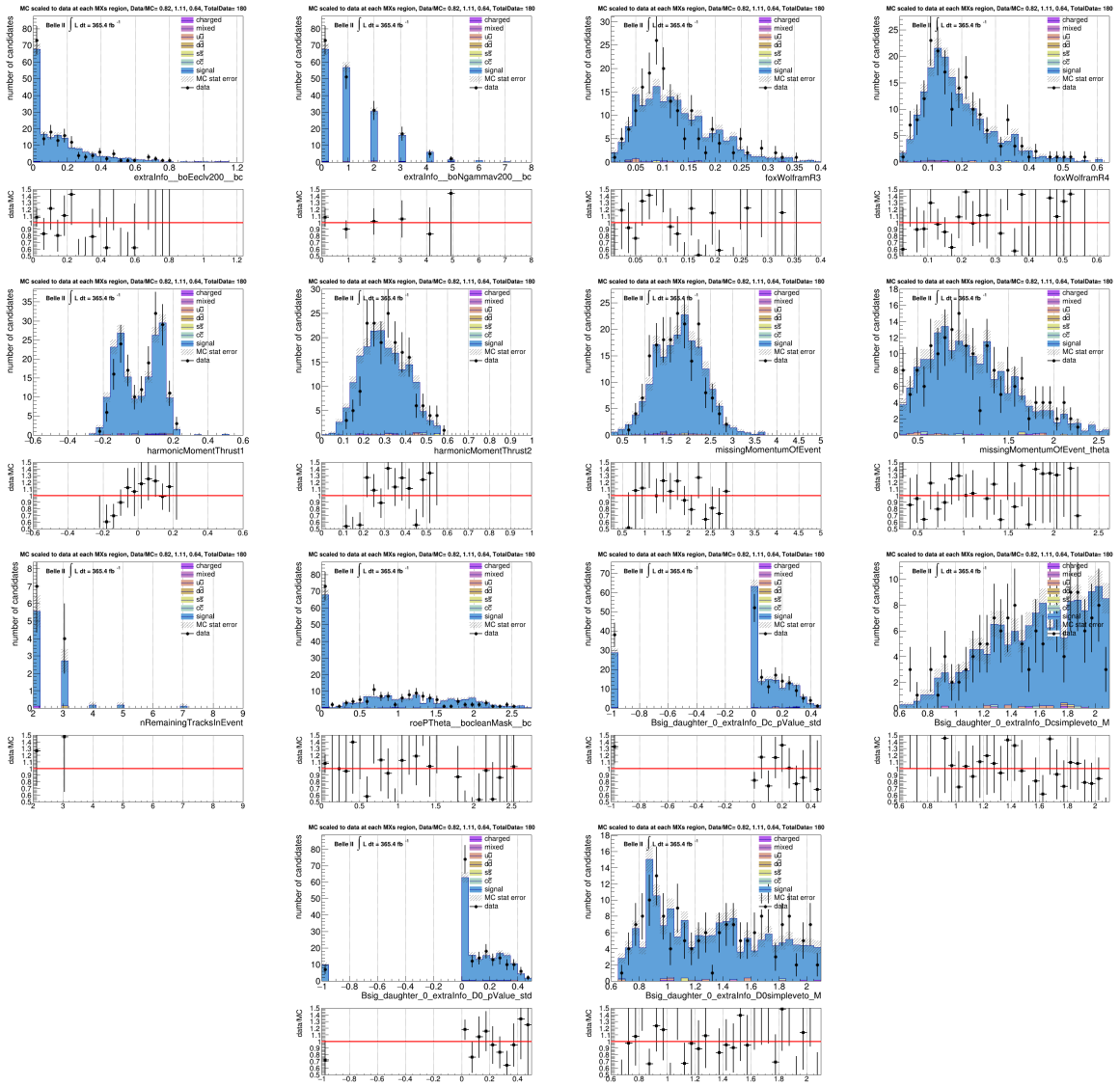


Figure F.2: FastBDT input variables.

Appendix G

PID correction table

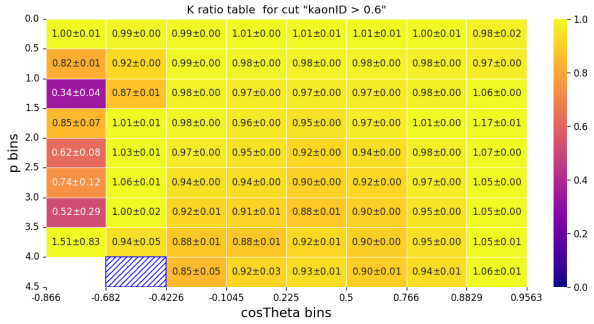


Figure G.1: Data-to-MC ratio for KaonID efficiency. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

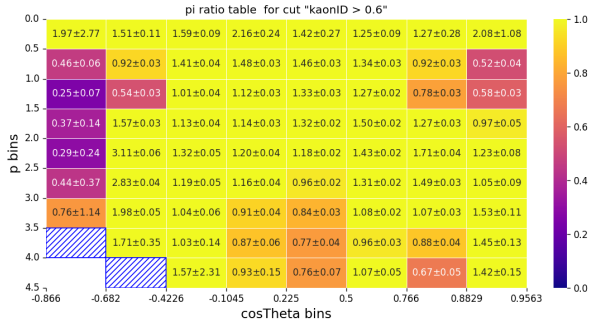


Figure G.2: Data-to-MC ratio for the rate at which pions are misidentified as kaons. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

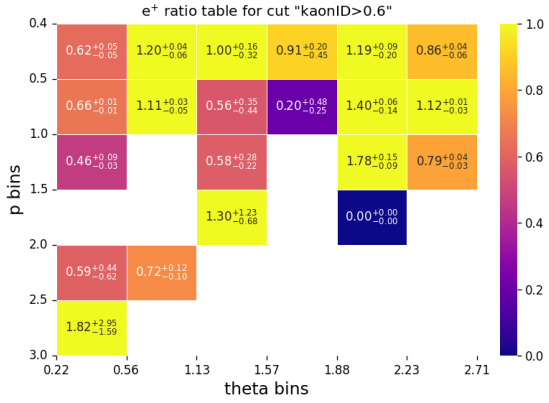


Figure G.3: Data-to-MC ratio for the rate at which e^+ candidates are misidentified as kaons. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

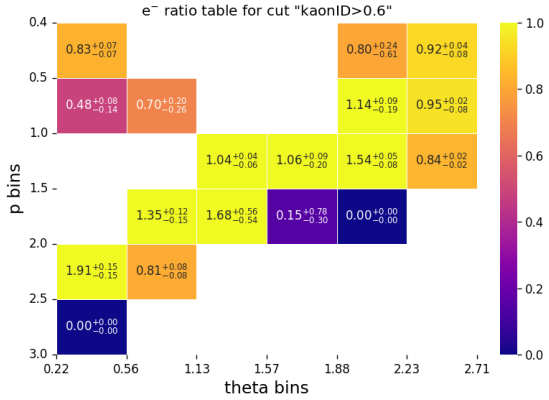


Figure G.4: Data-to-MC ratio for the rate at which e^- candidates are misidentified as kaons. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

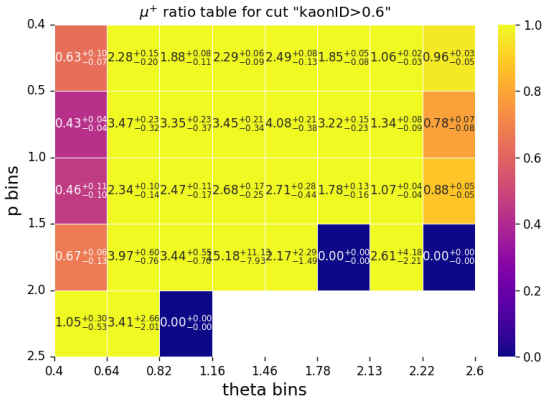


Figure G.5: Data-to-MC ratio for the rate at which μ^+ candidates are misidentified as kaons. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

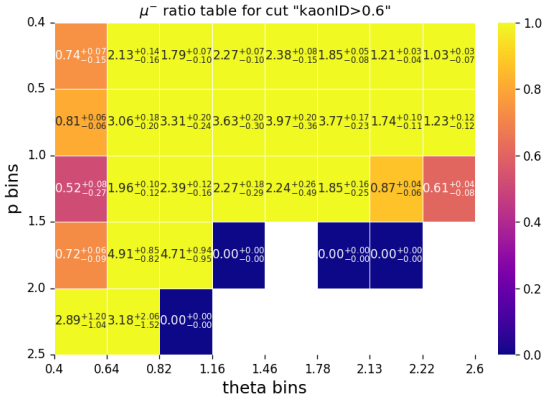


Figure G.6: Data-to-MC ratio for the rate at which μ^- candidates are misidentified as kaons. The requirement `kaonID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

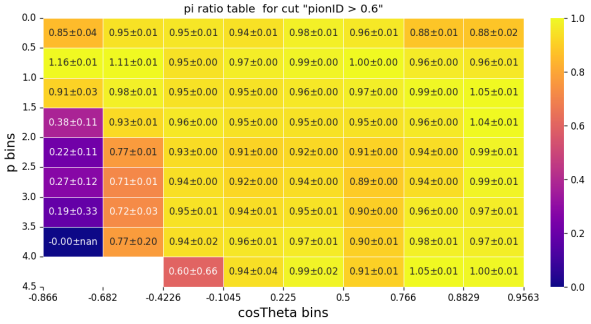


Figure G.7: Data-to-MC ratio for PionID efficiency. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

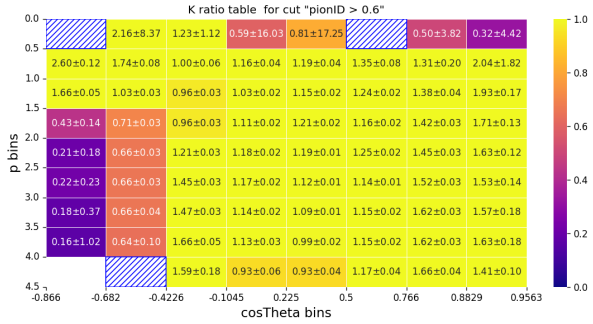


Figure G.8: Data-to-MC ratio for the rate at which kaons are misidentified as pions. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

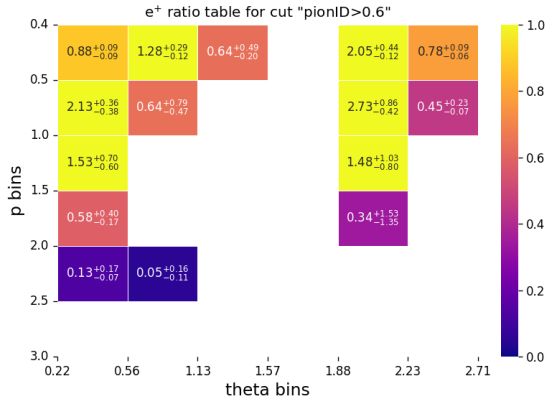


Figure G.9: Data-to-MC ratio for the rate at which e^+ candidates are misidentified as pions. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

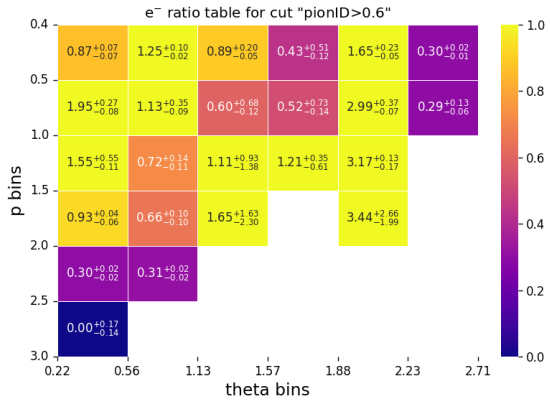


Figure G.10: Data-to-MC ratio for the rate at which e^- candidates are misidentified as pions. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

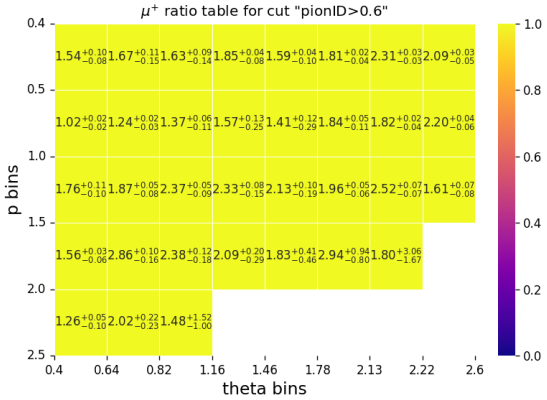


Figure G.11: Data-to-MC ratio for the rate at which μ^+ candidates are misidentified as pions. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

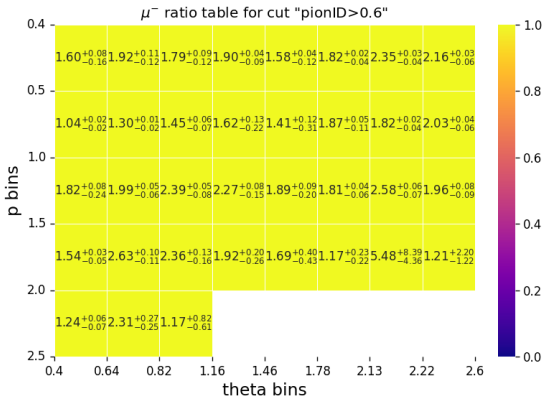


Figure G.12: Data-to-MC ratio for the rate at which μ^- candidates are misidentified as pions. The requirement `pionID` > 0.6 is applied. `nCDCHits` > 20 and `nPXDHits` > 0 are applied as preselection criteria.

Appendix H

Composition of hadron candidates

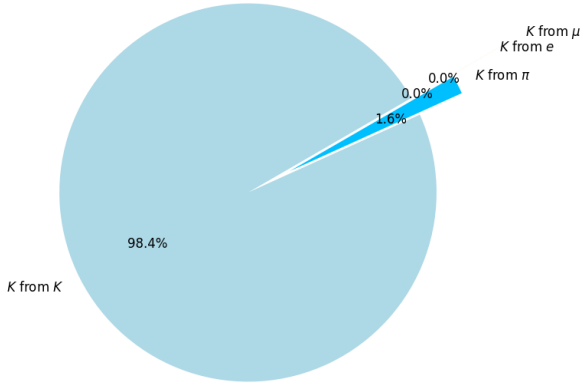


Figure H.1: Composition of kaon candidates in the signal sample. The fractions are calculated after all selection requirements are applied.

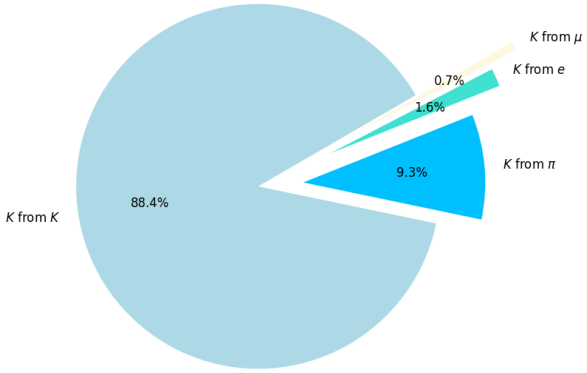


Figure H.2: Composition of kaon candidates in the background sample. The fractions are calculated after all selection requirements are applied.

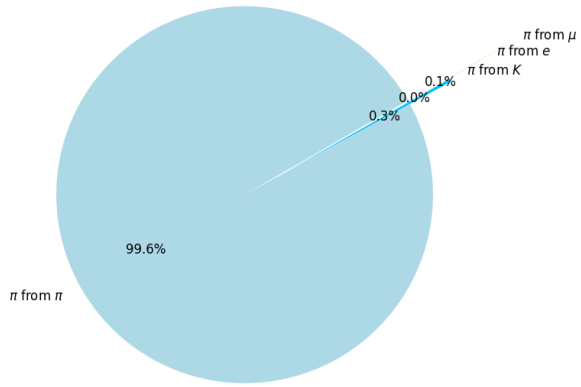


Figure H.3: Composition of pion candidates in the signal sample. The fractions are calculated after all selection requirements are applied.

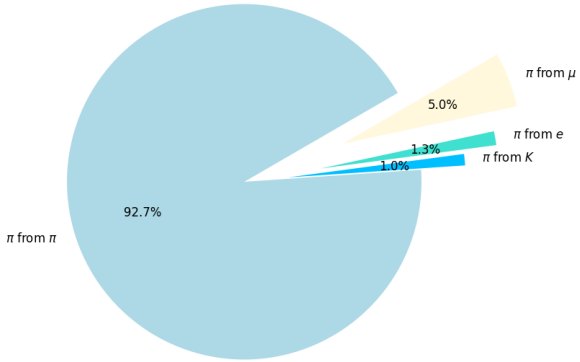


Figure H.4: Composition of pion candidates in the background sample. The fractions are calculated after all selection requirements are applied.

Appendix I

Comparison between $K_1(1270)\nu\bar{\nu}$ and $K\pi\pi\nu\bar{\nu}$ generated with PYTHIA

In the default PYTHIA settings, resonances such as $K_1(1270)$ are not produced. To investigate this effect, the $B \to K_1(1270)\nu\bar{\nu}$ decay is compared with $B \to K\pi\pi\nu\bar{\nu}$ decays generated with PYTHIA. Here, the $B \to K_1(1270)\nu\bar{\nu}$ sample is generated using a flat phase-space model. Figures I.1, I.2, I.3, and I.4 show the efficiency distributions as functions of M_{X_s} for several final states.

The efficiencies are found to be consistent between the two samples. In addition, the efficiency mainly depends on the multiplicity of X_s, as shown in Figure 4.9. Consequently, for the mismodelling associated with PYTHIA, only the systematic uncertainty from the fragmentation among the $nK\,m\pi$ modes is considered.

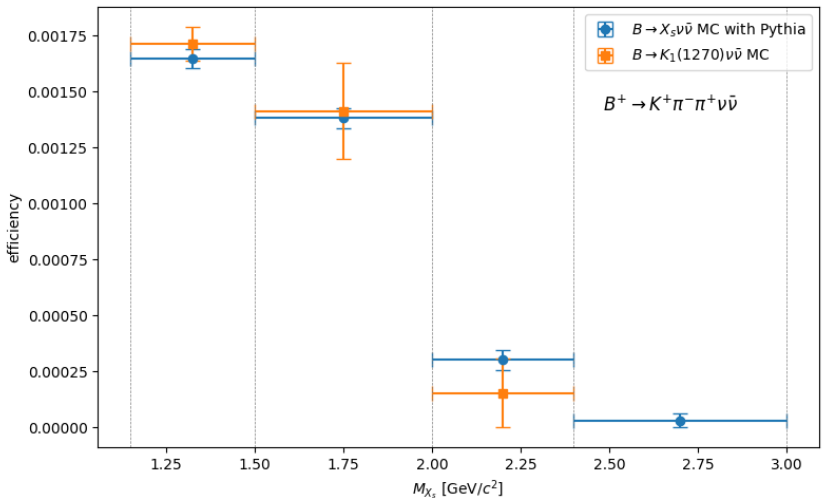


Figure I.1: Efficiency as a function of M_{X_s} for $B^+ \to K^+\pi^-\pi^+\nu\bar{\nu}$ decays. The blue points show the efficiency for the sample generated with PYTHIA. The orange points show the efficiency for the $B \to K_1(1270)\nu\bar{\nu}$ sample.

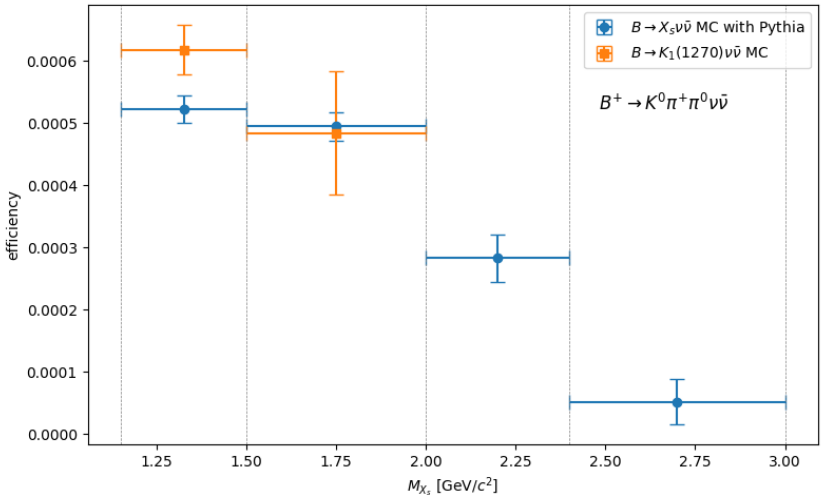


Figure I.2: Efficiency as a function of M_{X_s} for $B^+ \to K^0\pi^+\pi^0\nu\bar{\nu}$ decays. The blue points show the efficiency for the sample generated with `PYTHIA`. The orange points show the efficiency for the $B \to K_1(1270)\nu\bar{\nu}$ sample.

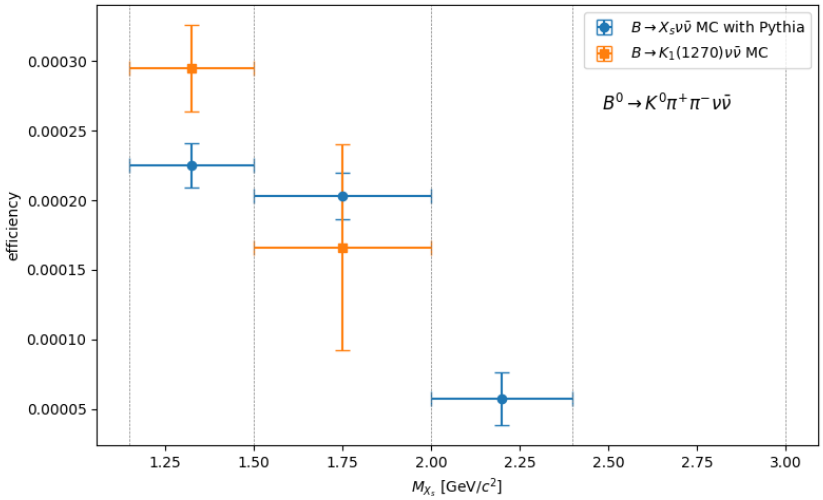


Figure I.3: Efficiency as a function of M_{X_s} for $B^0 \to K^0\pi^+\pi^-\nu\bar{\nu}$ decays. The blue points show the efficiency for the sample generated with `PYTHIA`. The orange points show the efficiency for the $B \to K_1(1270)\nu\bar{\nu}$ sample.

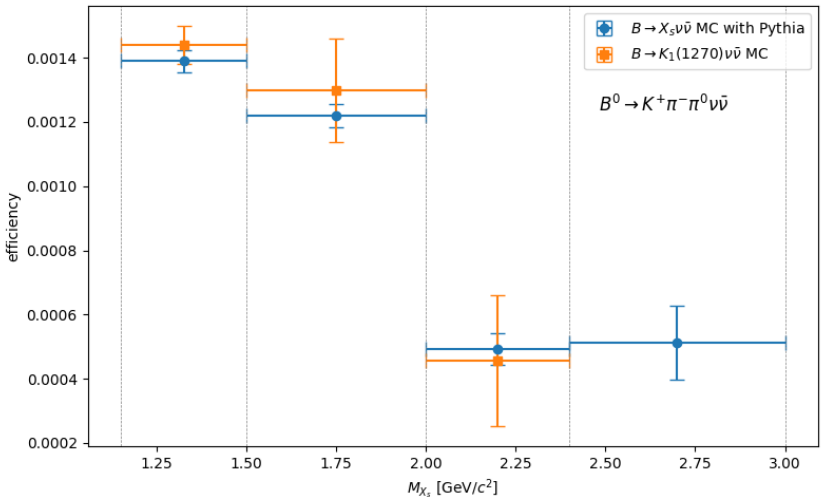


Figure I.4: Efficiency as a function of M_{X_s} for $B^0 \to K^+\pi^-\pi^0\nu\bar{\nu}$ decays. The blue points show the efficiency for the sample generated with `PYTHIA`. The orange points show the efficiency for the $B \to K_1(1270)\nu\bar{\nu}$ sample.

Appendix J

Impact of the S-wave component in $K\pi\nu\bar{\nu}$ decays

In this analysis, only the $B \to K^*\nu\bar{\nu}$ sample is generated in the mass region $0.6 < M_{X_s} < 1.0$ GeV/c^2. However, an S-wave component may also contribute in this mass region. In this section, the impact of the S-wave component is studied.

To estimate the impact of the S-wave component, an S-wave MC sample is generated. The branching fraction of the S-wave component is estimated using the LHCb study in Ref. [48]. The S-wave fraction F_S is used for this estimate. Assuming no interference between the S-wave and P-wave components, the S-wave fraction is written as

$$F_S = \frac{BR(\text{S-wave})}{BR(\text{P-wave}) + BR(\text{S-wave})}. \tag{10.1}$$

The relative fraction of the P-wave and S-wave components is obtained from this relation. The S-wave MC sample is generated using the $B \to K\nu\bar{\nu}$ model. Figure J.1 shows the fit-variable distributions for the P-wave-only sample and the sample including both P-wave and S-wave components in $0.6 < M_{X_s} < 1.0$ GeV/c^2. A difference in the shape is observed. Therefore, a fit is performed including this shape difference as a systematic uncertainty. However, the fit result does not change at the first decimal place:

$$BR(B \to X_s\nu\bar{\nu}) = \begin{cases} [0.3^{+0.8}_{-0.8}(\text{stat})^{+0.9}_{-0.7}(\text{syst})] \times 10^{-5} & (0.0 < M_{X_s} < 0.6 \text{ GeV}/c^2), \\ [3.5^{+2.6}_{-2.5}(\text{stat})^{+3.1}_{-2.6}(\text{syst})] \times 10^{-5} & (0.6 < M_{X_s} < 1.0 \text{ GeV}/c^2). \\ [5.1^{+9.2}_{-8.8}(\text{stat})^{+12.9}_{-11.0}(\text{syst})] \times 10^{-5} & (1.0 \text{ GeV}/c^2 < M_{X_s}). \end{cases}$$

Therefore, the impact of the S-wave component is concluded to be negligible, and it is not included in the nominal analysis.

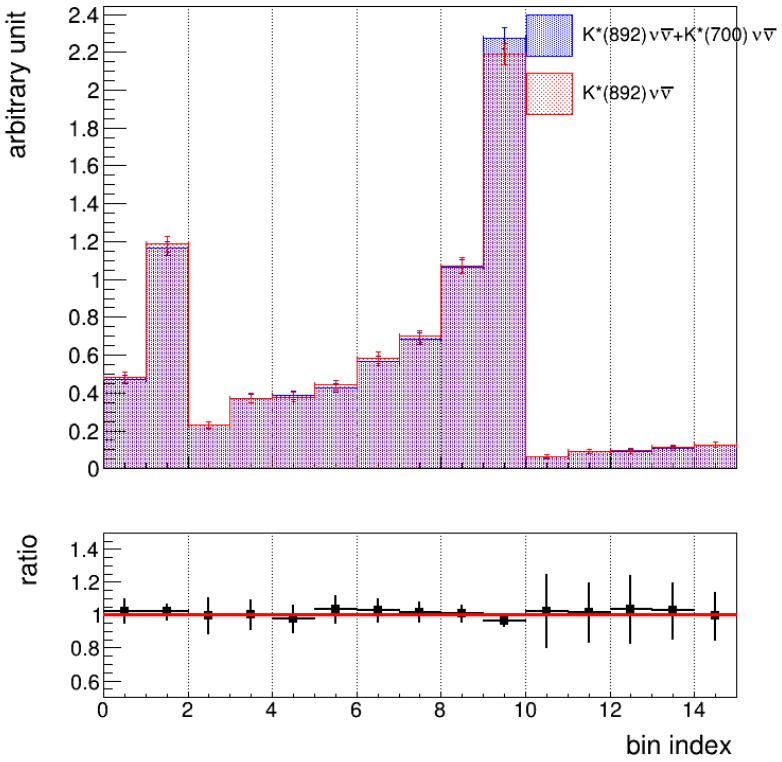


Figure J.1: Fit-variable distributions for the P-wave-only sample and the sample including both P-wave and S-wave components in $0.6 < M_{X_s} < 1.0$ GeV/c^2. The distribution including both components is normalized to the P-wave-only distribution. The uncertainty in the ratio is not meaningful because the same P-wave sample is used in both distributions.

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